DG CLIMA

Technical Assessment Paper (TAP) for long-term temporary biogenic carbon storage in buildings

(with Certification Rules Report)

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VIEGAND MAAGØE

Report:	**Technical Assessment Paper (TAP) for long-term temporary biogenic carbon storage in buildings (with Certification Rules Report)**
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Executive Summary

This Technical Assessment Paper supports the development of the EU certification methodology for biogenic carbon storage in buildings under Regulation (EU) 2024/3012 (the CRCF Regulation). It refines the 2024 draft by integrating expert feedback, recent regulatory developments, and alignment with EU standards and policies. The TAP establishes the technical basis for future Certification Rules and a delegated act defining the official EU methodology.

Purpose and Scope

The TAP focuses on quantifying, certifying, and ensuring the environmental integrity of biogenic carbon stored in durable construction products and buildings with service lives of **≥ 35 years**. It aims to provide consistent methods for determining the net carbon removal benefit, covering quantification, additionality, storage/liability, and sustainability dimensions in line with the CRCF's QU.A.L.ITY principles.

Regulatory and Policy Linkages

The methodology must interface with key EU frameworks:

- **Energy Performance of Buildings Directive (EU) 2024/1275:** mandates life-cycle GWP assessment of new buildings from 2028/2030, enabling future CRCF baselines.
- **Construction Products Regulation (EU) 2024/3110:** introduces harmonised product rules for declarations of performance and conformity (DoPCs) including life cycle assessments for GWP-biogenic, GWP-total, and GWP-LULUC impacts.
- **EN 15804, EN 15978, EN 16485, EN 16449:** provide standardised LCA rules and carbon-content calculations for wood-based materials.
- **EU Taxonomy, and potentially RED (Art. 29):** define "Do No Significant Harm" and sustainable land management criteria applicable to biomass sourcing and building activities.

Quantification (Article 4)

- **Eligible materials**: primarily structural wood and biobased insulation products meeting ≥ 35 year lifespans.
- **Building scope**: potentially all buildings so long as they have a suitable lifetime, relevant baseline and comply with the sustainability DNSH criteria. Can apply to new construction or renovation projects.
- **GHG-associated impacts**: count all GWP components (biogenic, fossil, LULUC) across modules A1–A5.
- **Baseline units**: separate standardised baselines are recommended: in units of kg CO_2/m^2 for structural wood and in units of % biobased share for insulation.
- **Standardised baseline**: generally preferred to activity-specific baselines, and can initially be built using data from DG GROW studies or other sources and then be refined as data from life cycle assessments becomes available from EPBD obligations at national level.
- **Uncertainty**: should be handled conservatively at product level using PEF, EN 15804 or Level(s) data quality indicators.

Additionality (Article 5)

Two baseline approaches are proposed:

- Standardised baselines for typical typologies to ensure comparability and simplicity.
- Activity-specific baselines for projects with unique design or material performance or for common building typologies when a standardised baseline is not yet available.
- Worked examples are provided to illustrate how these baselines could be calculated in theory.

Storage, Monitoring and Liability (Article 6)

- **Liability mechanisms**: evaluation of buffer pools and insurance options to address potential carbon reversal.
- **Issuance models**: comparison between upfront and termed issuance and how termed issuance reduces liability exposure.
- **Beyond-35-year storage**: certification schemes are free to issue certificates for periods longer than 35 years, but this will increase the required monitoring period and associated liability. If a certified unit for a 35-year

storage period is still stored beyond this point, it will expire and be cancelled in the certification registry unless the operator and certification scheme commit to further monitoring.

Sustainability (Article 7)

- Translates EU Taxonomy DNSH and Technical Screening Criteria into simplified requirements for CRCF projects.
- Assessment of the potential applicability of the RED Article 29 sustainability requirements, with a possible flexible approach considered for demonstrating compatibility with RED Article 29.

Key Recommendations

- Prioritise **structural wood and biobased insulation** as eligible elements.
- Eventually develop **harmonised, standardised baselines** linked to EPBD whole-life-carbon data.
- Ensure **consistency** among EPBD, CPR, and EN standards to guarantee credible quantification.
- Integrate **uncertainty management** at product level and **sustainability safeguards** at both product and building level.
- Allow for liability-free **termed certificate issuance** and require robust liability mechanisms for upfront issuance of certificates.

Content

1 Introduction

This Technical Assessment Paper (TAP) on long-term storage of biogenic carbon in buildings replaces an earlier draft prepared in September 2024 by Partners for Innovation and Wageningen University & Research. The initial TAP reviewed existing certification methodologies and identified methodological challenges. This version refines key concepts, incorporates expert feedback, and reflects regulatory and standardisation developments up to mid-2025. It is intended to serve as the technical basis for the associated Certification Rules Report (CRR) and draft Certification Methodology (CM) under the EU Carbon Removals and Carbon Farming Certification (CRCF) Regulation.

1.1 Brief overview of Regulation (EU) 2024/3012 and activity types

Regulation (EU) 2024/3012, adopted on 27 November 2024, establishes a Union certification framework for permanent carbon removals, carbon farming, and carbon storage in products (hereafter referred to as the CRCF Regulation). More specifically, these certifiable activity groups refer to:

- Permanent carbon removals, which are practices that capture and store carbon for several centuries, including chemically bound carbon in products.
- Carbon farming, which covers land-based practices that store carbon or reduce soil emissions over a minimum of five years.
- Carbon storage in products, which covers practices that store atmospheric or biogenic carbon in long-lasting products for at least 35 years, with on-site monitoring throughout the period.

Once CO2 is removed from the atmosphere through photosynthesis into plant-based biomass, different CRCF activities may utilise the same biomass feedstock, which requires clear boundaries in certification. The main competing options are to generate:

- Permanent carbon removal units by harvesting the biomass and converting it into biochar, then incorporating the biochar into soil.
- Permanent carbon removal units by harvesting the biomass, burning it, and capturing the exhaust gas CO2 using carbon capture and storage (BioCCS) technologies.
- Carbon farming (CF) units through continued biomass growth, and possibly through increases in soil carbon stocks by simply leaving the trees or plants in place to grow further.
- Carbon storage in product units by harvesting the biomass and converting it into durable construction materials that are likely to last at least 35 years in buildings.

All activities must adhere to a standard methodology (denoted QU.A.L.ITY), which is based on the four principles listed below:

- **QUantification**: how to calculate net carbon removals for any specific activity (explained in Article 4 of the CRCF Regulation).
- **Additionality**: how to demonstrate that a particular activity exceeds the status quo or baseline (explained in Article 5 of the CRCF Regulation).
- **Liability**: how to demonstrate that the activity continues to meet the requirement for permanent or temporary storage of net removed carbon over potentially long periods of time (explained in Article 6 of the CRCF Regulation).
- **sustainabilITY**: how to demonstrate that the activity does no significant harm in other areas (explained in Article 7 of the CRCF Regulation).

Parallel TAPs are under development for other CRCF activities (e.g. soil carbon, peatlands, biochar, BECCS, DACCS[1]), providing a coordinated foundation for delegated acts specifying EU certification methodologies.

[1] BECCS stands for BioEnergy with Carbon Capture and Storage; DACCS stands for Direct Air Capture with Carbon Capture and Storage.

1.2 Purpose of this TAP

The updated version of this TAP concerning the activity of temporary storage of biogenic carbon in buildings aims to:

- Clarify conceptual aspects identified as unclear in the 2024 draft and in expert feedback.
- Provide practical examples to improve consistency of interpretation.
- Incorporate any recent developments since the draft TAP was published (Sept. 2024).
- Include regulatory, policy, and standardisation developments since late 2024.
- Present more concrete proposals on aspects that remained unresolved in the draft TAP.
- Serve as the technical foundation for the Certification Rules Report (CRR), which will underpin a formal EU Certification Methodology (CM) for this activity.

Annexes provide supplementary detail and references to keep the core of this TAP more focused and concise.

1.3 General concept of temporary storage of biogenic carbon in buildings

Under the CRCF Regulatory framework, biogenic carbon storage in buildings refers to atmospheric CO_2 that has been captured through photosynthesis, incorporated into biomass, and subsequently embedded in construction products with a minimum expected service life in buildings of 35 years.

Figure 1 illustrates the overall system boundary: carbon uptake during biomass growth (green arrows) and associated upstream emissions from harvesting, processing, transport, and installation (red arrows).

- **Carbon uptake**: The amount of CO_2 removed depends on the mass of biomass utilised and its carbon content. When forestry and agricultural activities are managed sustainably - avoiding the depletion of standing biomass, soil carbon loss, or excessive use of fertilisers - this stage results in a large net removal.
- **Upstream emissions from processing and installation**: Converting raw materials into wood- or bio-based products involves energy consumption, process inputs (e.g. water), transportation to site, and auxiliary materials (e.g. adhesives, packaging) and produces waste to be disposed of, wastewater to be treated and emissions to air. These processes all generate embodied emissions, which must be accounted for.

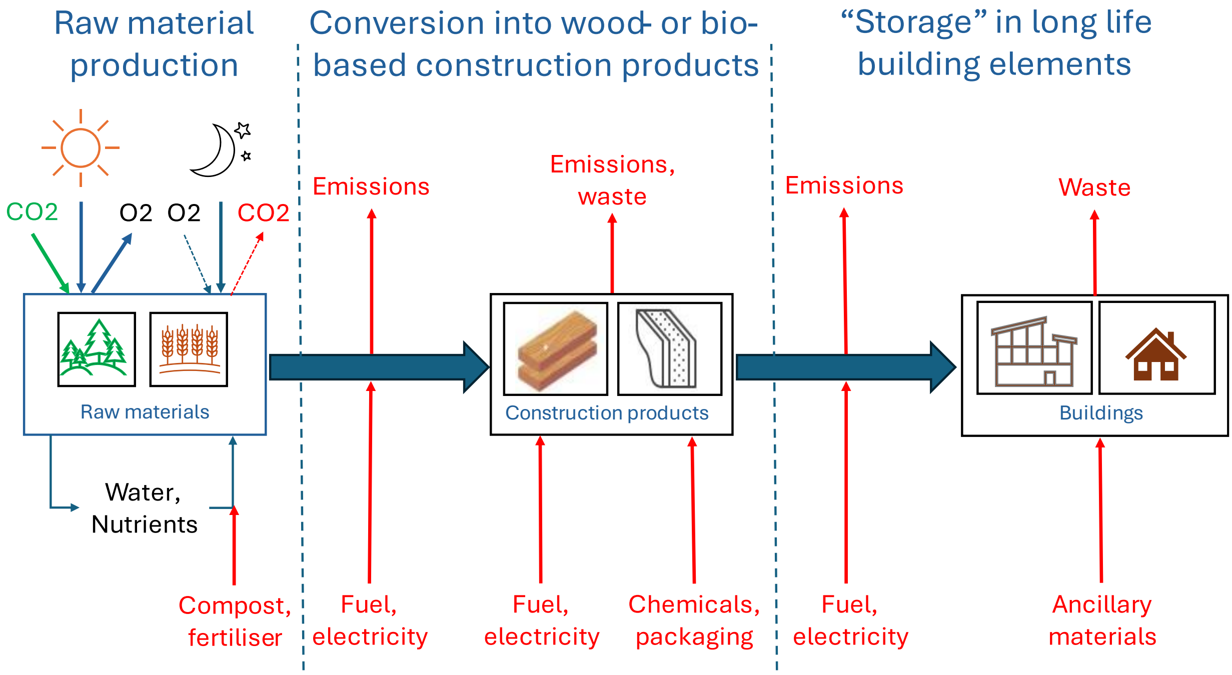


Figure 1. Conceptual system boundary of biogenic carbon storage in buildings (green denotes carbon removal, while red denotes potential sources of carbon emissions, whether embodied or direct)

The activity qualifies as biogenic carbon storage in buildings only when the following conditions are met:

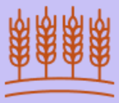
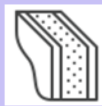
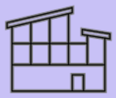
1. Net negative cradle-to-installation balance: The life cycle inventory up to installation (modules A1–A5) must demonstrate a net removal of CO_2.
2. Minimum storage period: The stored carbon must be securely retained within the building for at least 35 years, consistent with CRCF definitions for carbon storage in products.

These two criteria ensure that only durable, genuinely net-removing materials contribute to certification under the CRCF framework.

1.4 Most relevant EN technical standards and EU policy and legislation

The CRCF methodology for biogenic carbon storage in buildings must be compatible with existing EU legislation, policies and technical standards. Table 1 provides a mapping of these instruments to raw materials, products, and buildings.

Table 1. Overview of potentially relevant EU legislation, policy, technical standards and certification schemes

	Raw materials: wood	Raw materials: other biobased materials	Construction products	Buildings
			 	 
CRCF publications	TAP forestry (here)	TAP agricultural land (here)	-	Previous TAP buildings (here)
EU Directives	RED 2018/2001 (amended in 2024)		-	-
	-	-	-	EPBD 2024/1275
EU Regulations	-	-	CPR 2024/3110 (esp. Annex II)	-
	LULUCF 2018/841 (amended in 2023)		-	LULUCF 2018/841
	Taxonomy 2021/2139	-	-	Taxonomy 2021/2139 and 2023/2486
EU policy	-	-	Product Environmental Footprint (2021)	
	-	-	-	Level(s)
EN standards	-	-	EN 15804	EN 15978
	-	-	EN 16449	
	-	-	EN 16485	
Third party "sustainability" certification	FSC	Schemes such as: Better biomass; ISCC PLUS; REDcert; RSB etc.	FSC	BREEAM; DGNB; HQE; VERDE; LEED, etc.
	PEFC		PEFC	

The sources listed above primarily relate to the Quantification and Sustainability dimensions of the QU.A.L.ITY framework. When relevant, additional detail is provided in later sections of this report.

An understanding of the impact of the recast EPBD, the new CPR, and the relevant EN standards is vital. These instruments influence how carbon storage in buildings can be assessed, reported, and verified. Ensuring consistency and coordination between EPBD requirements, CPR data, and EN-based LCA methods within the CRCF certification process is essential to ensure the credibility, compatibility, and robustness of the methodology.

1.4.1 Energy Performance of Buildings Directive (EPBD)

The recast EPBD, adopted in 2024, introduces numerous new requirements for both new and existing buildings. These aim to enhance energy performance and reduce greenhouse gas emissions across the EU building stock. Improved energy performance will be achieved through phasing in minimum energy performance class requirements for existing buildings. From 2030 onwards, new buildings will be required to be "zero-emission buildings". To lower lifecycle greenhouse gas emissions, mandatory requirements will be introduced to measure the global warming potential of new buildings. This applies from 2030, or from 2028 if the new building has more than 1,000 m² of useful floor area. Member States must also set mandatory limits, in terms of kgCO2eq./m² for lifecycle global warming potential from 2030.

There are several clear connections between the revised EPBD requirements and the CRCF activity of biogenic carbon storage in buildings. The most relevant parts relating to life cycle assessments are:

- In Article 7(2), where it says: "*2. Member States shall ensure that the life-cycle GWP is calculated by Annex III and disclosed in the energy performance certificate of the building:*
  (a) from 1 January 2028, for all new buildings with a useful floor area larger than 1 000 m2;
  (b) from 1 January 2030, for all new buildings."
- In Article 7(3), where it says: "*The Commission is empowered to adopt delegated acts in accordance with Article 32 to amend Annex III to set out a Union framework for the national calculation of life-cycle GWP with a view to achieving climate neutrality. The first such delegated act shall be adopted by 31 December 2025*".
- In Article 7(5), where it says: "*By 1 January 2027, Member States shall publish and notify to the Commission a roadmap detailing the introduction of limit values on the total cumulative life-cycle GWP of all new buildings and set targets for new buildings from 2030, considering a progressive downward trend, as well as maximum limit values, detailed for different climatic zones and building typologies […]*".
- In Annex III, where it says: "*For the calculation of the life-cycle GWP of new buildings pursuant to Article 7(2), the total life-cycle GWP is communicated as a numeric indicator for each life-cycle stage expressed as kgCO2eq/(m2) (of useful floor area) calculated over a reference study period of 50 years. The data selection, scenario definition and calculations shall be carried out in accordance with EN 15978 (EN 15978:2011 Sustainability of construction works. Assessment of environmental performance of buildings. Calculation method) and taking into account any subsequent standard relating to the sustainability of construction works and the calculation method for the assessment of environmental performance of buildings. The scope of building elements and technical equipment is as defined in the Level(s) common EU framework for indicator 1.2. Where a national calculation tool or method exists or is required for making disclosures or for obtaining building permits, that tool or method may be used to provide the required disclosure. Other calculation tools or methods may be used if they fulfil the minimum criteria established by the Level(s) common EU framework. Data regarding specific construction products calculated in accordance with Regulation (EU) No 305/2011 of the European Parliament and of the Council shall be used when available*".

And the most relevant parts relating to biogenic carbon storage in buildings are:

- In Article 7(6), where it says: "*Member States shall also address carbon removals associated to carbon storage in or on buildings.*"
- In Annex V (2c), where it says: "*In addition, the energy performance certificate may include the following indicators: [...] the temporary storage of carbon in or on buildings*".

Assessing the GWP of a building requires detailed information about all construction products and materials used. This includes materials that might be eligible for CRCF certification. Consequently, there is clear potential for synergies if national methodologies or the Union framework itself set requirements for identifying data related to CRCF-eligible materials. The identified data could then be further processed to produce a result for kg biogenic CO2 storage per m2 of usable floor area.

As per Annex V (2c) quoted above, the requirement to communicate CRCF biogenic carbon storage results in energy performance certificates (EPCs) is optional. Some Member States may choose this option, while others might not. If reporting is adopted, it would serve as a valuable tool for assessing data for future baselines because the data would be included in a national database for EPCs.

One key provision in Annex III states:" *Data regarding specific construction products calculated in accordance with Regulation (EU) No 305/2011 of the European Parliament and of the Council shall be used when available*". First, it should be clarified that this reference now applies to the new CPR (i.e. Regulation (EU) No 305/2011 is replaced by Regulation (EU) 2024/3110). This means that whenever a construction product has its life cycle impacts declared in accordance with the Construction Products Regulation (CPR) requirements, this data must be used for the building LCA instead of any other available data. This order of data preference in the EPBD should also be applied to CRCF calculations, ensuring that the life cycle data declared under the CPR functions as the reference for both LCA and CRCF calculations, ensuring consistency and reliability in any future standardised baselines.

1.4.2 Construction Products Regulation (CPR)

The CPR creates a common technical language for describing the essential technical characteristics of construction products. This is achieved through harmonised rules for placing construction products on the EU market. A CE marking and a declaration of performance and conformity denote compliance with the CPR for products harmonised under it.

It begins to apply from 8 January 2026 and includes a lengthy transition period until 2040, during which the industry can gradually adopt ambitious new standards for assessing and disclosing the life cycle impacts of construction products. Specifically:

- Recital 7 says: "*[...] necessary to establish, [...] new environmental obligations and to lay the ground for the development and the application of an assessment method for the calculation of the environmental sustainability of construction products. The calculations should cover the life cycle of the product using the methods established through standardisation. For new products the calculated life cycles should include all stages of a product's life, from raw material acquisition or generation from natural resources to its final disposal, including potential benefits and loads outside the boundary limits.*"
- Article 1(1a) says that the CPR will establish: "*harmonised rules on how to express the environmental and safety performance of construction products in relation to their essential characteristics, including on life cycle assessment*".
- Article 15, specifically 15(2) and 15(3), require that life cycle performance be included in the declaration of performance and conformity (DoPC), at least in terms of the essential characteristics listed in Annex II.

The scope of the CPR applies to construction products, including used products, key parts of products and in some cases, parts or materials intended to be used for relevant products[2]. Of the 36 product families listed in Annex VII, the most pertinent for the CRCF are:

- Code 4: Thermal insulation products. Composite insulating kits/systems (standardisation request to be adopted in 2026).
- Code 13: Structural timber products/elements and ancillaries (standardisation request to be adopted in 2026).
- Code 14: Wood-based panels and elements (standardisation request to be adopted in 2026).
- Code 19: Floorings (standardisation request to be adopted in 2027).

The Commission, with support from the CPR Acquis expert group, has developed a working plan that sets priorities for standardisation committee work on complementary product category rules built upon EN 15804. This will ensure compatibility with the CPR environmental declarations. A complete list of the CPR product families, along with their priority and status, is provided in Appendix I.

Annex II lists 19 different LCA impact categories, which are based on EN 15804 and aligned with the latest Product Environmental Footprint methodology recommended by the Commission for conducting LCAs. According to Article

[2] According to the CPR, Article 3(1), 'construction product' means any formed or formless physical item, including 3D-printed products, or a kit that is placed on the market, including by means of supply to the construction site, for incorporation in a permanent manner into construction works or parts thereof with the exception of items that need first to be integrated into a kit or another construction product prior to being incorporated in a permanent manner into construction works;

15(3)(a), from 8 January 2026 onwards, the essential characteristics of any new harmonised standards for construction product LCAs will include climate change impacts, not only in terms of total climate change but also in terms of the GWP-biogenic, GWP-total, and GWP-LULUC components of this impact.

The influence of the CPR on the CRCF can be summarised as follows:

- CPR has established legal requirements for developing harmonised standards and LCA calculation rules. These include families of bio-based products most relevant for the CRCF building certification.
- The harmonised rules will be progressively developed for different product families. New standards for the product families relevant to CRCF will not be issued for several years. Moreover, upon publication, their application remains voluntary for the first year before becoming mandatory.
- Any additional product category rules that would be valuable for the CRCF methodology should be highlighted for consideration in future harmonised standards.
- Once CPR-based harmonised product category rules are established, they will amend any existing rules that were previously built on the general EN 15804 standards.

1.4.3 EN standards

Until the harmonised standards under CPR are established, the quantification of net carbon removal and biogenic carbon storage should rely on relevant methods outlined in suitable EN standards, where available. This involves considering calculation methods at both the whole building level and the level of construction products. A summary of the most relevant aspects of related EN standards is provided in the table below.

Table 2. Relevant EN standards for the quantification of carbon assessments in construction products and buildings

Standard	Remarks
EN 15978: Sustainability of construction works: Assessment of environmental performance of buildings - Requirements and guidance	Originally published in 2011, with a new version expected by the end of 2025. This standard defines all life cycle stages and modules for the building life cycle, which are fully aligned with EN 15804 for construction product LCAs. It requires reporting on GWP-total, GWP-biogenic, GWP-fossil, and GWP-LULUC as core environmental indicators. However, it does not include any requirements to report on stored biogenic carbon[3]. This standard will underpin the Union framework for mandatory life cycle carbon assessments of new buildings starting in 2028 or 2030, as required by the EPBD Article 7(2) and Annex III.
EN 15804: Sustainability of construction works: Environmental product declarations - Core rules for the product category of construction products	Initially published in 2012, amended in 2013 (+A1) and again in 2019 (+A2). EN 15804 establishes common rules for calculating environmental impacts of all construction products. Like EN 15978, it is necessary to report on GWP-biogenic, GWP-fossil, GWP-LULUC, and the sum of these three, which is GWP-total. It requires reporting on life cycle modules A1-A3, C1-C4, and D as a minimum, ensuring that product data can be directly integrated into building-level LCAs in accordance with EN 15978. Section 5.4.3 explicitly states that temporary storage of carbon within products shall not be included in the calculations of GWP impacts for construction products. However, this does not preclude the separate presentation of this figure within the EPD or DoPCs.
EN 16485: Round and sawn timber: Environmental Product Declarations - Product category rules for wood and wood-based products for use in construction	Published in 2014, this standard builds on the core rules of EN 15804, providing product-specific guidelines for structural wood products relevant to CRCF certification. It includes provisions for managing forest carbon pools and clarifies when they can be excluded, such as under proven sustainable forest management.

[3] Stored biogenic carbon is the atmospheric carbon that is incorporated into biomass during its growth and remains in biomass-based products. Biogenic GWP is a broader concept, encompassing both the potential storage of carbon in biomass and the emissions to the atmosphere from its direct combustion or decomposition (see illustration in section 1.5.1).

Standard	Remarks
	The standard also provides guidance on the -1/+1 approach to CO2 emissions, assuming carbon neutrality of biomass. Unlike EN 15804, it explicitly states the separate reporting of the amount and duration of biogenic carbon storage.
EN 16449: Wood and wood-based products: Calculation of the biogenic carbon content of wood and conversion to carbon dioxide	Published in 2014, the standard provides formulas for calculating biogenic carbon content based on carbon fraction, moisture content, density, and volume. It supplies default values (e.g. 50% carbon fraction of dry mass, 12% moisture content) in the absence of specific values and example calculations, e.g. for glulam.

Overall, there is a clear set of established EN standards for calculating stored biogenic carbon in wood-based construction products and reporting LCA impacts at the building level. These standards require that biogenic carbon storage be reported separately from aggregated GWP indicators.

For CRCF purposes, this separate reporting is essential: it provides a quantifiable metric for certifying the net amount of biogenic carbon durably stored in buildings. CRCF methodologies must therefore ensure that this value is systematically and transparently extracted, even though it is not used directly in life-cycle GWP calculations.

While EN 15804 and EN 16485 provide solid guidance for wood-based products, there is no comparable level of standardisation for other bio-based materials, especially insulation products. Addressing this methodological gap with default assumptions will help ensure a more consistent approach to CRCF calculations for non-wood biobased construction products.

1.5 Carbon storage in the context of building life cycles

The EN 15804 and EN 15978 standards follow a common structure of life-cycle stages and modules (Figure 2). For CRCF purposes, the most significant stages are A1–A5 (cradle-to-installation), B4 (replacement), and C1–C4 (end-of-life).

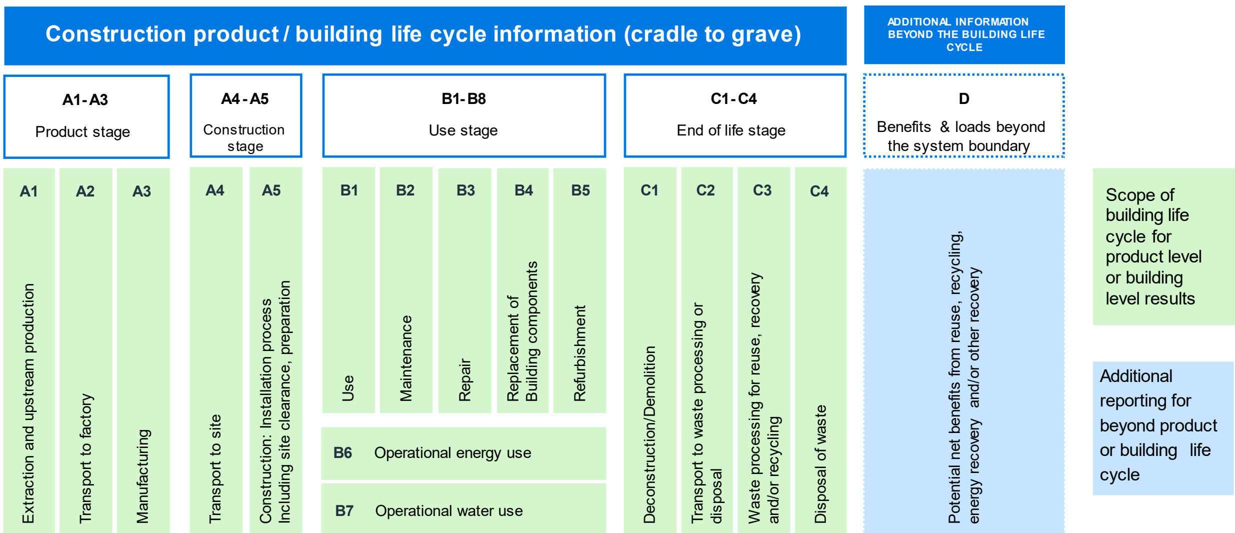


Figure 2. The life cycle stages and modules that are common to EN 15804 and EN 15978.

Life cycle module reporting

Any certification of temporary carbon storage in buildings currently depends on information from construction product EPDs or, in the future, on Declarations of Performance and Conformity (DoPCs) once harmonised standards or new European Assessment Documents (EADs) under the CPR are established. Both types of documents rely on data and assumptions provided by the manufacturer of the construction product.

Table 3. Overview of life cycle module reporting requirements in EN 15804 for construction product EPDs

Modules	Remarks
A1-A3	Mandatory. Covers raw material supply, transport, and manufacturing. Packaging included. Data often aggregated to protect supplier confidentiality.
A4	Currently optional; will be mandatory under new CPR. Requires transparent transport assumptions.
A5	Currently optional; will be mandatory under new CPR. Requires rules on waste rates, packaging disposal, and ancillary materials.
B1-B7	Optional. Typically only B2 (maintenance) reported at product level. For B4 (replacement), expected service life may be declared to enable LCA practitioners to model replacement cycles.
C1-C4	Mandatory. Based on representative end-of-life scenarios. If product lifetime < building lifetime, C1–C4 impacts are assigned to B4 of the building LCA in EN 15978. Replacement products reintroduce A1–A5 impacts at building level.
D	Mandatory, but reported separately. Represents benefits/loads beyond system boundaries.

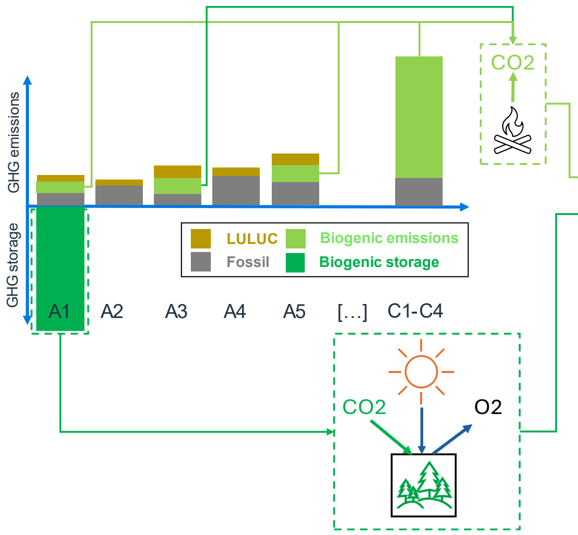
Stored carbon vs. GWP indicators

EN 15804 and EN 15978 require reporting of GWP-total, GWP-biogenic, GWP-fossil, and GWP-LULUC. These indicators do not result in any declaration of stored biogenic carbon (as stated in section 5.4.3 of EN 15804). However, sections 6.4.4 and 7.2.5 of EN 15804 make provision for the separate declaration of stored biogenic carbon in the product and in the packaging (in units of kgC). Such a declaration is mandatory unless the mass of biogenic carbon is less than 5% of the total product mass or packaging mass.

To understand how stored biogenic carbon should be counted, and how the numbers look in a CRCF content, consider the concepts and example below:

- Biogenic carbon uptake = only the amount of biogenic CO_2 removed from the atmosphere into biomass during module A1.
- Product-stored biogenic carbon = Total biogenic carbon uptake remaining in product after module A3.
- Net CRCF GWP-biogenic storage = the net sum of biogenic CO_2 uptake (negative number) plus the biogenic CO_2 emissions released (positive number) during modules A1-A5 of the life cycle.

As shown in Figure 3, biogenic carbon uptake occurs in A1 and is later balanced in C1–C4, following the -1/+1 approach in EN 16485. In full life-cycle LCAs, this means storage does not alter the net GWP outcome.



	GWP (kgCO2.eq.)						
	A1	A2	A3	A4	A5	[…]	C1-C4
Biogenic GWP	0	0	+7	0	+7		+100
Biogenic uptake	-100	n/a	n/a	n/a	n/a		n/a
Fossil GWP	+6	+8	+5	+10	+8		+9
LULUC GWP	+2	+2	+3	+2	+3		0
Net GWP	**-92**	**+10**	**+15**	**+12**	**+18**		**+109**
EPD product stored carbon	**-93** (100-7)						
CRCF GWP-biogenic storage (net A1-A5)	**-37** (sum of A1 -A5 net GWPs)						
Total GWP (net A1 -C4)	**+72** (sum of A1 -C4 net GWPs)						

Figure 3. Illustration of the different types of GWP emission and biogenic carbon storage over the life cycle of a wood- or bio-based construction product (numbers are arbitrary and columns not to scale).

In the example above, wood containing 100 kg of biogenic CO2eq. is grown and harvested (A1), which resulted in a total of 8 kg CO2eq. being emitted. Transport to the factory (A2) results in a further 10 kgCO2eq. emissions. During processing (A3), 7% (or 7 kgCO2eq.) of the wood carbon is wasted and burned in a factory boiler to generate useful heat. Other consumables during production add a further 8 kgCO2eq. emissions. Transport of the construction product to the construction site (A4) is assumed to emit a further 12 kgCO2eq. At the construction site (A5) a further 7 kgCO2eq. of biogenic GWP is emitted due to wastage of the construction product during installation, it is assumed that the waste wood is incinerated for energy recovery in this case. A further 11 kgCO2eq. emissions are also assumed during installation.

The numbers for the terms above would be:

- Biogenic carbon uptake = 100 kgCO2eq.
- Product-stored biogenic carbon = 93 kgCO2eq.
- Net CRCF carbon storage = 37 kgCO2eq.

For CRCF certification, the separately reported product storage values in EPDs are critical: they provides the quantifiable basis for certifying the net amount of carbon durably stored in buildings. Current EPD reporting often combines A1–A3 data, hiding the original uptake value from A1. In these cases, it is not clear if there were any biogenic GWP emissions in A3 (or even in A2 if transport energy is directly or indirectly biobased). CRCF methodologies must therefore either refer to detailed product category rules such as EN 16485 (wood) and EN 16449 (carbon content calculation) or establish clear default rules and assumptions where such standards are not available. This guarantees that the stored biogenic carbon is consistently and transparently identified.

Open questions for stakeholders:

Q1. Opinions about the distinctions between the terms “biogenic carbon uptake” and “product-stored biogenic carbon”?

Q2. If an EPD reports on the A1-A3 aggregate, and the declared product-stored biogenic carbon level is higher than net A1-A3 biogenic GWP, the difference should be due to biogenic GWP emissions during A1-A3 that were unrelated to wastage or combustion of the raw biobased material being used to make the product. Should an explanation for any such discrepancies be required in the EPD/DoPC?

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2 Quantification

2.1 Introduction to the basic elements of quantification

Quantification under the CRCF Regulation is the foundation for assessing the impact of carbon storage activities. Article 4 of the Regulation lays out the quantification equations and principles. The provisions that are directly related to the activity of carbon storage in buildings are reproduced below:

"*5. A carbon storage in product activity shall provide a temporary net carbon removal benefit, which shall be quantified using the following formula:*

$$Temporary\ net\ carbon\ removal\ benefit = CR_{baseline} - CR_{total} - GHG_{associated}$$

where:

- $CR_{baseline}$ *is the amount of carbon removals under the baseline;*
- CR_{total} *is the total amount of carbon removals of the activity;*
- $GHG_{associated}$ *is the increase in direct and indirect greenhouse gas emissions over the entire lifecycle of the activity which are attributable to its implementation, including indirect land use change, calculated, where applicable, in accordance with the protocols set forth in the 2006 IPCC Guidelines for National Greenhouse Gas Inventories and any further refinement to these 2006 IPCC Guidelines.*

6. Quantities referred to in paragraphs 1 to 5 shall be attributed a negative sign (–) if they are net greenhouse gas removals and a positive sign (+) if they are net greenhouse gas emissions; they shall be expressed in tonnes of CO2 equivalent.

7. Permanent carbon removals, temporary carbon removals through carbon farming and carbon storage in products, soil emission reductions and associated greenhouse gas emissions shall be quantified in a relevant, conservative, accurate, complete, consistent, transparent and comparable manner, in accordance with the latest available scientific evidence. The monitoring shall be based on an appropriate combination of on-site measurements with remote sensing or modelling in accordance with the rules set out in the applicable certification methodologies.

8. The baseline referred to in paragraphs 1, 2 and 5 shall be highly representative of the standard performance of comparable practices and processes in similar social, economic, environmental, technological and regulatory circumstances and take into account the geographical context, including local pedoclimatic and regulatory conditions ('standardised baseline').

9. The standardised baseline shall be established by the Commission in the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8.

The Commission shall review at least every five years and update, where appropriate, the standardised baseline in light of evolving regulatory circumstances and of the latest available scientific evidence. The updated standardised baseline shall apply only to an activity for which the activity period starts after the entry into force of the applicable certification methodology.

10. By way of derogation from paragraph 8, where duly justified in the applicable certification methodology, including due to the lack of data or the absence of sufficient comparable activities, an operator shall use a baseline that corresponds to the individual performance of a specific activity ('activity-specific baseline').

11. The activity-specific baselines shall be periodically updated, at the beginning of each activity period, unless otherwise stated in the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8.

12. The quantification of permanent carbon removals, temporary carbon removals through carbon farming and carbon storage in products, and soil emission reductions shall account for uncertainties in a conservative manner and in accordance with recognised statistical approaches. Uncertainties in the quantification of carbon removals and soil emission reductions shall be duly reported."

To qualify for certifiable units, the final result from the equation in Article 4(5) must be overall positive (i.e. >0). The same equation and principles can be expressed in more specific terms that relate more easily to the activity of temporary storage of biogenic carbon in buildings.

$$Temporary\ net\ carbon\ removal\ storage = CRS_{baseline} - CRS_{total} - GHG_{associated}$$

where:

- CRS-baseline is the amount of biogenic Carbon Removals Stored in eligible building elements under the baseline;
- CRS-total is the amount of biogenic Carbon Removals Stored in eligible building elements of the activity;
- GHG-associated is the increase in direct and indirect greenhouse gas emissions over the defined lifecycle of the activity, which are attributable to its implementation, including land use change, fossil and biogenic carbon emissions in accordance with the EN 15804 and EN 15978 standards.

When applying this methodological concept to building activities, several aspects require clarification to ensure consistent assessments. These aspects are listed below and are discussed in more detail in subsequent sub-sections in this report.

- Section 2.2: "CR-total": Scope of building elements.
- Section 2.3: "CR-total": Scope of biobased materials.
- Section 2.4: "CR-total": Scope of building typologies and project types.
- Section 2.5: "GHG-associated": Scope of life cycle modules and GWP categories.
- Section 2.6: "CR-baseline" – separate baselines for wood-based and other biobased materials.
- Section 2.7: Accounting for uncertainties in data.

2.2 "CR-total": scope of building elements

Purpose: The eligibility of building elements for CRCF certification must be clearly defined. Building elements are broader than individual products and may consist of multiple parts, materials, products or construction products. Buildings include many elements such as doors, windows, internal walls, floors and structural frames. Each has a specific purpose and characteristics that influence their eligibility for CRCF-certified units. This section explores which building elements could be included in the CRCF methodology.

Aspect	Remarks about 2.2: scope of building elements / construction products
Definition	The scope of construction products refers to which construction products or building elements are included when measuring the net carbon removals stored in the building, both during the activity and in the baseline.
Issue	Under Recital 28 and Article 8(2) of the CRCF Regulation, priority is given to wood-based and other bio-based construction products capable of storing carbon for at least 35 years (as required by Article 2(11)). Including more product categories increases potential storage but also adds complexity to baseline setting and monitoring.
Objective	Clearly define which construction products are eligible for temporary storage of biogenic carbon certificates and explain why.
Existing, proven certification methodologies	The scope of construction products and building elements in various certification schemes and initiatives includes: • **Rainbow**: includes, but is not limited to, wood framing, wood panels, hempcrete (concrete containing hemp fibres), and cellulose thermal insulation. Types of raw materials include, but are not limited to, wood (timber/lumber), bamboo, hemp, straw, recycled paper, and flax. Rainbow allows for two distinct certification paths: one for building, and another one for biobased construction products manufacturers (at factory level). • **C-sink standard (v1.0)**: includes "*only materials that will be preserved for as long as the building is preserved*". More specifically, it includes "*all statically relevant structures such as walls, beams, pillars, foundations, roof trusses, and the roof cover. It further includes the insulation material, the fixed flooring (but not the floor cover), ceilings, and permanent interior walls. However, it does not include quickly exchangeable or mobile materials such as wallpaper, movable walls, decoration, furniture, parquet, carpets, doors, window frames, and sills*". • **Climate Cleanup Foundation certification protocol (v1.0)**: eligible products must meet the QUALITY criteria, which means a minimum 35 year lifetime and that products like cladding, floor covers and doors are excluded. The excel-based calculation tool includes references to a list of product types (floors, ceilings, frames, insulation, roofs, boards, planks and "other"). • **Verra VCS**: Still under development, the draft methodology for "Mass Timber Constructions" is currently considering all engineered wood products used in primary loadbearing structures and, if present, in any bracing or secondary loadbearing structures. • **SNK** (in Dutch only): in the method "hemp for long-term carbon storage" refers to the use of hemp in hempcrete and in insulation materials. There is also a generic protocol for all biobased products with a lifespan of at least 50 years. • **Label Bas Carbone**: all bio-based materials with an application coefficient depending on the lifespan of the material (coefficient = 1 in 100 years, 0.5 if 50 years, 0.3 if 30 years).

<table>
<tr><th>Aspect</th><th>Remarks about 2.2: scope of building elements / construction products</th></tr>
<tr><td>Options</td><td>From the full list of building elements in the Level(s) indicator 2.1 user manual, those with the highest potential suitability for carbon storage certification are assessed (see the full list in Appendix II).
<table>
<tr><th>Type of building element</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Loadbearing structural frame (including roof structure)</td><td>• Potentially high quantities involved.
• Has a very long lifespan.
• Will not be removed except via heavily regulated renovation or refurbishment activities.
• Quantities should be accurately known due to the need to carry out loadbearing calculations.
• No GHG-associated for maintenance.</td><td>• Baseline will vary significantly depending on building typology.
• Baseline will vary significantly based on regional climate and available materials.</td></tr>
<tr><td>External walls, cladding, shading devices, windows, and doors</td><td rowspan="2">• Potentially high quantities involved.
• Some of these elements have a long lifespan.
• Some of these elements (e.g. insulation) will not be removed except via regulated renovation or refurbishment activities.
• Quantities of some of these elements should be accurately known due to clear unit values.</td><td rowspan="2">• High risk of many elements not having a sufficient lifetime (35 years).
• Many elements might be removed via DIY actions or other unregulated renovation or refurbishment activities.
• Certification of certain elements could encourage deliberate overdesign (e.g. extra cladding or shading devices).
• With wall systems, counting the whole system might lead to excessive GHG-associated.
• Quantities of loose fill insulation need to be estimated based on packing density and cavity volume occupied – both numbers with real-life inaccuracies.</td></tr>
<tr><td>Internal walls, partition walls, doors, stairs, and ramps</td></tr>
<tr><td>Fittings and furnishings, including floor coverings and decking</td><td>• Quantities could be significant.</td><td>• Quantities could be made up of lots of small contributions – difficult to sum up and to monitor.
• High risk of many elements not having a sufficient lifetime (35 years).
• Many elements might be removed via DIY actions or other unregulated renovation or refurbishment activities.
• GHG-associated for maintenance (e.g. cleaning).</td></tr>
</table>
</td></tr>
<tr><td>Technical conclusions</td><td>Based on the analysis presented, the following decisions are recommended for inclusion or exclusion in the certification methodology for temporary biogenic carbon storage in buildings:</td></tr>
</table>

Aspect	Remarks about 2.2: scope of building elements / construction products
	1. Include: All wood-based elements of the loadbearing structural frame (beams, columns, and slabs) and roof. 2. Include: Wood fibre-based or other biobased insulation materials, but not the entire wall systems they are part of. 3. Exclude: Cladding, shading devices, window frames, doors, walls, stairs, and ramps - unless they are part of the structure of the building (and become part of point 1). 4. Exclude: Other fittings and furnishings, including floor coverings due to the difficulty of monitoring. 5. Exclude: Biobased plastic construction products due to the difficulty of initial verification, probability of high GHG-associated and subsequent monitoring.
Next steps	Create a list of eligible products based on their technical definitions or typology. This could involve consulting relevant technical committees and other experts familiar with the categorisation of wood-based structural elements, insulation materials, and biobased materials used in construction products. The relevant technical committees would include at least: CEN/TC 88 on thermal insulation; CEN/TC 112 on wood-based panels; CEN/TC 124 on timber structures; CEN/TC 175 on round and sawn timber; and possibly CEN/TC 134 on floorings. Wood-based materials are relatively straightforward to define, but biobased materials encompass a potentially broad scope, with some being more relevant than others in terms of current insulation products and innovation.

2.3 "CR-total": Scope of biobased materials

Purpose: To establish which bio-based materials are suitable for CRCF certification, it is necessary to ensure that their stored carbon can be consistently and transparently quantified.

At present, simplified calculation rules are only available for wood products. **EN 16449** provides an equation for estimating stored biogenic carbon based on carbon fraction, density, moisture content, and product volume (see equation below). **EN 16485** requires wood-based EPDs to apply this rule, with default values available where product-specific data is lacking.

Example of wood (EN 16449):

$$P_{CO2} = \frac{44}{12} \, x \; cf \; x \; \frac{\rho_{\omega} \; x \; V_{\omega}}{1 + \frac{\omega}{100}}$$

where:

- P_{CO2} is biogenic carbon released as CO_2 at end of life (kg);
- c_f is the carbon fraction of oven-dry wood (default 0.5);
- ω is the moisture content of the product (%);
- ρ_{ω} is the density of woody biomass at that moisture content;
- V_{ω} is the volume of the product at that moisture content (for not-solid wood-based products, V_{ω} = VP x percentage of wood, where VP is the gross volume of the wood-based product).

Equivalent rules for other biobased materials are not yet in place. Until such standards are developed, CRCF should define default values or simplified calculation methods to maintain fair, consistent, and credible accounting across products. However, one useful reference study is that of Van den Oever *et al.* (2024), which provides the following values:

- Wood: 495-520 g/kg.
- Fibrous lignocellulose: 430-440 g/kg.
- Woody lignocellulose: 460-495 g/kg
- Acetylated, thermally treated wood: 518-520 g/kg
- Sheep wool: 473 g/kg
- Mycelium: 485 g/kg
- Cork (expanded): 626-685 g/kg

Biobased materials in insulation products: A review of approximately 70 EPDs for biobased insulation products on the revalu.io platform illustrates the diversity of materials currently in use:

- Wood (solid, chips, sawdust, fibres): ~27%
- Cellulose from recycled paper: ~16%
- Cotton fibres (mainly pre-consumer waste): ~15%
- Straw: ~10%
- Grass fibre: ~7%
- Cork: ~6%
- Others: hemp, flax, bamboo fibres, seagrass, driftwood, wool, mycelium (~19%)

This variety of materials underscores the necessity for CRCF methodologies to differentiate between materials with evident photosynthetic carbon uptake and those without. For instance, wool and mycelium do not involve direct photosynthetic carbon fixation and should consequently be excluded from the scope.

The situation becomes more complicated with recycled materials. Post-consumer paper and textiles may contain non-biogenic carbon via inks, fillers, adhesives, or synthetic fibres blended with natural fibres. These fractions pose challenges for accurately attributing stored biogenic carbon. In such cases, CRCF would need to develop clear methodological rules to determine what proportion of the material can be recognised as biogenic.

Another issue is packaging. Biogenic content in packaging, such as cardboard or wooden pallets, has been found to contribute anywhere from 0 to 6.3 percent[4] of the total reported biogenic carbon in some product EPDs (see Appendix III for more details). While this represents only a minor share, it introduces variation that must be consistently addressed within the methodology, either by requiring separate reporting or by setting thresholds for inclusion.

Overall, these findings emphasise that while wood-based materials already have robust accounting methods through EN standards, other bio-based materials require default values and harmonised approaches if they are to be included in CRCF certification. Without this, comparability across EPDs and product categories cannot be guaranteed.

<table>
<tr><th>Aspect</th><th>Remarks about 2.3 on simplified reporting for stored biogenic carbon</th></tr>
<tr><td>Issue</td><td>There are currently no standardised rules for estimating the biogenic carbon content of non-wood bio-based materials. By contrast, EN 16449 provides a clear equation and default values for wood. Any new CRCF rules for other bio-based materials should follow a similar approach, using a fixed equation with default values to ensure comparability. However, due to the wide diversity of bio-based materials, tailored defaults will be needed for different material groups.</td></tr>
<tr><td>Objective</td><td>To identify the most relevant bio-based insulation materials, evaluate their suitability for CRCF certification, and define standardised rules for estimating their stored biogenic carbon content.</td></tr>
<tr><td>Existing, proven certification methodologies</td><td>The scope of construction products and building elements in various certification schemes and initiatives includes:
• Rainbow: Requires an EN 15804 EPD with biogenic carbon content but provides no calculation rules. If no EPD is available for a project, then a similar document may be used instead, given that it includes the above information, is independently verified, and follows ISO 14025.
• C-sink standard (v0.91): Requires EN 15804 EPDs. Defines eight feedstock classes (annual crops, perennials, forest biomass, residues, etc.).
• Climate Cleanup Foundation certification protocol (v1.0): Uses EN 15804 EPD values where available; otherwise refers to EN 16449 for wood. For other materials, section 2.1.3 provides a series of approaches for calculating quantification depending on the available information about biogenic carbon storage. The excel-based calculation tool includes references to a list of material types (i.e. timber, lumber, CLT, LVL, bamboo, wood, fibre, straw fibre, grass/bamboo fibre, hemp/flax fibre, composite wood, composite concrete, mycelium and “other”).
• Verra VCS (draft): Appears to rely on EPDs and third-party databases. Requirements for explicit biogenic content reporting remain unclear.
• Label Bas Carbone: all biobased materials with an application coefficient depending on the lifespan of the material (Coefficient = 1 if 100years, 0.5 if 50 years, 0.3 if 30years). The material must have an FDES sheet, otherwise a DED can be used.

Observation: Most schemes depend heavily on EN 15804 EPDs but lack harmonised rules for non-wood bio-based materials, underscoring the need for CRCF to provide clear guidance.</td></tr>
<tr><td>Options</td><td>The main options are to either set default carbon content rules for bio-based materials (except for wood, which is already covered by EN 16449) in the CRCF methodology that: (i) must always be used; (ii) must be used as a back-up only in the case of a lack of specific values, or (iii) are not set at all.

<table><tr><th>Option</th><th>Pros</th><th>Cons</th></tr><tr><td>Obligatory default stored carbon content rules</td><td>• Ensures consistency
• Avoids large errors
• Simplifies reporting for manufacturers.</td><td>• Requires significant upfront research
• May not cover new materials immediately.</td></tr></table></td></tr>
</table>

[4] For six structural wood products, the biogenic carbon in packaging was 0%, 0%, 0%. 0,05%, 0.2% and 0.7% of the carbon stored in the product. For six bio-based insulation products, these figures were 0%, 0%, 0%, 0.3%, 4.3% and 6.3%.

<table>
<tr><th>Aspect</th><th colspan="3">Remarks about 2.3 on simplified reporting for stored biogenic carbon</th></tr>
<tr><td rowspan="2"></td><td>Back-up default stored carbon content rules</td><td>• Most flexible approach.
• Manufacturers can avoid burdensome calculations and assumptions.
• Well-informed manufacturers can use specific data.</td><td>• Risks inconsistency
• Assumptions often undocumented in EPDs.</td></tr>
<tr><td>No default stored carbon content rules</td><td>• Minimal development effort.</td><td>• Fully reliant on supplier data
• Inconsistent results
• Heavier workload for EPD developers
• Higher risk of errors</td></tr>
<tr><td>Technical conclusions</td><td colspan="3">The CRCF scope for biobased materials should initially be limited to those that:
• Have a demonstrable photosynthesis-based carbon removal mechanism, and
• Have default calculation rules and values for stored biogenic carbon content in place.
This ensures consistency and credibility in certification.</td></tr>
<tr><td>Next steps</td><td colspan="3">• Engage experts and EPD developers to consult on default values for key biobased materials.
• Prioritise materials with proven technical relevance: wood chips, fibres, sawdust, hemp fibre, hemp shives, straw, grass fibre, cork.
• Address methodological challenges for recycled paper and pre-consumer recycled cotton, which are widely used but require clear rules on biogenic vs. non-biogenic fractions.</td></tr>
</table>

Open questions for stakeholders:

Q3. How should recycled feedstocks (e.g. paper, cotton, textiles) in insulation products be addressed? Are recycled materials compatible with the principle of “certifiable carbon removals,” or should only the verifiable biogenic fraction from virgin material be counted?

Q4. Current practice is inconsistent in the choice of functional unit for insulation materials (examples in Appendix III show reporting in kg, m², and m³). Should there be a standard functional unit be for reporting insulation products in EPDs and Declarations of Performance and Conformity? Or is it enough to require that all relevant physical information (e.g. density and thickness) is provided to allow it to be calculated back to common units?

2.4 "CR-total": Scope of building typologies and project types.

Purpose: The eligibility of building typologies and project types is central to CRCF certification. The methodology should prioritise building categories where standardised baselines can be established, where monitoring and verification are practical, and where certification would be cost-effective.

CRCF certification must distinguish between new construction projects and renovation projects. The considerations are outlined below.

2.4.1 Considerations for new construction

New construction projects are generally suitable for CRCF certification as design data is available at the beginning of the project and relatively high quantities of materials are involved. Wood for the building structure and wood for other purposes can be well distinguished. Accurate information on material quantities is already needed for costing purposes, logistics, structural design and thermal/acoustic performance of the envelope and floors.

Building typology classification

There is no official European classification of building typologies. However, examples of different typologies can be found in the Common Procurement Vocabulary (CPV) codes provided in Regulation (EC) No 213/2008 or in the EN ISO 52000 series of standards (see Appendix IV):

1. CPV codes: A highly detailed list of building categories. Useful for procurement, but probably too fragmented for practical CRCF baseline setting.
2. EN ISO 52000: Provides fewer but broader categories of building typologies. More practical for CRCF because it focuses on the most common building types, which is where work on standardised baselines should be prioritised.

The EN ISO 52000 categories offer an additional benefit due to their direct connection to EPCs, which already classify buildings using these or similar terms found in national classification systems. Additionally, life cycle assessments will be mandatory for most new buildings from 2030 onwards, establishing an aligned dataset for both energy performance and potential carbon storage. Due to consistent design and permitting requirements, new construction projects will eventually have a large pool of data from which to develop future standardised baselines for different building categories.

2.4.2 Considerations for renovation

Renovation projects are highly diverse in scope and complexity, covering activities such as (i) change of use, internal space alterations or building extensions, (ii) energy performance upgrades, interior redesigns, or modernisation of older buildings and (iii) combinations of these with repair works.

When wood is used, it is difficult to determine whether the elements will meet the minimum 35-year lifespan. Most renovation projects do not involve replacing existing structures with new wooden ones. This variation creates challenges in establishing standardised baselines, as renovations differ in terms of affected floor areas, façade areas, project costs, and technical scope. Reference conditions are inconsistent across projects, making comparability and standardised baseline development more difficult.

Different options for dealing with renovation baselines are considered in the table below and are further developed in **Sections 2.6 and 3.3**.

<table>
<tr><th>Aspect</th><th>Remarks about 2.4 - scope of building typologies and projects</th></tr>
<tr><td>Issue</td><td>Baselines directly affect net carbon storage results. Their robustness depends on:
Building typology (e.g. residential vs. office).
Project type (new construction vs. renovation).
Nature of the renovation activity (extensions, energy retrofits, modernisations, etc.).
Because of this variability, it is essential to identify building/project types where meaningful and standardised baselines are possible.</td></tr>
<tr><td>Objective</td><td>To adapt the CRCF scope so that only projects with feasible and credible baselines are eligible for certification.</td></tr>
<tr><td>Existing, proven certification methodologies</td><td>The scope of building typologies and project types in various certification schemes and initiatives includes:
<u>Rainbow</u>: Includes both new construction and renovation of permanent buildings. It also includes the manufacture of materials where permanence is ensured upon production e.g. hempcrete (with proof that products are sold to building companies).
C-sink standard (v1.0): Extends beyond buildings to infrastructure (e.g. bridges or scaffolding for agrivoltaics) if biomass-derived carbon is durably embedded.
<u>Label Bas Carbone</u>: New permanent buildings > 500 m², or grouped projects with a total surface area > 500 m² (excluding individual houses, which are not eligible). A prerequisite is that carbon footprint has been measured by the BBCA (or another label approved by the BBCA) and is equivalent to or better than the RE2020 threshold for year 2028.
<u>Climate Cleanup Foundation certification protocol</u> (v1.0): Covers construction, renovation, or alteration of buildings aimed at long-term carbon storage (Section 1.1).
<u>Verra VCS</u> (draft): Expected to include most building types, but excludes single-family homes, agricultural buildings and any buildings where wood accounts for more than 20% of the materials (not sure if this is only structural materials and if it is % by mass or volume).</td></tr>
<tr><td>Options</td><td>Two options are presented for new construction projects and three options are presented for renovation projects, as shown in the table below.
<table>
<tr><th>Options</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Include all new construction projects for CRCF certification</td><td>Greatest potential for CRCF certification.
Strong and consistent market signal to producers of eligible wood-based and biobased construction products.</td><td>Difficult to generate sufficiently nuanced standardised baselines.
Linked to first point, some projects will be favoured artificially, and others penalised.
Monitoring may be problematic in some building types.</td></tr>
<tr><td>Only include new EPC-certified construction projects</td><td>Still applies to the majority of new buildings.
Easier to establish standardised baselines.
Baseline synergy with EPBD.
Possible synergies with sustainability CCM criteria.
Possible synergies with monitoring.</td><td>Misses some opportunities with non-EPC buildings.</td></tr>
<tr><td>Include all renovation projects for structural wood and insulation</td><td>Greatest potential for CRCF certification.
Strong and consistent market signal to producers of eligible wood-based and biobased construction products.</td><td>Impossible to make suitable standardised baseline, especially for structural wood.
Activity-specific baselines would be cumbersome and difficult to verify.</td></tr>
<tr><td>Include all renovation projects but for insulation only</td><td>Feasible standardised baseline* (EU market-average bio-based share)
Large renovation market, boosted by EPBD requirements.</td><td>Misses structural wood opportunities.</td></tr>
<tr><td>Exclude all renovation projects</td><td>Avoids standardised baseline complexities with structural wood.</td><td>Misses a large share of potential storage (especially insulation).</td></tr>
</table>
* The standardised baseline for biobased insulation materials is currently close to zero and instead of trying to set it in the same units as structural wood (i.e. kgCO2eq./m2), which would be difficult to calculate in renovation projects, a much better</td></tr>
</table>

Aspect	Remarks about 2.4 - scope of building typologies and projects
	standardised baseline would be to count the average (plant-based) biogenic carbon content in insulation materials sold in the EU in a given year (see section 3.2.2 for more details).
Technical conclusions	We recommend that the scope of building projects be as follows: • New construction: including both structural wood and bio-based insulation materials. • Renovation: applicable to all renovation projects but using an activity-specific baseline for any structural wood.
Next steps	Collaborate with insulation producers to develop an average biobased content for insulation materials used in construction – leading to a standardised baseline at EU or national market level. Review background data and assumptions to establish a default standardised baseline for structural wood in different building typologies and regions (see section 2.6.5 and Appendix V for some initial details).

Q5. What are your opinions about how to deal with renovation projects? Should there be a minimum scale defined in the methodology, or should that be left up to certification schemes?

2.5 ”GHG-associated”: scope of building life cycle stages and GWP categories

Purpose: To demonstrate at a fundamental level how the different components of GHG-associated emissions are counted in a full building life cycle assessment. Then, to clarify why it is necessary to narrow the scope of life cycle modules for CRCF quantification. Finally, to compare the relative significance of various life cycle stages and GWP categories in selected examples of real wood-based and bio-based construction product EPDs.

2.5.1 Illustration of CRCF quantification elements over the entire building life cycle

A hypothetical example illustrating biogenic carbon removal, associated GHG emissions, and storage of net removed carbon throughout a building's lifecycle is shown in the figure below. Green indicates the removal of atmospheric carbon into biomass via photosynthesis. Red signifies GHG emissions for various reasons across the entire building lifecycle. Blue denotes the storage of net removed carbon during the use phase.

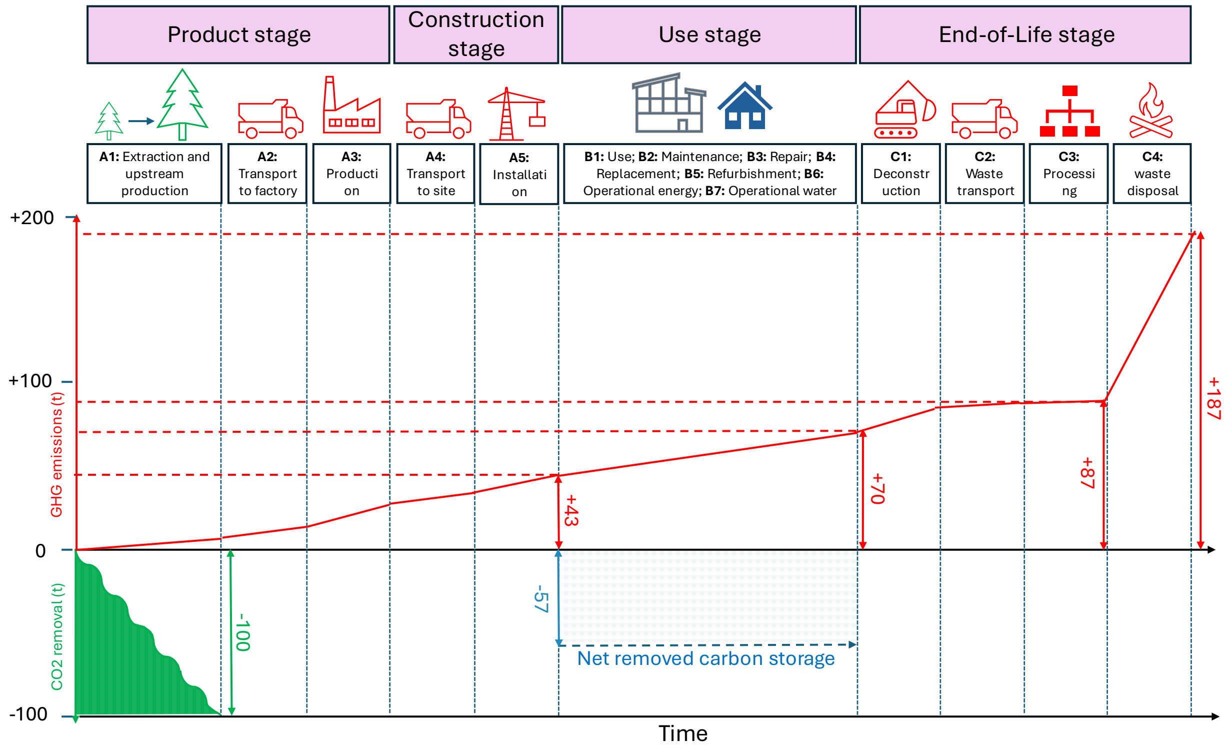

Figure 4. Hypothetical illustration of carbon removal, GHG-associated emissions, and storage of net removed carbon over the full building life cycle (numbers are arbitrary but proportionate to each other)

In the example above, the eligible wood-based or bio-based construction products used in the building start in module A1 with 100 tonnes of CRCF carbon removals stored in their biomass. The necessary processes to install these products in the building, covering modules A1 to A5, generate GHG emissions of 43 tonnes. Therefore, the net carbon removed and stored in the building would be 57 tonnes (100 minus 43).

As more building life cycle modules are taken into account, GHG emissions can only rise, and net removed carbon storage would theoretically decrease, even if the biobased materials remain untouched in the building. In the use stage (B modules), emissions may originate from various sources, such as cleaning, replacing other building materials, and energy consumption in the building. Additional emissions also occur at the end-of-life stage (C modules), particularly in module C4, where it is generally assumed that all stored biogenic carbon is released back into the atmosphere, typically through direct combustion. Using the figures in the diagram above, GHG emissions (shown in red) will eventually surpass the original carbon removals (shown in green).

Following a chronological order, it is logical to count emissions only from modules A1 to A5, as this covers everything up to the use phase, when the certified carbon storage activity begins. Because we propose to limit the scope to biobased materials that have no maintenance requirements, it is deemed appropriate to ignore all B-module emissions.

2.5.2 Considerations on limiting the scope of GHG emissions within modules A1-A5

The three types of GWP (biogenic, fossil, and LULUC) must be reported in construction product EPDs according to EN 15804, but modules A4 and A5 are optional. Harmonised standards under the CPR will also require reporting on the three types of GWP in construction product declarations of performance and conformity (DoPCs), but modules A4 and A5 will be compulsory.

The main issues with GHG emission reporting in modules A4 and A5 are:

- It relies on assumed average transport and installation scenarios. These scenarios will never exactly match the actual situation in a given building project, but serve as an approximation only.
- Some projects may wish to replace the assumed A4 and A5 data with project-specific data because they generated fewer GHG emissions in the real project.
- Many current construction product EPDs and generic datasets do not provide information on modules A4 and A5 because this was not required by EN 15804.
- Failing to report on modules A4 and A5 favours construction options that have higher onsite impacts (e.g. ready-mix concrete that is poured onsite versus precast concrete delivered to the site).

Based on the above points, reporting on modules A4 and A5 should be mandatory within the CRCF methodology. However, there are different options for how this requirement could be implemented.

Regarding the three types of GWP, different positions are taken by the main certification schemes. For example, GWP-LULUC is not directly included in the Climate Cleanup Foundation, Global C-Sink, or Rainbow standards. Instead, a risk-based approach is used that aims to minimise potential land-use change emissions (e.g. only using virgin material from sustainably managed forests). The C-Sink standard requires that all fossil GWP is offset by purchasing permanent carbon removal units. To ensure compatibility with EN 15804, the PEF methodology defined in Commission Recommendation (EU) 2021/2279, and the new CPR, all three types of GWP should be reported in the CRCF methodology.

However, the option in EN 15804 to omit declaring GWP-LULUC values if they contribute less than 5% to the total GWP, while still counting them as part of the total GWP, can cause confusion. This situation will no longer apply in upcoming CPR-harmonised product rules, where GWP-LULUC will also be declared separately. In CRCF calculations, GWP-LULUC should ideally always be included and separately declared, even if it is a relatively small contribution.

Table 4. General considerations of potential contributions of GHG emissions for different types of GWP in modules A1-A5 for wood-based or bio-based construction products

Life cycle module	Potential sources of:		
	GWP-biogenic	**GWP-fossil**	**GWP-LULUC**
A1: Extraction and upstream production	From the burning of any residual biomass from forestry or agricultural production.	Use of fertilisers, pesticides or fuel in heavy machinery and chainsaws.	Conversion of forest land into access roads. Conversion of forest land into agricultural land. Deforestation via clear-cutting.
A2: Transport to factory	Not expected unless vehicles run on biodiesel or electricity generated by biomass combustion.	Combustion of conventional diesel or petrol in transport vehicles.	Negligible unless new transport infrastructure that is rarely used for other purposes is required to get to the factory.
A3: Manufacturing	Combustion of biomass or production residues to generate heat or combined heat & power. May be embedded in any chemicals used.	Consumption of electricity, combustion of fossil fuels and embodied in any chemicals used.	Negligible unless the specific factory is built on a greenfield site and has a low production intensity. May be embedded in very specific chemicals used (e.g. palm oil-based resins).

A4: Transport to site	Not expected unless vehicles run on biodiesel or electricity generated by biomass combustion.	Combustion of conventional diesel or petrol in transport vehicles.	Negligible unless new transport infrastructure that is rarely used for other purposes is required to get to the market.
A5: Construction and installation processes	Waste from off-cuts or biogenic packaging materials that is not recycled.	Consumption of electricity, combustion of fossil fuels in heavy machinery and tools, and embodied in any chemicals or ancillary materials used.	Negligible unless the specific building is built on a greenfield site and is low-rise.

2.5.3 Review of GWP numbers in selected EPDs for wood-based and bio-based construction products

A selection of EPDs related to wood-based or bio-based construction products was reviewed. The aim of the exercise is to identify the figures corresponding to GHG emissions and to determine where the most significant emissions generally originate. A more detailed analysis is provided in Appendix III.

All twelve EPDs reviewed declared on the three types of GWP. Only nine had complete reporting across modules A1 to A5. Furthermore, seven of these nine EPDs only provided A1 to A3 data as single numbers. A comparison of the relative contributions to A1-A5 impacts for each product is provided in the table below.

Table 5. Relative contributions to GHG emissions for selected products by life cycle module and GWP type (contributions >50% in red, contributions of 20-50% in orange and contributions of 5-20% in yellow)

<table>
<tr><th></th><th>Product</th><th>GWP type</th><th>A1</th><th>A2</th><th>A3</th><th>A4</th><th>A5</th><th>Total A1-A5 GHG-associated emissions</th></tr>
<tr><td rowspan="13">Structural wood-based products</td><td rowspan="3">Nordlund (planed timber)</td><td>Fossil</td><td colspan="3">85.6%</td><td>9.0%</td><td>1.9%</td><td rowspan="3">51.9 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">0.0%</td><td>0.0%</td><td>3.4%</td></tr>
<tr><td>LULUC</td><td colspan="3">0.1%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td rowspan="3">Holzwerk (Glulam)</td><td>Fossil</td><td colspan="3">95.9%</td><td>2.9%</td><td>0.8%</td><td rowspan="3">164.8 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">0.1%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td>LULUC</td><td colspan="3">0.3%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td rowspan="3">Stora Enso (Glulam)</td><td>Fossil</td><td colspan="3">43.1%</td><td>55.6%</td><td>0.9%</td><td rowspan="3">185.3 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">0.0%</td><td>0.0%</td><td>0.1%</td></tr>
<tr><td>LULUC</td><td colspan="3">0.3%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td rowspan="3">KLH (CLT)</td><td>Fossil</td><td colspan="3">57.9%</td><td>30.9%</td><td>8.8%</td><td rowspan="3">147.4 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">1.0%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td>LULUC</td><td colspan="3">1.4%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td colspan="2">Avg. shares by stage</td><td colspan="3">71.4%</td><td>24.6%</td><td>4.0%</td><td>100%</td></tr>
<tr><td rowspan="16">Bio-based insulation materials</td><td rowspan="3">Isocell</td><td>Fossil</td><td>21.5%</td><td>23.8%</td><td>21.8%</td><td>25.7%</td><td>10.0%</td><td rowspan="3">8.8 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td>0.0%</td><td>0.0%</td><td>-32.5%</td><td>0.0%</td><td>28.5%</td></tr>
<tr><td>LULUC</td><td>0.0%</td><td>0.4%</td><td>0.2%</td><td>0.5%</td><td>0.0%</td></tr>
<tr><td rowspan="3">Cyclin</td><td>Fossil</td><td>27.9%</td><td>2.9%</td><td>5.3%</td><td>1.9%</td><td>2.9%</td><td rowspan="3">93.4 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td>55.8%</td><td>0.0%</td><td>1.2%</td><td>0.0%</td><td>2.0%</td></tr>
<tr><td>LULUC</td><td>0.1%</td><td>0.0%</td><td>0.0%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td rowspan="3">Holzwerk</td><td>Fossil</td><td colspan="3">94.3%</td><td>1.9%</td><td>1.5%</td><td rowspan="3">83.2 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">0.0%</td><td>0.0%</td><td>1.0%</td></tr>
<tr><td>LULUC</td><td colspan="3">0.3%</td><td>0.0%</td><td>1.0%</td></tr>
<tr><td rowspan="3">Havenens Haender</td><td>Fossil</td><td colspan="3">66.3%</td><td>14.7%</td><td>3.6%</td><td rowspan="3">37.7 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">0.0%</td><td>0.0%</td><td>15.4%</td></tr>
<tr><td>LULUC</td><td colspan="3">0.0%</td><td>0.0%</td><td>0.0%</td></tr>
<tr><td rowspan="3">FASBA</td><td>Fossil</td><td colspan="3">78.0%</td><td>8.6%</td><td>6.2%</td><td rowspan="3">23.7 kgCO2/m3</td></tr>
<tr><td>Biogenic</td><td colspan="3">4.2%</td><td>0.0%</td><td>0.2%</td></tr>
<tr><td>LULUC</td><td colspan="3">1.5%</td><td>0.2%</td><td>0.9%</td></tr>
<tr><td colspan="2">Avg. shares by stage</td><td colspan="3">74.6%</td><td>10.7%</td><td>14.7%</td><td>100%</td></tr>
</table>

The data in the table above show that contributions to GHG-associated emissions are generally dominated by the A1-A3 modules and by GWP-fossil emissions.

Regarding GWP types, emissions from GWP-LULUC ranged from 0.0% to 2.6% for products. For GWP-biogenic contributions, the values were around 1% or less for six of the nine products, but could be significantly higher in some biobased insulation products (4.4%, 15.4%, or 59.0%). Concerning life cycle modules, the A4 and A5 modules, on average, contributed about 25 to 30% of the total GHG-associated emissions. This share was similar, on average, for both wood-based and biobased products.

Overall, the review indicated that:

- it would be relevant to include modules A4 and A5 in the scope of GHG-associated calculations,
- that GWP-biogenic can sometimes be significant, and
- that GWP-LULUC would generally fall below the 5% threshold, allowing it to remain undeclared in EN 15804-compliant EPDs.

To minimise potential confusion associated with GWP-LULUC, we recommend that future product calculation rules for structural wood or biobased insulation always declare GWP-LULUC regardless of its percentage share. Currently, under EN 15804, it is permitted to omit GWP-LULUC as separate information if its contribution is less than 5% of GWP over the declared modules in life cycle stages A to C.

Only modules A1-A3 need to be reported under EN 15804, and it is not mandatory to declare stored biogenic content for all bio-based construction products or packaging under EN 15804. The sector is moving towards harmonised standards for construction products, where reporting on modules A4 and A5 will become compulsory, and rules may be introduced for mandatory declarations of stored biogenic carbon content. Regarding this, it should also be considered to separately declare any biogenic carbon that is only stored in packaging.

Aspect	Remarks about 2.5 GHG-associated
Issue	The scope of CRCF calculations for GHG-associated should cover modules A1-A5. However, many relevant wood-based or biobased construction products on the market lack data for modules A4 and A5. The first issue is how quickly to move towards requiring data for A4 and A5. The second issue concerns the extent to which common rules and scenarios should be used for reporting on modules A4 and A5.
Objective	To present the advantages and disadvantages of different ways to address these issues, with a view to identifying the best approach.
Existing, proven certification methodologies	The approaches to the calculation of GHG-associated include the following: • **Rainbow**: Has assessments at both construction product level and building project level. At product and building level, modules A1 to A3 are included as part of the embodied carbon calculations. Modules A4 and A5 are considered equivalent in both baseline and project scenarios. • **C-sink standard (v1.0)**: In section 4.2, states that modules A1 to A5 are covered. Section 4.1 states that GWP-fossil emissions, plus N2O emissions from biomass cultivation must be offset by permanent carbon storage certificates. Section 4.2 makes reference to emissions from modules B4 and C1-C4 as well and section 4.3 mentions scope 1, scope 2 and scope 3 emissions. • **Label Bas Carbone**: the method only considers the stored carbon, and does not directly consider the emissions. However, the prerequisites (BBCA and RE2020 – 2028) make sure that buildings are low carbon at the level of the whole building LCA. • **Climate Cleanup Foundation certification protocol (v1.0)**: Section 2.1.4 clearly explains that modules A1 to A5 are counted for GWP-LULUC, GWP-fossil and, for A4 and A5 only, for GWP-biogenic. • **Verra VCS**: Still under development, the draft methodology appears to focus on modules A1-A3 and on GWP-fossil and GWP-biogenic only. Emissions from A4 transport can be counted if the transport distance is more than 200km. Emissions from A5 are considered as negligible.

<table>
<tr><th>Aspect</th><th>Remarks about 2.5 GHG-associated</th></tr>
<tr><td></td><td></td></tr>
<tr><td>Options</td><td>
<table>
<tr><th>Type of building element</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Initially, only count modules A1-A3, then later extend to modules A1-A5.</td><td>• Less burden for manufacturers.
• All current EPDs provide sufficient data.
• Time for industry to update EPDs.</td><td>• Overly optimistic assessment.
• Less incentive for EPD producers to report on A4 and A5 today.
• Will create an artificial disparity between activities today and in the future.</td></tr>
<tr><td>Require reporting on modules A1-A5 with the possibility to overwrite manufacturer scenario rules for A4 and A5.</td><td>• No extra burden for the construction company if they only want to use supplier scenarios.
• Option for scenarios to be tailored to the specific building project context.
• Option to be more accurate.
• Option to reward good installation practice and local sourcing.</td><td>• Many existing EPDs with no A4 and A5 data are unsuitable.
• Requires LCA expertise from the constructor to modify A4 and A5 assumptions.
• More complicated for any external auditors to assess and verify the underlying data and assumptions.</td></tr>
<tr><td>Require reporting on modules A1-A5 with no possibility to overwrite scenarios for modules A4 and A5.</td><td>• More consistent calculations.
• Simpler for any external auditors to assess and verify underlying data and assumptions.</td><td>• Many existing EPDs with no A4 and A5 data are unsuitable.
• A4 and A5 data for the same product will be of variable accuracy in different building projects.
• No option to improve accuracy.
• No option to incentivise better installation practice or local sourcing.</td></tr>
<tr><td>Variation on each of the options above: to require a separate declaration of stored biogenic carbon in the product and in the packaging.</td><td>• Provides more precise detail because packaging is not eligible for CRCF units, but the product is.
• Even more critical when module A1-A3 data is reported together.
• Will explain any A5 biogenic emissions for waste packaging.</td><td>• Some existing EPDs will not be compliant.</td></tr>
</table>
</td></tr>
<tr><td>Technical conclusions</td><td>Based on the arguments presented, the following actions are recommended:
1. Report on all three types of GWP in CRCF calculations (biogenic, fossil and LULUC).
2. Include modules A4 and A5 in GHG-associated accounting in line with the scope of EPDs or CPR product rules.
3. Require separate reporting of stored biogenic carbon in a) the product, and b) the packaging.
4. Define clear options for scenario assumptions with modules A4 and A5 in relevant harmonised standards under the CPR.
5. That assumed A4 and A5 emissions in the EPD or DoPC should be unaltered when doing building level calculations.</td></tr>
</table>

Aspect	Remarks about 2.5 GHG-associated
	6. To ensure better accuracy in point 5, EPDs or DoPCs should include multiple transport scenarios for different destination countries in the EU.
Next steps	Consult with relevant technical committees on the state of the art concerning product rules under harmonised standards for the LCA of construction products, especially bio-based insulation materials. Consult with manufacturers about current approaches to EPDs, especially regarding the best ways to deal with A1-A5 GWP-biogenic emissions and storage.

2.6 "CR-baseline": structural wood vs bio-based insulation and new build vs renovation

Purpose: This section explains why separate baseline figures for structural wood and biobased insulation materials are deemed beneficial and how they can be incorporated into the quantification equation. It also points out differences depending on whether the project involves new construction or renovation.

2.6.1 Key points from the CRCF Regulation

A net carbon removal benefit is achieved only if carbon removals exceed the baseline, as stated in recital 17 of the CRCF Regulation: "*An activity delivers a net carbon removal benefit when the carbon removals above the baseline exceed any increase in greenhouse gas emissions associated with the implementation of that activity.*". If the activity's carbon removal is equal to or less than the baseline, it is ineligible for certified units. This remains true even if GHG-associated is zero.

A standardised baseline should be used if available. Article 4(8) of the CRCF Regulation states: "*8. The baseline referred to in paragraphs 1, 2 and 5 shall be highly representative of the standard performance of comparable practices and processes in similar social, economic, environmental, technological and regulatory circumstances and take into account the geographical context, including local pedoclimatic and regulatory conditions ('standardised baseline')*."

Furthermore, this standardised baseline has to be reviewed and updated at least every five years according to Article 4(9), which says: "*The Commission shall review at least every five years and update, where appropriate, the standardised baseline in light of evolving regulatory circumstances and of the latest available scientific evidence. The updated standardised baseline shall apply only to an activity for which the activity period starts after the entry into force of the applicable certification methodology.*".

However, provision is made for the use of activity-specific baselines in some instances. Article 4(10) states: "*By way of derogation from paragraph 8, where duly justified in the applicable certification methodology, including due to the lack of data or the absence of sufficient comparable activities, an operator shall use a baseline that corresponds to the individual performance of a specific activity ('activity-specific baseline').*". The requirement to update activity-specific baselines is less explicitly stated than for standardised baselines. Article 4(11) states: "*The activity-specific baselines shall be periodically updated, at the beginning of each activity period, unless otherwise stated in the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8.*".

2.6.2 Separate standardised baselines for structural wood and biobased insulation

The previous draft TAP suggested that any net carbon removal should be expressed in units of kgCO2eq/m2 of useful floor area. The same units should therefore be used for the baseline. However, quantifying a standardised baseline for biobased insulation materials in this unit is not so straightforward. Arguments supporting a separate approach to the standardised baseline for biobased insulation materials are summarised in the table below.

Table 6. Reasons for separate standardised baselines for structural wood and biobased insulation material

Reason	Further explanation
Use of structural wood varies considerably by building type and region. There are technical and market reasons behind this. These reasons do not apply to biobased insulation.	High-rise buildings face technical limitations because wood structures are less capable of providing the necessary lateral wind resistance. Buildings in southern Europe, due to hot climates, limited proximity to forest reserves, and a lack of cultural tradition, do not often utilise wooden structures.
At the building level, the amounts of stored biogenic carbon in structural wood can be estimated more easily and accurately than in biobased insulation.	This is due to the greater consistency of wood materials and established rules for wood (e.g. EN 16449). Biobased materials encompass a diverse range of materials, whether virgin or recycled, and can incorporate carbon from various sources, some of which may not involve plant-based photosynthesis (see section 2.3). Furthermore, quantities of insulation,

Reason	Further explanation
	especially loose-fill insulation, can be difficult to accurately quantify in a building.
A market-share-based baseline for biobased insulation is feasible because products are consistently used for purposes that are within the CRCF scope. This is not always the case for harvested wood products.	Insulation products serve a clear market segment for buildings. While they may take various forms (e.g. loose-fill or rigid panels), serve different applications (e.g., wall, floor, or roof), and have different primary purposes (e.g. acoustic or thermal insulation), they all fall within the CRCF scope. However, not all timber products suitable for building structures are actually used for that purpose, and some of these materials will end up in applications outside the CRCF scope. Trying to apply a market-based approach for wood in all building structures would be highly uncertain. Data from the DG GROW study was not sufficiently disaggregated.
The market share of biobased insulation materials is expanding rapidly. Production can scale up much faster than for harvested wood products.	Biobased insulation materials have progressed from niche markets to an average market share of 6.5% in the EU[5] and as high as 12% in some countries. Because raw materials can be sourced from agricultural by-products or paper waste, the scale of production can be quickly increased if there is demand. Demand can change swiftly since insulation materials are largely interchangeable. With harvested wood products, demand for structural wood is less volatile. Decisions on structural materials generally follow established and proven practices unless there is a clear demand from the client or other factors influence the choice. Furthermore, the skills and experience needed in construction workers remain the same when switching from conventional insulation to biobased insulation. But they change when switching from one structural material to another (e.g. concrete to wood).

Based on the reasons above, the calculation of a standardised baseline should involve combining two separate calculations, one for biobased insulation and one for structural wood. The equation from section 2.1 could be adapted as follows:

$$Temporary\ NCR\ storage = \begin{array}{c} \boxed{(CRS_{baseline} - CRS_{total} - GHG_{associated})[for\ insulation]} \\ + \\ \boxed{(CRS_{baseline} - CRS_{total} - GHG_{associated})[for\ structural\ wood]} \end{array}$$

The final result on the left-hand side of the equation must be in units of tonnes of CO2eq. To get there for insulation, using a market-based approach to the baseline, the following steps apply:

- Estimate the total mass of insulation used in the project (e.g. 4 tonnes).
- Define the standardised market baseline (e.g. average of 6.5% of insulation being biobased, by mass).
- Baseline for project would be: 4 tonnes x (6.5/100) = 0.26 tonnes biobased insulation.
- Estimate the actual quantity of biobased insulation used in the building project (e.g. 2 tonnes).
- Calculate a net standardised additionality of 1.74 tonnes (i.e. 2.0t – 0.26t).
- Multiply by the biogenic carbon content (e.g. 0.42 tCeq/t material) to get 0.7308 tC.
- Multiply up from C to CO2 (i.e. 0.7308 tC x 44/12 = 2.68 tCO2eq.).

[5] According to a public extract from a market report on biobased insulation materials published by Ceresana. Data for the following EU Member States was included: AT, BE, CZ, FR, DE, IT, PL, ES and NL, plus an overall analysis for the Rest of Europe. Data is also broken down into "new construction" & "renovation", and into "residential" & "non-residential". The average was estimated based on data from See here: https://ceresana.com/en/produkt/biobased-insulation-material-market-report

- Subtract the module A1-A5 GHG-associated with the production of 1.74 tonnes of biobased insulation material (e.g. 0.17 tCO2eq).[6]
- Final result for insulation would be a Temporary NCR storage of **2.51 tCO2eq** (i.e. 2.68 – 0.17).

For structural wood, using different units for the standardised baseline, the calculation would follow these steps:

- Define the standardised baseline for structural wood (e.g. 50 kg/m2 useful floor area).
- Estimate the mass of structural wood used in the building project (e.g. 75 tonnes, or 75,000 kg).
- Estimate the useful floor area of the building (e.g. 600 m2).
- Derive the project-specific normalised mass of structural wood (i.e. 75000 / 600 = 125 kg/m2).
- Subtract the standardised baseline from the project-specific value (i.e. 125 – 50 = 75 kg/m2).
- Multiply back out by the useful floor area (i.e. 75 x 600 = 45,000 kg or 45 t “additional” structural wood).
- Calculate the biogenic carbon content of the “additional” structural wood (e.g. 0.50 tC/t wood) to get 22.5 tC.
- Multiply up from C to CO2 (i.e. 22.5 tC x 44/12 = 82.5 tCO2eq.).
- Subtract the module A1-A5 GHG-associated with the production of 45t “additional” structural wood (e.g. 1.85 tCO2eq).[7]
- Final result for structural wood would be a Temporary NCR storage of **80.65 tCO2eq** (i.e. 82.5 – 1.85).

Combining the figures for biobased insulation and structural wood, a total of **83.16 tCO2eq** (i.e. 80.65 + 2.51) would be applicable.

2.6.3 Separate baselines for new construction and for renovation projects

Establishing a standardised baseline for new construction projects involves estimating the quantities of structural wood in an average new building typology within a defined geographical region. This can be done using various methods and levels of detail (e.g. by more or less specific building typologies or regions). The analysis requires two pieces of information: (i) the total quantity of structural wood used, and (ii) the useful floor area.

In renovation projects, setting a standardised baseline for structural wood is much more complicated. The main issues are:

- Each renovation project is unique and may vary in size depending on the client's available budget, the size of the building being renovated, and the depth of the renovation required.
- For structural wood, the baseline unit is kg/m2. Significantly different results will arise depending on whether the m2 corresponds to the entire useful floor area or just the area being renovated.

However, these issues apply only to structural wood baselines. The market-based approach to the standardised baseline for biobased insulation avoids any concerns with the floor area. There are two main options for dealing with structural wood in renovation projects: (i) exclude it from the methodology, or (ii) allow for activity-specific baselines to be used.

The issues with baselines for structural wood and biobased insulation (section 2.6.2) and for new construction and renovation projects (this section 2.6.3) are presented in the table below.

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[6] Note: We do not subtract the GHG-associated with the full 2 tonnes of biobased insulation because we assume that any GHG-associated with the baseline biobased insulation is the same as the biobased insulation used in the project, so they cancel out.
[7] Note: We do not subtract the GHG-associated with the full 75 tonnes of structural wood because we assume that any GHG-associated with the baseline structural wood is the same as the structural wood used in the project, so they cancel out.

<table>
<tr><th>Aspect</th><th>Remarks about 2.6a setting baselines</th></tr>
<tr><td>Issue</td><td>A single default EU standardized baseline for all new buildings is unlikely to be practical because of the high degree of heterogeneity at regional levels and in terms of building typologies within the same region. Furthermore, it is known from the implementation of the EPBD, that floor areas for buildings are not defined in the same way in different Member States, so a baseline in units of kgCO2eq./m2 could be problematic if not nuanced at national level at least.
Since the need for a number of regional or national baselines is foreseen, two key issues to consider are raised in this table:
• Baseline approach to structural wood and biobased insulation.
• Baseline approach to new construction and renovation.</td></tr>
<tr><td>Objective</td><td>To recommend part of the approach to CR baselines.</td></tr>
<tr><td>Existing, proven certification methodologies</td><td>The approaches to baselines include the following:
• Rainbow: The baseline is activity-specific and the conditions or practices that would occur in the absence of the project. For example, what materials would have been used if not biobased ones? Then it must be considered how the quantities of alternative materials needed to provide the same performance as the activity-specific choices may vary (e.g. may need replaced during the certification period or may need thicker or thinner layers for a given insulation performance).
• C-sink standard (v1.0): Does not appear to use a baseline approach. A baseline of effectively zero applies when the fossil component of GHG-associated is offset by the purchase of permanent removal units.
• Label Bas Carbone: the scenario developed is based on the average wood volumes for each construction method (timber frame, CLT, etc.) provided in the 2019 BIPE FCBA Ademe France Bois Forêt study. Projects may only claim credit under the Low Carbon label for the difference between their actual carbon stock and the reference stock.
• Climate Cleanup Foundation certification protocol (v1.0): Section 2.1.2 provides a standardised baseline for certain building typologies in the Netherlands (in units of kgCO2eq./m2 GFA). Other buildings have to use an activity-specific baseline based on a reference scenario that is peer-reviewed and verified. There is also some possibility to use the activity-specific baseline in cases where the standardised baseline is not considered appropriate for that individual project.
• Verra VCS: Still under development, the draft methodology uses activity-specific baselines where the reference building is a “mineral twin”.</td></tr>
<tr><td>Options</td><td>Many of the options to be discussed here are common to those that will be faced by Member States when developing national methods for life cycle GWP assessment of new buildings under Article 7(2) of the recast EPBD. For this reason, it is recommended that those national experts also be consulted on this topic. Any data provided or statements signed should be at the as-built stage of a building, when the exact wood products used are known and any alterations to building design during the construction phase can be accounted for.
<table>
<tr><th>Standardised baselines</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Single combined baseline units of kg/m2</td><td>• Simpler when running the quantification equation.</td><td>• Difficulties reconciling average biobased insulation with average m2 during baseline development.
• Hides potentially large differences between numbers for structural wood and for biobased insulation.
• Difficult to apply in renovation projects.</td></tr>
</table></td></tr>
</table>

Aspect	Remarks about 2.6a setting baselines		
	Separate baselines, both in units of kg/m2	• Shows the differences between structural wood and biobased insulation.	• Difficulties reconciling average biobased insulation with average m2. • Requires two equations as part of the quantification calculation. • Difficult to apply in renovation projects
	Separate baselines, with structural wood in kg/m2 and insulation as % market share that is biobased	• Shows the differences between structural wood and biobased insulation. • Easier to define and update biobased insulation baseline. • Biobased insulation baseline is easy to apply in renovation projects.	• Requires two equations as part of the quantification calculation. • Still difficult to apply structural wood in renovation projects.
	Renovation project baseline	**Pros**	**Cons**
	Exclude structural wood from standardised baseline.	• Problems are completely avoided.	• Opportunity to incentivise use of structural wood in renovation is lost.
	Include structural wood in standardised baseline only for extension projects with a new floor area.	• Incentivises the use of structural wood in renovation. • Floor area can be clearly and consistently defined (i.e. new floor area in extension).	• Baseline for extension projects may already be very high. • Extra research effort needed to quantify these separate baselines in the first place.
	Include structural wood and use an activity-specific baseline.	• Incentivises the use of structural wood in renovation.	• Activity-specific baselines bring additional considerations as per Article 5(1). • For example, would need to demonstrate that the extension is not financially viable without incentive effect of CRCF. • May need to also bring biobased insulation to an activity-specific baseline to be consistent. • Need to define a reference project that would not use wood or biobased insulation (or that would use less).

Aspect	Remarks about 2.6a setting baselines
Technical conclusions	Based on the arguments presented above, it is highly recommended to: • Have separate standardised baselines for structural wood and biobased insulation. • Either focus solely on biobased insulation in renovation projects (using a market-based standardised baseline), or to allow an activity-specific approach for all renovation projects with structural wood.
Next steps	Liaison with the consultants behind the DG GROW study on GHG emissions from the EU building stock to develop a default EU standardised baseline for new construction projects.

2.6.4 Synergies with EPBD in baseline development

From 2030 onwards, the EPBD will require the assessment of life cycle GWP of new buildings. Results are to be expressed in kgCO2eq./m2 of useful floor area. Data from these assessments can help inform future standardised baselines for new build projects. To ensure synergies between the EPBD and the CRCF, buildings under both policies should use the same floor area definition. Experience with the EPBD has shown that the floor area definitions vary between Member States. Just this one aspect alone is sufficient to justify the need for different standardised baselines between different Member States.

As national methodologies are being developed for the assessment of life cycle GWP of new buildings, the input data and processing should be adapted so that it is straightforward to identify the inputs in the bill of materials that would qualify for baseline calculations (i.e. structural elements and insulation). Each input in these categories should have a check-box to indicate whether it is CRCF-eligible as “structural wood” or as “biobased insulation”.

An alternative or complementary approach would be to require a signed declaration from the architect and/or structural engineer for the project that clearly states:

- The total quantity of structural wood.
- The total quantity of biobased insulation.
- The total quantity of non-biobased insulation.
- The relevant useful floor area of the building.

These considerations lead to several options regarding the advantages and disadvantages of different methods for verifying quantities and/or defining floor areas (in the event that national methodologies for EPBD have not yet made a decision on this matter). These considerations are equally relevant to future baseline development and to actual quantification exercises in certified buildings.

Aspect	Remarks about 2.6b on data confirmation and floor area definitions in data gathering exercises to develop standardised baselines in buildings
Issue	A number of standardized baselines for new buildings of different typologies and in different Member States will be necessary to help facilitate CRCF calculations. However, before this can be done, clear positions on how to confirm data and which floor areas to use require some consideration.
Objective	To recommend part of the approach to CR baselines (and potentially to project quantification exercises).
Existing, proven certification methodologies	The approaches to baselines include the following: • **Rainbow**: Uses one of a series of reference buildings to act as activity-specific baselines. The reference buildings should have a comparable floor area, region and use (e.g. commercial, housing, collective housing…). An average LCA is provided using a public database of registered LCAs in France (this is possible only in France). • **C-sink standard (v1.0)**: Does not appear to use a baseline approach. A baseline of effectively zero applies when the fossil component of GHG-associated is offset by the purchase of permanent removal units. • **Label Bas Carbone**: uses “SDP”, which is equivalent to Gross Floor Area.

<table>
<tr><th>Aspect</th><th>Remarks about 2.6b on data confirmation and floor area definitions in data gathering exercises to develop standardised baselines in buildings</th></tr>
<tr><td></td><td>

Climate Cleanup Foundation certification protocol (v1.0): Section 2.1.2 uses the gross floor area (which is the area to the external face of outer walls and including areas occupied by building elements). It was not explained how the standardised baseline values were developed because the studies were confidential, but it include both structural and non-structural wood elements with lifetimes of >35 years.
Verra VCS: Still under development, the draft methodology seems to use activity-specific baseline. It is not clear yet how these baselines are developed.

</td></tr>
<tr><td>Options</td><td>
Many of the options to be discussed here are common to those that will be faced by Member States when developing national methods for life cycle GWP assessment of new buildings under Article 7(2) of the recast EPBD. For this reason, it is recommended that those national experts also be consulted on this topic. Any data provided or statements signed should be at the as-built stage of a building, when the exact wood products used is known and any alterations to building design during the construction phase can be accounted for.
<table>
<tr><th>Proof of wood quantities</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Signed statement by structural engineer and architect.</td><td>Reputable professionals who need to know these quantities accurately.
Could be applied retroactively to existing buildings that were recently constructed.</td><td>Additional paperwork.
Only gives raw wood quantities, further assumptions still needed on moisture and carbon contents.
Care needs to be taken when converting volumes into mass.</td></tr>
<tr><td>Bill of materials for the project</td><td>Offers the chance to cross-check that the products used line up with any associated EPDs.</td><td>A lot more extra paperwork.
Often not straightforward to pick out the relevant data.
Higher potential for mistakes or abuse.
Cannot be applied retroactively.
Care needs to be taken when converting volumes into mass.</td></tr>
<tr><th>Definition of building area</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Use the entire gross floor area</td><td>Most accurate representation of the building.
Minimum possibility for error.
Quickest area to check when looking at building drawings.
Probably already registered in national databases for land tax purposes.
Would be comparable to buildings in other countries.</td><td>Might not be compatible with floor area used in future national methods for LCGWP assessments under Article 7(2) of EPBD.
Not compatible with the floor area displayed on EPCs.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Aspect</th><th>Remarks about 2.6b on data confirmation and floor area definitions in data gathering exercises to develop standardised baselines in buildings</th></tr>
<tr><td></td><td>
<table>
<tr><td>Use whatever useful floor area is used in national building regulations</td><td>• Should be compatible with floor area used in future national methods for LCGWP assessments under Article 7(2) of EPBD.
• Definitely already registered in national databases.</td><td>• Probably not compatible with the floor area displayed on EPCs.
• Might not be easy to determine when looking at building drawings.
• Not comparable to buildings in other countries.</td></tr>
<tr><td>Use the EPC floor area (which is often just the area where temperatures are regulated)</td><td>• Consistent floor area used on the EPC (for both energy consumption and carbon stored).
• Definitely already registered in national databases.</td><td>• Not comparable to buildings in other countries.
• No direct relationship to the building structure (could get some strange numbers when low or high shares of total floor area are thermally conditioned).</td></tr>
</table>
</td></tr>
<tr><td>Technical conclusions</td><td>Regarding the contents of structural wood, the best option is to obtain a signed declaration from the architect and structural engineer. These are both highly reputable professionals who need to know these figures very precisely for load-bearing calculations and floor layouts.
Concerning the floor area, although a consistent, EU-wide definition for floor area is preferred (for example, gross internal floor area to the outer edge of the building envelope), it is advisable to use the floor areas specific to each Member State as they implement Article 7(2) of the recast EPBD. This is crucial for establishing future standardized baselines at the national level.</td></tr>
<tr><td>Next steps</td><td>To discuss with relevant Member State representatives their floor area definitions for EPBD lifecycle GWP assessments and potential synergies with the CRCF.</td></tr>
</table>

2.6.5 Creation of default standardised baselines based on data from DG GROW study

An extensive study has been published by DG GROW regarding the analysis of life cycle GHG emissions (and removals) of EU building stock. The DG GROW study is broken down into the following parts:

- Existing sources of data, gaps or shortcomings in data, and ways to collect or generate new data.
- Mapping of the most promising carbon reduction and removal strategies, taking into account national contexts.
- Report with quantitative baseline figures for whole life carbon and carbon removals.
- Methodology for collecting and generating new data.

As part of the research behind this TAP, it has been investigated whether the report in the third point above could be used as a starting point for calculating an average stored biogenic carbon content in structural wood and biobased insulation for certain building typologies. Although the data presented therein offer valuable insights, methodological limitations prevent its direct use as a CRCF baseline because aggregated values include out-of-scope products like flooring, window frames, and doors, thus inflating removal estimates.

For this reason, the KU Leuven Scenario Explorer was used for deriving default values as it provides more granular regional and building-type data, even though it retains similar limitations. These limitations have been addressed by estimating the typical quantity of wood contained in standard structural timber assemblies for external walls and timber roofs (pitched and flat). These estimations provide approximate values in kg of structural wood mass per m^2 of external wall or roof area. Because the Scenario Explorer reports wood mass per square metre of useful floor area (kg/m^2), reasonable ratios to translate the roof or external wall area to useful floor area for the relevant building types are applied, allowing a realistic allocation of wood to structural components only.

The resulting default values are summarised in the table below. Although data for most EU Member States was available, only results for the higher quality datasets are presented below, with Germany (DE) being considered as representative of Central Europe, Italy (IT) being considered as representative of Southern Europe and Sweden (SE) being considered as representative of Northern Europe. No representation of Eastern Europe is included at this stage due to doubts about data quality for all of those relevant Member States.

Table 7. Approximate estimates of stored carbon in different geographical regions and building typologies

Country	Building type	Building element[8] and normalised masses (kg/m2)						Total wood mass (kg/m2)	CRCF-eligible wood mass (kg/m2)[9]
		Floor storeys	External walls	Common walls	Roofs	Stair-cases	Sub-structure		
DE	SFH	0.0	0	0.0	9.5	1.0	0.0	10.5	8.0
	MFH	0.0	0.0	0.0	0.0	0.0	0.0	0	0
	OFF	0.0	0.0	0.0	0.0	0.0	0.0	0	0
IT	SFH	0.0	0.0	0.0	1.4	1.5	0.0	2.9	0
	MFH	0.0	0.0	0.0	0.9	0.0	0.0	0.9	0
	OFF	0.0	0.0	0.0	0.0	0.0	0.0	0	0
SE	SFH	0.00	26.4	0.0	17.6	0.8	14.2	85.4	30.0
	MFH	9.2	9.3	0.0	5.9	0.0	4.7	38.3	12.0
	OFF	0.0	0.0	0.0	0.0	0.0	0.0	0	0

Each of the elements listed in the table above can contain CRCF-eligible parts and non-eligible parts, except for sub-structure, which should always be eligible.

For single-family homes (SFH) in Central Europe (represented here by Germany), a default baseline of around **8 kg/m²** is appropriate, reflecting structural wood primarily in roof elements. For multi-family homes (MFH) and office buildings (OFF), the standardised baseline would be zero. In Southern Europe (represented here by Italy), a zero-baseline assumption for all three building types (SFH, MFH and OFF) is appropriate. In Northern Europe (represented here by Sweden), the use of wood in residential buildings was much more intensive, especially in SFH and inexternal walls. For SFH, baseline values in Sweden were around 4 times higher than in DE. Multi-family homes (MFH) in Sweden show lower wood intensities due to shared roofs and external wall areas relative to useful floor area. Despite the higher values in residential SE, office building baselines were basically zero. The full analysis is included in Appendix V.

2.6.6 Summary of aspects to consider with baselines

To conclude this section, an overall summary of baseline considerations is presented below.

Table 8. Summary of considerations relating to different aspects related to baselines

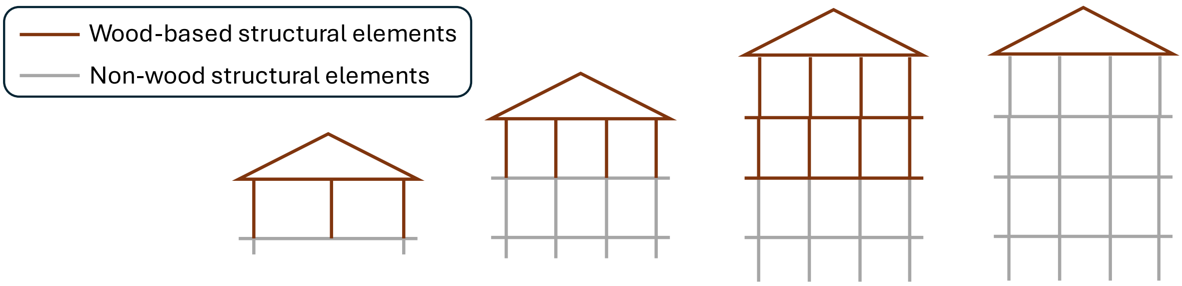
Aspect	Structural wood		Biobased insulation	
	New build	Renovation	New build	Renovation
Proposed baseline metric	$kgCO_2eq$ / m² (use national EPBD LCA floor area unit)	Either exclude or use an activity-specific baseline	% biobased content by mass (compared to market average)	
Baseline scope	Regional / national, typology-sensitive baselines required (EU-wide average inappropriate)	To an applicable reference renovation project	Single market-average % bio-based content figure for insulation sold in reference year	

[8] As defined in the KU Leuven data explorer. See more details in Appendix V.
[9] For further details about how these numbers were derived, see Appendix V, specifically Table 28 and Table 29.

Aspect	Structural wood		Biobased insulation	
	New build	Renovation	New build	Renovation
Data proof / source	Signed declaration by architect + structural engineer (as-built stage)	Same as new build, if included	Product data + quantity installed	
Floor area definition	Use national LCA floor area metric (aligned with EPBD Art 7(2))	Same as new-build, if included	N/A	
Risk of gaming / overdesign?	Low if properly verified by professionals	High unless reference project is reasonable	Low — cost plus space constraints limit overuse	
Main advantage	Tailored to building type and region; credible quantification	Captures valid wood use in structural additions	Simple, comparable baseline; easy to monitor	
Main challenge	Need regional/typology baselines; national coordination	Difficult to define meaningful standardised baseline	Requires reliable data on market bio-based content	
Open questions	How granular should regional/typology baselines be? Should hybrid baselines (region + typology) apply?	Should structural wood be included?	Should bio-based % be averaged EU-wide or national? How often update (e.g. 5 years)?	
Next step	Develop regional/typology baseline framework; align with national EPBD Art 7(2) LCA methods	Decide if only activity-specific baselines are permitted for structural wood in renovations	Collect market data on bio-based content; propose baseline figure	

2.6.7 How to generate a standardised baseline for structural wood

The method for determining the baseline for structural wood-based materials is illustrated below, using a group of four related buildings that can be classified within the same typology (e.g. single-family houses).



	One storey	Two storey	Three storey	Three storey
Useful floor area	100m2	190m2	285m2	285m2
Roof	1 x 2000kg wood	1 x 2000kg wood	1 x 2000kg wood	1 x 2000kg wood
Floor slabs	0kg wood	0kg wood	2 x 1000kg wood	0kg wood
Column sections	9 x 100kg wood	12 x 100kg wood	24 x 100kg wood	0kg wood
Raw wood baseline	3000kg/100m2	3390kg/190m2	6685kg/285m2	2285kg/285m2
Specific wood baseline	30kg/m2	17.8kg/m2	23.5kg/m2	8.0kg/m2
Carbon content (kg/kg)	0.5	0.45	0.49	0.45
Specific C baseline	15.0kgC/m2	8.0kgC/m2	11.5kgC/m2	3.6kgC/m2
Specific CO2 baseline	55.0kgCO2/m2	29.4kgCO2/m2	42.1kgCO2/m2	13.2kgCO2/m2
Average CO2 baseline	(55.0 + 29.4 + 42.1 + 13.2) / 4 = **35.0 kgCO2/m2**			

Figure 5. Example of how numbers would be used to develop a low-resolution standardised baseline for structural wood (masses and areas are purely arbitrary, but consistent across the four hypothetical buildings).

The images in the figure above are front-facing section diagrams in 2D. It is assumed that the column sections are repeated three times in each building along the third dimension. When examining the table from top to bottom, the first aspect to consider is the useful floor area. This should be based on a consistent definition of floor area that is used for all buildings within a given baseline development exercise. Next, it is necessary to estimate the mass of wood-based structural materials used. This may require a careful review of technical drawings and documentation to distinguish between eligible elements (e.g. beams, columns, floor slabs, and roofs) and those that are not eligible (e.g. cladding, floor coverings, solar shading devices, etc.).

Technical documents might only specify the volume of structural wood-based products. However, these volumes can easily be converted to mass using the densities and moisture contents provided in the relevant product EPDs.

The total mass of structural wood divided by the useful floor area provides an initial baseline in kg wood/m2. This baseline can then be converted to kgC/m2 by multiplying by the elemental carbon content of the dry wood mass – this carbon content should be sourced from the product EPD if available and may vary slightly among different types of wood. Multiplying by 44/12 then converts the baseline to kgCO2/m2. These final figures can be averaged across all individual buildings in the baseline exercise to produce a standardised average for that building typology (35.0 kgCO2/m2 in the example above).

2.6.8 How to generate a standardised baseline for bio-based insulation materials

While the approach to structural wood is a building-based method, which requires examining individual buildings to establish a baseline, a market-based approach is recommended for bio-based insulation materials. The latter relies on sales figures and should provide an average percentage of bio-based content for insulation materials sold within the EU market. Market shares can be expressed in different ways, as outlined in the table below.

Table 9. Market-based approach for calculating the standardised baseline at EU level for bio-based insulation materials in a given year.

Insulation material	Bio-based content (%)	EU27 sales (m3/yr)	Density (kg/m3)	Sold mass (kg)	% share by m3	Average bio-based content by m3	% share by kg	Average bio-based content by kg
A	0%	1900	30	57000	55.2%	1.67%	54.0%	2.15%
B	0%	850	32	27200	24.7%		25.8%	
C	0%	625	30	18750	18.2%		17.8%	
D	50%	15	35	525	0.4%		0.5%	
E	100%	50	40	2000	1.5%		1.9%	
TOTALS		**3440**		**105475**	**100.0%**		**100.0%**	

The data in the table above (the numbers are purely arbitrary) shows how data for specific insulation materials could be combined and converted into market shares by volume and by mass. In all cases, supporting information about the insulation products would be required, specifically the average bio-based content and average density. If only monetary sales data were provided, then an average unit price would also be required to calculate values per m3 and per kg.

In the table above, it can be seen that the average percentage of bio-based content in EU insulation material is 1.67% by volume sold (m3), or 2.15% by mass sold (kg). To establish a standardised baseline for insulation materials, the following unit preferences are considered:

- Percentage by monetary value should not be used, as changes in unit price can cause unwanted volatility in the baseline.
- Percentage by volume is acceptable, but variations in packing densities and in-situ use densities might lead to some discrepancies between the baseline assumption and actual use in a certified building.
- Percentage by mass is the preferred unit because the mass sold and used remains consistent, even when packed more or less densely. It is also easier to measure unit mass accurately in a project than volume.

Sales data could be provided directly by relevant EU industry associations or at least verified by them for accuracy. While sales of major insulation materials are likely to be well known and stable, the share of bio-based insulation, starting from a low proportion, could change quite rapidly. Therefore, engaging with producers of bio-based insulation materials, who may be entirely separate from mainstream insulation producers, is equally important.

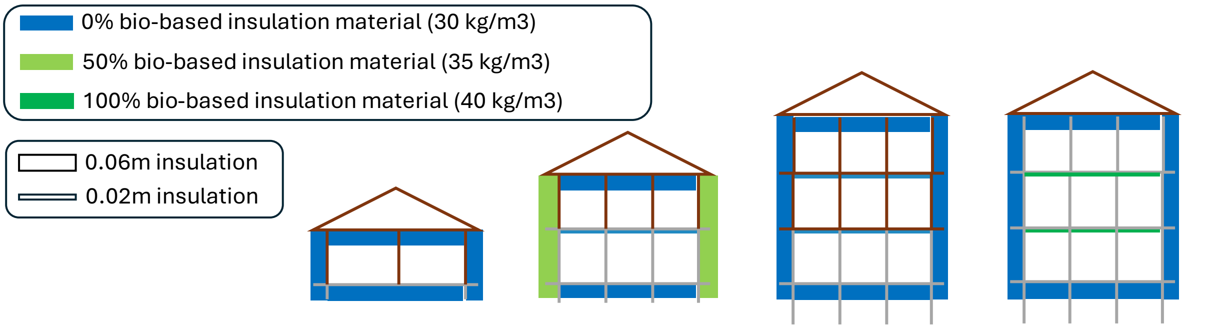
We believe that a standardised EU baseline for bio-based insulation materials could be justified because these can be easily traded and transported within the Union's single market. There are no definitive cultural, climatic or architectural differences across Europe that would make one region more or less likely to use bio-based insulation than other types of insulation. However, at present, shares of biobased insulation in individual EU countries could range from 0 to 12% due to different degrees of market penetration of the biobased insulation producers[10].

2.6.9 Example calculations of additionality for biobased insulation in specific buildings

Using the 2.15% by mass standardised baseline for biobased insulation from section 3.2.2, and applying the same hypothetical buildings as in section 3.2.1, the method for evaluating additionality in biobased insulation materials is illustrated below. In this example, there are three different types of insulation, each with varying biobased content, depths, and densities. Although many more varieties of insulation will exist in reality, each will feature its own bio-based content and density. Any reduction below 100% in a biobased insulation material indicates the presence of non-biobased carbon and/or non-carbon materials. Potential reasons for this may include the use of fossil-based resins, partially fossil-based polymers, composite layering with non-biobased layers, or the addition of inorganic flame retardants or fillers.

The diagrams below consider four main types of insulated surfaces: roof slab, ground floor slab, external wall, and mid-floor insulation. These examples can apply to both renovation projects and new construction.

[10] Based on freely published excerpts from a market report published by Ceresana titled: "Biobased Insulation Material Market Report – Europe". See here: https://ceresana.com/en/produkt/biobased-insulation-material-market-report



	One storey	Two storey	Three storey	Three storey
Baseline (% bio)	Average bio -based content of building insulation material sold in EU is: **2.15%**			
Useful floor area	100m2	190m2	285m2	285m2
Roof slab insulation	100m2 x 0.06m = 6m3 **6m3 @0% bio (180kg)**	95m2 x 0.06m = 5.7m3 **5.7m3 @0% bio (171kg)**	95m2 x 0.06m = 5.7m3 **5.7m3 @0% (171kg)**	95m2 x 0.06m = 5.7m3 **5.7m3 @0% (171kg)**
Ground floor slab insulation	100m2 x 0.06m = 6m3 **6m3 @0% (180kg)**	95m2 x 0.06m = 5.7m3 **5.7m3 @0% (171kg)**	95m2 x 0.06m = 5.7m3 **5.7m3 @0% (171kg)**	95m2 x 0.06m = 5.7m3 **5.7m3 @0% (171kg)**
Wall insulation	4x (3.5m x 10m x 0.06m) = **8.4m3 @0% (252kg)**	4x (6.5m x 10m x 0.06m) = **15.6m3 @50% (546kg)**	4x (9.5m x 10m x 0.06m) = **22.8m3 @0% (684kg)**	4x (9.5m x 10m x 0.06m) = **22.8m3 @0% (684kg)**
Mid-floor insulation	0m2 x 0.02m = 0m3 **0m3 @0% (0kg)**	95m2 x 0.02m = 1.9m3 **1.9m3 @0% (57kg)**	2x (95m2 x 0.02m) = 3.8m3 **3.8m3 @0% (114kg)**	2x (95m2 x 0.02m) = 3.8m3 **3.8m3 @100% (152kg)**
Average bio-based insulation (% by mass)	(612kg x 0.0)/612kg = **0%**	(546kgx0.5))/945kg = **28.89%**	**0%**	(152kgx1.0))/1178kg = **12.90%**
Additionality (%)	**0%**	28.89-2.15 = **26.74%**	**0%**	12.90-2.15 = **10.75%**
Total biobased products (kg)	**0 kg**	**546 kg (or 273kg)**	**0 kg**	**152 kg**
Biogenic C content (kg/kg)	**0.00**	**0.25 (or 0.50)**	**0.00**	**0.50**
Total biogenic C (kg)	**0 kg**	**136.5 kg**	**0 kg**	**76 kg**
Additionality adjusted total biogenic C (kg)	0kg x (-2.15/0.00) = **0 kg C**	136.5kg x (26.74/28.89) = **126.34 kg C**	0kg x (-2.15/0.00) = **0 kg C**	76kg x (10.75/12.90) = **63.33 kg C**
Additionality adjusted total biogenic CO2 (kg)	0kg x (44/12) = **0kg CO2**	126.34kg x (44/12) = **463.25 kg CO2**	0kg x (44/12) = **0 kg CO2**	63.33kg x (44/12) = **232.22 kg CO2**
Normalised by floor area	**0 kgCO2/m2**	**2.44 kgCO2/m2**	**0 kgCO2/m2**	**0.81 kgCO2/m2**

Figure 6. Assessing biobased insulation additionality in specific buildings against a standardised baseline.

The data in Figure 6 shows that buildings 1 and 3 do not have biobased insulation material, while buildings 2 and 4 do. Since the baseline is based on a % biobased content, it is essential to account for all insulation material used in any assessed building and to determine the average biobased content of all the insulation used.

To do this, each of the four main types of insulation is then quantified and multiplied by their respective volumes, unit densities, and biobased contents. An average percentage of biobased content is then determined for the entire building. This overall percentage is subsequently compared to the market's average standardised baseline.

A more detailed walkthrough of the calculations in rows 3 to 7 for the two-storey building is shown below for clarity:

- Mass of roof slab insulation = 95m2 x 0.06m = 5.7m3; 5.7m3 x 30kg/m3 = **171kg**
- Of which is biobased insulation = 171kg x 0% = **0kg**.
- Mass of ground floor slab insulation = 100m2 x 0.06m = 5.7m3; 5.7m3 x 30kg/m3 = **171kg**
- Of which is biobased insulation = 171kg x 0% = **0kg**.
- Mass of wall insulation = 4 x (6.5m x 10m x 0.06m) = 15.6m3; 5.7m3 x 35kg/m3 = **546kg**
- Of which is biobased insulation = 546kg x 50% = **273kg**.
- Mass of mid-floor insulation = 95m2 x 0.02m = 1.9m3; 1.9m3 x 30kg/m3 = **57kg**
- Of which is biobased insulation = 171kg x 0% = **0kg**.
- Total insulation mass = 171kg + 171kg + 546kg + 57kg = 945kg
- Total biobased insulation mass = **0kg + 0kg + 273kg + 0kg = 273kg**
- % biobased insulation used = 273 / 945 = **28.89%**

As shown in the seventh row of the table, both buildings 2 and 4 are above the standardised baseline[11] and therefore qualify for certified units. At this stage, the calculation returns to the total mass of insulation with any bio-based content, and this quantity is multiplied by the average biogenic C content (expressed here as kg biogenic C per kg insulation material). Ideally, this information should be directly included on insulation product EPDs. In the example above, a ratio similar to that used for wood is applied to the 100% bio-based insulation material (0.50), while a lower ratio (0.25) is used for the 50% bio-based insulation material. The lower ratio accounts for the fact that around half of the carbon in that product will not be biogenic carbon or is inorganic. This information should ideally also be provided in the product EPDs or DoPCs. A total biogenic C content is then calculated for the insulation material within the building.

2.7 Accounting for uncertainties in CR-total and GHG-associated

Article 4(12) of the CRCF Regulation states: "*12. The quantification of permanent carbon removals, temporary carbon removals through carbon farming and carbon storage in products, and soil emission reductions shall account for uncertainties in a conservative manner and in accordance with recognised statistical approaches.* ***<u>Uncertainties in the quantification of carbon removals and soil emission reductions shall be duly reported.</u>***".

As far as quantification of carbon storage in buildings is concerned, uncertainties are based on how accurate and precise the data used for the calculations is. There are three areas where uncertainty needs to be considered:

- Uncertainty in the biogenic carbon storage in products (CR-total).
- Uncertainty in the GHG-associated in products.
- Uncertainty in the baseline (CR-baseline).

The CR-total and GHG-associated uncertainties can apply at both product and building level. However, any approach to uncertainties must also consider the balance between practicality, administrative burden and credibility.

2.7.1 Uncertainties at product level based on PEF dataset quality evaluation

At construction product level, the approach to uncertainty is based on a qualitative assessment of data quality rather than on quantitative ranges for accuracy or precision. The main data quality framework for construction products is defined in Annex E to EN 15804, which aligns with the PEF methodology. This should be applied at product level quantification by operators who are producing the EPDs for construction products.

Table 10. Data quality level and criteria from the PEF Category Rules (as reproduced in Annex E to EN 15804)

Quality level	Geographical representativeness	Technical representativeness	Time representativeness
Very good	The processes included in the data set are fully representative for the geography stated in the "location" indicated in the metadata	Technology aspects have been modelled exactly as described in the title and metadata, without any significant need for improvement	Data are not older than 0 years as expressed in the ILCD field ("data set valid until" and the difference between the "valid until" and the "reference year" is not higher than 8 years)
Good	The processes included in the data set are well representative for the geography stated in the "location" indicated in the metadata	Technology aspects are very similar to what described in the title and metadata with need for limited improvements. For example: use of generic technologies' data instead of modelling all the single plants	Data are not older than 3 years as expressed in the ILCD field ("data set valid until" and the difference between the "valid until" and the "reference year" is not higher than 8 years)
Fair	The processes included in the data set are sufficiently	Technology aspects are similar to what described in the title and	Data are not older than 6 years as expressed in the ILCD field ("data

[11] Which, following from the example in table in section 3.2.2, is 2.15%.

Quality level	Geographical representativeness	Technical representativeness	Time representativeness
	representative for the geography stated in the “location” indicated in the metadata. E.g. the represented country differs but has a very similar electricity grid mix profile	metadata but merits improvements. Some of the relevant processes are not modelled with specific data but using proxies	set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years)
Poor	The processes included in the data set are only partly representative for the geography stated in the “location” indicated in the metadata. E.g. the represented country differs and has a substantially different electricity grid mix profile	Technology aspects are different from what described in the title and metadata. Requires major improvements.	Data are not older than 10 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years, confirmed by the reviewer(s))
Very poor	The processes included in the data set are not representative for the geography stated in the “location” indicated in the metadata.	Technology aspects are completely different from what described in the title and metadata. Substantial improvement is necessary.	Data are older than 10 years as expressed in the ILCD field (“data set valid until” and the difference between the “valid until” and the “reference year” is not higher than 8 years)

It is the quality-of-life cycle datasets used in LCA modelling that is assessed and how well the production processes are covered by those datasets and the model. A minimum level of quality could be specified for EPDs used in CRCF calculations (e.g. fair or better). However, this does not result in any quantification of uncertainties.

2.7.2 Quantification of dataset quality as defined in Level(s) indicator 1.2

The Level(s) framework also introduces a potentially relevant method for quantifying dataset quality. The method is described in the user manual for Level(s) indicator 1.2 and as briefly explained here. The approach is similar to that described in the table above, but also adds an uncertainty factor and allows for an overall single score to be generated. The four factors are:

- Technological representativeness of data (TeR)
- Geographical representativeness of data (GR)
- Time-related representativeness of data (TiR)
- Uncertainty of data (U)

A rating level is to be evaluated for each parameter according to the matrix in the table below. The overall rating is equal to the Data Quality Index (DQI), which can be calculated from the individual ratings as follows:

$$DQI = \frac{\left[\frac{(TeR + GR + TiR)}{3 + U}\right]}{2}$$

Table 11. Data quality evaluation matrix (copied from Level(s) indicator 2.1 user manual)

Rating aspect	*Brief description of each aspect*	Rating score			
		0	1	2	3
Technological representativeness (TeR)	Degree to which the dataset reflects the true population of interest regarding technology (e.g. the technological	No evaluation made	The data used **does not reflect satisfactorily** the technical characteristics of the system (e.g.	The data used **reflects partially** the technical characteristics of the system (e.g. Portland Cement	The data used **reflects** the technical characteristics of the system (e.g.

Rating aspect	Brief description of each aspect	Rating score			
		0	1	2	3
	characteristics, including operating conditions)		Portland Cement, without other specifications)	type II, without further specifications)	Portland Cement type II B-M)
Geographical representativeness (GR)	Degree to which the dataset reflects the true population of interest regarding geography (e.g. the given location/site, region, country, market, continent)	No evaluation made	The data used refer to **a totally different geographic context** (e.g. Sweden instead of Spain)	The data used refers to a **similar geographic context** (e.g. Italy instead of Spain)	The data used refers to **the specific geographic context** (e.g. Spain)
Time-related representativeness (TiR)	Degree to which the dataset reflects the specific conditions of the system being considered regarding the time/age of the data (e.g. the given year compared to the reference year of the analysis)	No evaluation made	There are **more than 6 years** between the validity of the data used and the reference year to which the data applies.	There are **between 2 and 4 years** between the validity of the data used and the reference year to which the data applies.	There are **less than 2 years** between the validity of the data used and the reference year to which the data applies.
Uncertainty (U)	Qualitative expert judgment or relative standard deviation expressed as a percentage.	No evaluation made	**Modelled/similar data** is used. **Accuracy and precision of the data has been estimated qualitatively** (e.g. by expert judgment of suppliers and process operators)	**Modelled/similar data** is used which is considered to be **satisfactorily accurate and precise** with the support of a quantitative estimation of its uncertainty (e.g. representative data from trade associations for which a sensitivity analysis has been carried out).	**Site specific and validated data** is used which is considered to be **satisfactorily accurate and precise** (e.g. window system for which a verified EPD is available) The allocation hierarchy has been respected.

As shown above, there are four variables (TeR, GR, TiR and U) and each variable can have 0, 1, 2 or 3 points. The overall DQI can vary from a minimum of 0 to a maximum of 4.5 for the highest quality data (and least uncertainty). Level(s) requires that the DQI must be 2 or higher. In the CRCF methodology, one option would be to automatically apply an uncertainty penalty of 4.5% to the GHG-associated. Then this penalty could be fully or partially cancelled out by the DQI.

2.7.3 Uncertainties at product level with physical properties affecting stored carbon estimates

Wood and biobased materials are naturally occurring materials which have differences in physical properties based on the species in question and, to a smaller extent, within the same species but based on its specific age, local growth environment. Differences in subsequent processing and storage can further affect moisture content and density.

Default rules and assumptions are needed for moisture content, carbon content per unit of dry mass, and density. Moisture content might range from 12-20%, with an average of 16% used for the calculation. A conservative approach would then be to use a higher moisture content range. Similar adjustments could be made for other variable features such as wood density and elemental carbon content in which the result would be to make the estimate of stored biogenic carbon more conservative. Any conservative corrections applied here would detract from the CR-total result.

2.7.4 Uncertainties at building level

The EN 15978 uses building element data to calculate embodied carbon as part of broader life cycle assessments that will be mandatory for new buildings under the EPBD from 2028 or 2030 onwards. The EPBD refers to EN 15978, Level(s) and national calculation tools and methods but does not make any specific recommendation about how to deal with uncertainties or data quality.

EN 15978: The data needed for CRCF calculations is a small part of the larger EN 15978 calculations. But all this data is considered within the same uncertainty matrix for building-level analyses. The EN 15978 approach looks at uncertainty as a function of the building project stage. Uncertainty is highest at the beginning, becoming more certain as the project nears completion, as reflected in the four stages or types listed below:

- Type 1: Assessment using an early building model in concept design.
- Type 2: Assessment using a building model in building permit design.
- Type 3: Assessment using the "as designed" building model.
- Type 4: Assessment using a fully detailed "as designed" or "as built" building model.

To minimise uncertainties at building level, it is recommended that the issuance of any certified units cannot occur until the building project is completed (i.e. the "as-built" stage has been reached) and the quantities are already final. Even if quantities are accurately estimated in the detailed design stage, the contractor still needs to procure the materials, and the product(s) supplied can change right up until the point of delivery to site. Even if final products are identical in their function and properties, their GHG-associated will be different between different suppliers and their compliance with sustainability DNSH must be confirmed.

Nevertheless, these issues should not prevent a preliminary process involving the preparation of an activity plan, of a monitoring plan and the submission of an initial estimate of the eligible carbon units stored from taking place before the project is completed.

2.7.5 Uncertainties at baseline level

This is the part of the quantification equation where the largest uncertainties are likely to exist. The same principles listed in the tables above for product-level and building-level data quality assessment could also apply to baselines. Baselines will be more accurate and relevant when:

- They are based on similar building typologies, in terms of size and use (analogous to technological representativeness).
- They are based on buildings in the same region (analogous to geographical representativeness).
- They are based on recently constructed buildings (analogous to time representativeness).

Uncertainty in baseline data will decrease as data from buildings of similar types, in comparable regions, and constructed more recently are used and averaged. Larger datasets from more buildings with similar characteristics will typically improve confidence in the average baseline value. Currently, there are no proposals for managing uncertainty in baselines. Nevertheless, any future approach could employ a similar matrix structure as defined for Level(s), but using the three bullet points above instead.

2.7.6 Different options for dealing with uncertainty in the certification methodology

Aspect	Remarks about 2.7 on uncertainties
Issue	The CRCF Regulation, via Article 4(12), requires a conservative approach to uncertainties and requires these uncertainties to be quantified where suitable. Uncertainty exists at various levels in the activity of biogenic carbon storage in buildings. - **At product level**: Qualitative approach to assessing LCA dataset quality and completeness of modelled processes. Can also look at uncertainty in material properties (e.g. moisture content, density, carbon fraction) in a quantitative manner. - **At building level**: Accuracy of material quantity declarations. Accuracy of product EPDs for materials actually used. Quality assessment of those EPDs. - **At baseline level**: Representativeness and precision of baseline figures against which net carbon removal is calculated.

<table>
<tr><th>Aspect</th><th>Remarks about 2.7 on uncertainties</th></tr>
<tr><td>Objective</td><td>To establish a consistent, pragmatic approach for handling uncertainties that is conservative, but without excessive administrative or calculation burden.</td></tr>
<tr><td>Existing, proven certification schemes</td><td><u>Rainbow</u>: A discount factor of at least 6% is applied on all projects, as it considers uncertainties associated with the baseline scenario selection and the project EPD data (e.g. biogenic carbon content, emissions, and lifetime assumptions).
<u>Climate Cleanup Foundation certification protocol</u> (v1.0): Section 2.1.1 states that “Operators shall highlight any cases of uncertainty regarding the quantification of any of the elements of Formula (1) and shall make justifiably conservative estimates when exact quantities are not available. For example, when a range of values is available for the quantification of an element or sub-element of Formula (1), operators shall choose a value on the lower end of that range”.
<u>Label Bas Carbone</u>: A 10% discount is applied to account for the risk of non-permanence, due to factors such as flooding, fire, termite damage (region-dependent), and potential early deconstruction.</td></tr>
<tr><td>Options</td><td>Until approaches to baselines are clearly defined and datasets or reference scenarios are established, uncertainty in CR-baseline will remain high. Therefore, a specific approach to baseline uncertainty is not yet recommended. At building level, uncertainty should be reduced by requiring reporting of as-built data before the final calculation of certified units is completed. However, at product level, there are three main options, which are outlined below.

<table>
<tr><th>Option</th><th>Pros</th><th>Cons</th></tr>
<tr><td>Only require a minimum quality of data at product level (e.g. fair or better).</td><td>Aligns with current EPD practice (EN 15804).
No additional calculation or administrative burden.</td><td>Limited incentive for better quality data.
Might not be sufficient to comply with Article 4(12).</td></tr>
<tr><td>Convert product data quality ratings into scores (DQI) in line with Level(s) and use to compensate against a default uncertainty penalty on GHG-associated.</td><td>Works with EN 15804/PEF.
Synergies with Level(s).
Quantitative aspect is more in line with Article 4(12).</td><td>DQI scores probably not available on existing EPDs.
Requires additional calculation and administrative effort.
To make this practice standard, would need to amend product category rules via the CPR.</td></tr>
<tr><td>Require a conservative approach to product physical properties when estimating CR-total at product level.</td><td>Incentivises the collection and use of more accurate data.
Uncertainties are based on real variables, not arbitrary data quality ratings.</td><td>Requires a complete recalculation of EPD data.
Rules and default approaches would need to be codified in harmonized product category rules under the CPR.</td></tr>
</table></td></tr>
<tr><td>Technical conclusions</td><td>A pragmatic approach to uncertainty analysis at the product level is recommended to reduce administrative burdens on construction product manufacturers and building designers. Building-level analysis should be based on product-level data as much as possible.
The simplest method is the qualitative approach of the first option in the table above. However, if a quantitative approach to uncertainty is necessary, the second option in the table above is simpler than the third. Nevertheless, the second or third options would eventually need to be included in product category rules for relevant construction products so that this type of uncertainty assessment becomes a standard part of EPDs or DoPCs.</td></tr>
<tr><td>Next steps</td><td>To discuss details of the Level(s) DQI formula with relevant experts to make sure that the equation is correct. To discuss with Commission representatives about the continued relevance of the PEF quality rating system and if this will be used in future harmonised standards under the CPR. Also a potential discussion point in the focus group with certification scheme representatives.</td></tr>
</table>

3 Additionality

3.1 Introduction to the basic elements of additionality

The concept of additionality in the CRCF Regulation is mentioned in recitals 9, 11, and 14 and elaborated on in recitals 20 and 21 of the same Regulation. Additionality essentially relates to activities that deliver carbon removals (or soil emission reductions) beyond usual practice. What constitutes standard practice should be reflected in the development of standardised baselines (see Article 5(2)). If standardised baselines are not available or are unsuitable for a specific project, provisions are also made for activity-specific baselines in Article 5(1). The full text of Article 5 is reproduced below.

"*1. Any activity shall be additional. To that end, it shall meet both of the following criteria:*

(a) it goes beyond Union and national statutory requirements at the level of an individual operator;

(b) the incentive effect of the certification under this Regulation is needed for the activity to become financially viable.

2. Where a standardised baseline is used, additionality as referred to in paragraph 1 shall be considered to be complied with.

Where an activity-specific baseline is used, additionality as referred to in paragraph 1 of this Article shall be demonstrated through specific additionality tests in accordance with the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8."

These are the two options for additionality: (i) relative to an activity-specific baseline; and (ii) relative to a standardised baseline. As already mentioned in section 2.6.3, several certification schemes currently use activity-specific baselines. This situation is expected to evolve over time as illustrated below.

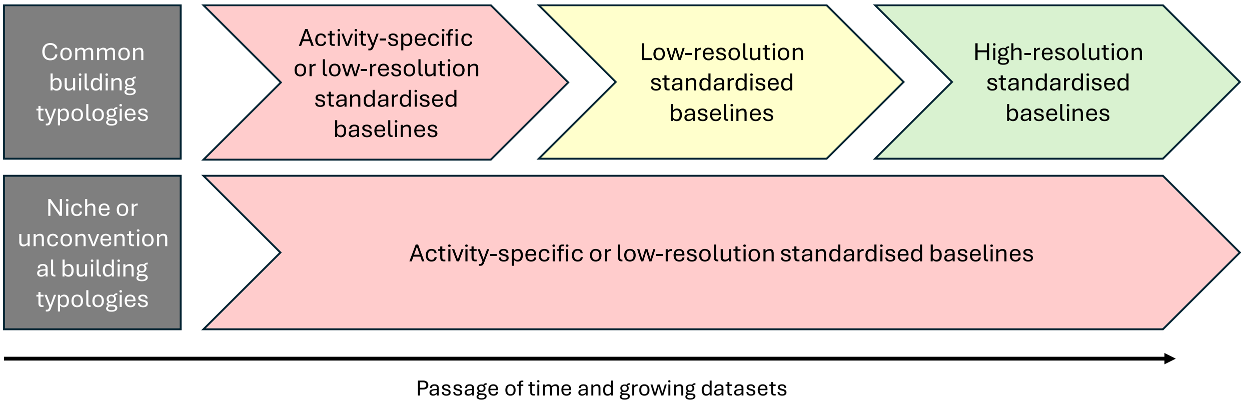


Figure 7. Expected evolution in baselines for the activity of carbon storage in buildings.

Both types of baselines are relevant today. While activity-specific baselines may become less common, they could still be valuable in the future for niche or unconventional building projects. Therefore, the following two sections explore how both types of baselines might be applied to the activity of carbon storage in buildings.

3.2 Additionality against an activity-specific baseline

The process of creating a standardised baseline requires assumptions about the broader market (for biobased insulation) and/or data from a significant number of similar building types (for structural wood). An activity-specific baseline should centre on a suitable reference scenario that can be directly compared to the individual building being assessed. Various possible approaches to building a reference scenario are discussed in this section.

However, before any activity-specific baseline can be applied, the two prerequisites stated in Article 5(1a) and 5(1b) of the CRCF Regulation must be met.

Regarding Article 5(1a), the authors are not aware of any Union or national statutory requirements that would require any individual building to be constructed with wood-based structural elements and/or bio-based insulation materials. Compliance with Article 5(1a) should therefore not pose an issue.

Regarding Article 5(1b), the argument about financial viability can only be considered practical if the scope of the cost comparison is limited to choosing structural wood over other structural materials or biobased insulation over conventional insulation. Some reasons for this include:

- As indicated in the International Cost Management Standard (ICMS-3)[12], the costs involved in a building project are highly complex.
- Material costs are just one of many expenses. Structural wood and biobased insulation are only two of the many materials used in buildings and would fit within ICMS-3 codes 2-03 and 2-04 (see footnote).
- At the building level, demonstrating financial viability, or the lack thereof, may require the declaration of certain profit margins or may shed light on cost data from suppliers that constructors prefer to keep confidential.

Another point to consider is the need to better understand what exactly it means in Article 5(1b) when it states: "*the incentive effect of the certification under this Regulation is needed for the activity to become financially viable*"? Should this imply that the revenue from certified units should at least offset the extra cost of wood or biobased materials? Or does it only need to contribute towards closing the gap to some specified or unspecified extent? The financial value of certified CRCF units (i.e., €/t CO2eq.) remains unclear and could vary over time, influencing whether or not Article 5(1b) is complied with.

Based on these arguments, the financial viability condition is deemed inappropriate for the activity of constructing or renovating buildings because the activity only applies to some of the many materials used in a project and that material costs overall are just one of many different costs in a project

Table 12. Article 5-related requirements for activity-specific baselines in relevant certification schemes

Certification scheme's approach to additionality	Remarks
Climate Cleanup Foundation certification protocol **(v1.0)**: In section 2.2, it states that carbon removals or carbon storage "**directly**" subsidised or required by law are not counted. By "directly", it means that the subsidy: (i) it is measured in terms of tCO2eq., and (ii) the only way to meet subsidy demands is to use biobased products.	Sounds like a sensible approach. The second condition provides greater legal certainty about which subsidy schemes can be disregarded.
Rainbow: The standard rules of Rainbow, under the additionality clause, state: "The mitigation activity would not have occurred without the revenues	A very comprehensive approach.

[12] The main upfront cost categories for new construction project in ICMS are: 1-01: Site acquisition; 1-02: Administration, finance, legal and marketing expenses; 2-01: Demolition, site preparation and formation; 2-02: Substructure; **2-03: Structure; 2-04: Architectural works / non-structural works;** 2-05: Services and equipment; 2-06: Surface and underground drainage; 2-07: External and ancillary works; 2-08: Preliminaries / Constructor´s site overheads / general requirements; 2-09: Risk allowances; 2-10: Taxes and levies; 2-11: Works and utilities off-site; 2-12: Production and loose furniture, fittings and equipment; 2-13: Construction-related consultants and supervision.

Certification scheme's approach to additionality	Remarks
from carbon finance". A 9-page template is then provided with targeted questions related to a regulatory analysis, an investment analysis and a barrier analysis.	Need to demonstrate that no other subsidies are being used to overcome the barriers identified. It's unclear what actions should be taken if existing subsidies only partially assist. Questions and information requested about the investment may require the sharing of commercially sensitive information.
Label Bas Carbone: Sworn statement by the project owner declaring that, to their knowledge, no local regulations such as the Territorial Climate-Air-Energy Plan (PCAET) or the Local Housing Program (PLH) exist related to the project's location that include mandatory measures and funding concerning carbon storage. It is generally recognised that using biobased materials is more expensive and more demanding than other more commonly used methods of building.	A declaration seems like a reasonable approach. It is also understandable that construction with wood is considered as more expensive, even if specific numbers are not provided.
C-sink standard (v0.91): Does not appear to use a baseline approach. Only C-sink materials can be used in C-sink constructions and section 5.2 of the standard on additionality, the risk of activity shifts relating to raw material production or fuel combustion or sub-optimal waste management need to be considered and be offset by geological C-sinks.	A very different approach to dealing with additionality. Does not mention any conditions that relate to article 5(1) of the CRCF regulation.

Open questions for stakeholders:
Q6. Are you aware of any national or Union statutory requirements that could oblige certain buildings to be constructed with structural wood or biobased insulation?
Q7. Are you aware of any national or regional subsidy schemes that "directly" support construction with structural wood or biobased insulation materials? ("directly" means that the only way to get the subsidy would be to use these materials in a building).
Q8. How should the statutory requirements be dealt with in the case that a biogenic carbon building is requested as part of mandatory requirements in a public procurement exercise? And what if biobased materials are not mandated but are encouraged via award criteria?
Q9a. Can you provide any reports, market data, rules of thumb or even anecdotal evidence of a **wooden structure being more expensive than a steel or concrete structure** for the same type of building? (ideally inputs here can be normalised to (EUR/m2 floor area).
Q9b. Can you provide any reports, market data, rules of thumb or even anecdotal evidence of **biobased insulation being more expensive than conventional insulation** for the same thermal conductivity and/or acoustic performance?

3.2.1 Level of resolution for structural materials

In a building project, the activity-specific baseline should be an alternative building project that uses conventional materials. This means the materials typically employed in the same particular region and for the same specific building typology as the actual building being assessed for certification.

The main steps to follow are:

- Categorise the building project subject to the certification activity (e.g. by region, floor area and building typology).
- Calculate the CR-total for the project.
- Define the corresponding reference building or reference building elements.
- Calculate the CR-total for the reference scenario.

- Subtract the CR-total (reference) from the CR-total (project).
- Subtract the corresponding GHG-associated for the project to get the net CR-total (project).

The third step in the list above requires further consideration. Expert judgement is necessary to identify which materials are typically used in the structure of the reference building for that region and type of building. The structure could also be a hybrid, with wood in the upper floors or roof, but concrete used in the lower floors, for example. The baseline reference scenario can also be viewed at different levels of resolution, as illustrated in the diagram below.

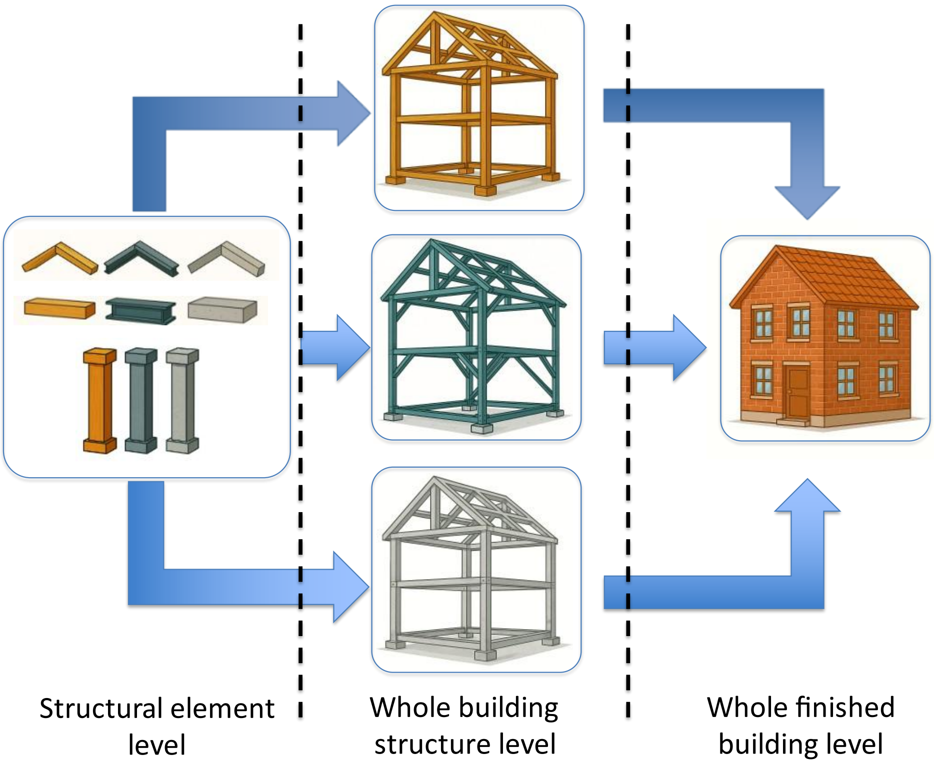


Figure 8. Different levels of resolution for activity-specific reference scenarios for structural wood (the same principle applies for biobased insulation).

Each of the levels has different advantages and disadvantages when it comes to defining a reference scenario for an activity-specific baseline, as summarised in the table below. The optimal balance is found in the middle approach, which considers the whole building structure level.

Table 13. Advantages and disadvantages of different levels of resolution for activity-specific baselines

Resolution level	Advantages	Disadvantages
Structural element	• Simplest cost comparison.	• Cannot determine if the comparison is fully appropriate without more information about the rest of the structure. • Does not allow for installation cost comparison.
Whole structure	• Accounts for the potential interactions between various structural elements that may influence the quantities required. • Structural engineers could apply some basic rules of thumb to create a useful baseline. • Avoids extra research needed to look at non-structural materials.	• Requires more research than just the comparison of individual elements.

Resolution level	Advantages	Disadvantages
	• The cost of installation can be compared as part of Article 5(1b) checks if deemed relevant.	
Finished building, including foundations (sub-structure).	• A heavier structure may require larger foundations and vice versa for a lighter structure. Therefore, this approach represents a more complete comparison.	• Requires a lot more research to construct a baseline (i.e. also non-structural materials). • Not cost-effective to tailor all these added details for each building project. • No added value looking at these extra materials either, because they will generally be common to both the specific and the reference project.

3.2.2 Comparative technical performance

In a building project, the activity-specific baseline should be an alternative building project that uses the conventional materials typically employed for the building structure and insulation and delivers a similar technical performance.

For example, consider a typical building usually constructed with a steel frame, but instead built with a wooden or even a concrete frame. If the same load-bearing capacity and floor area requirements are maintained, then a direct 1:1 substitution of steel for wood or concrete cannot be achieved. The materials have different mechanical properties and densities, which affect any comparison based on volume or mass of structural material.

Table 14. Comparison of physical properties of steel, wood or concrete, and approximate quantities of materials needed for a structure in a hypothetical three-story building of 1000m2 floor area.

Physical property	Steel	Wood (e.g. Glulam or CLT)	Concrete (reinforced)
Density	7850 kg/m3	400 to 600 kg/m3	2400 kg/m3 (incl. 100 kg/m3 steel rebar)
Compressive strength	250 to 400 MPa	30 to 60 MPa	25 to 60 MPa
Elastic modulus (E)	200 GPa	10 to 12 GPa	20 to 30 GPa
Strength to weight ratio	High	Medium	Low
Fire resistance	Needs protection	Chars	Excellent
Parameter	**Steel structure**	**Wood structure**	**Concrete structure**
Volume of structural material	20 to 30 m3	60 to 90 m3	120 to 150 m3
Weight of structural material	160 to 235 tonnes	30 to 55 tonnes	290 to 360 tonnes
Cross-section size (beams and columns)	Small (e.g. 200mm I-beams)	Large (e.g. 400 to 600mm)	Medium (e.g. 300 to 500mm)
Depths of floors/slabs	250 to 350mm (with steel deck)	250 to 400mm (with CLT floors)	300 to 500mm (with reinforced concrete slabs)
Volume ratio (versus steel)	1.0	Approx. 2.0 to 4.0	Approx. 5.0 to 6.0
Weight ratio (versus steel)	1.0	Approx. 0.25	Approx. 1.5
Formwork and curing needed	No	No	Yes (adds time and labour costs)
Biogenic carbon storage	Zero	Approx. 750 kg/m3	Zero

Comparisons with a purely steel or purely concrete structure seem unnecessary in the sense that they will always show a stored biogenic content of zero. However, some reference structures might be hybrid. For instance, a wooden structure on the upper floor(s) and roof could be used to create a lighter section of the building where lower load-bearing capacity is needed, while the lower elements are made of reinforced concrete, for example.

Most importantly, the quantities of structural materials in the baseline will need to be known in any case in order to do a cost comparison as per the requirements of Article 5(1b).

3.2.3 GHG-associated

The CRCF certification focuses on storing biogenic carbon in durable construction products. If the GHG emissions from the reference scenario are included in the activity-specific baseline calculation, it would add a source of variation. This could greatly change the number of units that qualify, depending on how GHG-associated is dealt with.

Article 4(5) uses "CR-baseline", where “CR" means Carbon Removal. According to Article 4(6), CR must be negative for net GHG removals and positive for net GHG emissions. The CR-baseline, due to its own GHG-associated, could be positive. Examples of how treating GHG-associated in different ways in the reference scenario can affect the overall NCRB (Net Carbon Removal Benefit) are provided in the table below.

Table 15. Examples of how different approaches to dealing with GHG-associated in activity-specific baselines affect certified units in two hypothetical building structures (1 and 2) each with two different baselines (a and b).

	Real project		Activity-specific reference scenario		Overall result (Net Carbon Removal Benefit)		
	CR-total	GHG-associated	CR-baseline	GHG-associated	NCRB (*)	NCRB (**)	NCRB (***)
Build-1a	-150 t	+25 t	0 t	+200 t	+125 t	+125 t	+325 t
Build-1b	-150 t	+25 t	-40 t	+150 t	+85 t	+125 t	+235 t
Build-2a	-50 t	+20 t	0 t	+35 t	+30 t	+30 t	+65 t
Build-2b	-50 t	+20 t	-10 t	+25 t	+20 t	+30 t	+45 t

*Assumes ignoring GHG-associated in the reference scenario. **Accounts for GHG-associated in the reference scenario, but not beyond the point of CR-baseline going above zero. ***Fully accounts for GHG-associated, even if it means CR-baseline goes positive.

In the table above, CR-baseline refers to the total eligible biogenic carbon stored in the reference scenario. It can either be zero or, in cases where some eligible wood or biobased insulations are assumed, it will be negative. The GHG-associated in the baseline scenario would be the module A1-A5 GWP of the conventional materials used in that scenario. These will always be positive.

To illustrate the different ways in which baseline GHG-associated values can be used, the complete quantification equations for the three options for building scenario 2b are laid out below:

General equation from Article 4(5): $(CR_{baseline} - CR_{total}) - GHG_{associated} = Temporary\ NCRB$
Building-2b* (ignore baseline GHG-associated): $((-10t + 0t) - -50t) - 20t = Temporary\ NCRB = +\mathbf{20t}$
Building-2b** (count baseline GHG up to a point): $((-10t + 10t) - -50t) - 20t = Temporary\ NCRB = +\mathbf{30t}$
Building-2b*** (count all baseline GHG): $((-10t + 25t) - -50t) - 20t = Temporary\ NCRB = +\mathbf{45t}$

Although the numbers chosen are arbitrary, the effects on the final results can be highly significant (e.g. going from 20t to 30t is +50%, and going from 20t to 45t is +125%).

Aspect	Remarks about 3.3 on activity-specific baseline GHG
Issue	The proposed approach to standardized baselines compares stored biogenic carbon with stored biogenic carbon. However, with activity-specific baselines, because it is necessary to compare costs, the whole building structure should be compared against a reference structure, and the full insulation system against a reference insulation system. The main issue is about how to deal with GHG-associated in the reference baseline scenario (i.e. CR-baseline). Depending on how it is dealt with, the final results can be very different.

Aspect	Remarks about 3.3 on activity-specific baseline GHG
Objective	To recommend what to do with GHG-associated in activity-specific baseline scenarios.
Existing, proven certification methodologies	The approaches to baselines include the following: • **Rainbow**: For removal credits, a series of reference buildings act as activity-specific baselines to provide references that are compatible with the floor area, region and use of a specific project. Comparisons with reference materials must be adjusted to account for any differences in primary function performance and lifetime. Any poorer performance of biobased materials in secondary performance is addressed in the risk evaluation section. GHG-associated is included in the baseline. • **C-sink standard (v0.91)**: Does not appear to use a baseline approach. A baseline of effectively zero applies when the fossil component of GHG-associated with the bio-based materials is offset by the purchase of permanent removal units. • **Climate Cleanup Foundation certification protocol (v1.0)**: Section 2.1.2 states that any activity-specific baseline reference scenario shall be found elsewhere or be developed and be subject to peer-review and verification. The calculations seem to focus purely on the stored carbon in the reference scenario and not any GHG-associated there. • **Verra VCS**: Still under development, the draft methodology seems to use activity-specific baseline. One of these baselines (the “emission reduction” baseline) will probably account for the GHG-associated in the reference scenario. With the other baseline (the “carbon removal” baseline) it is not so clear if GHG-associated is counted in the baseline.
Options	In the table below, the three options that exist for dealing with GHG-associated in reference scenarios from activity-specific baselines are considered.

	Pros	Cons
Ignore the GHG-associated in the reference scenario and focus only on stored carbon	• Simple approach. • Focuses only on stored carbon. • No variability introduced by changing non-biobased reference material assumptions or associated EPDs.	• Leads to the lowest result for certifiable units in a project. • May seem like an incomplete comparison, because GHG-associated is later subtracted for the biobased material, so why not also for the reference material.
Count the GHG-associated in the reference scenario, but only to the point that it might cancel out any stored biogenic carbon in that scenario.	• Real effect will be that most activity-specific baselines are zero. • Less drastic than the option below.	• More complicated than ignoring reference GHG-associated. • Some reality-check needed on assumptions with reference GHG-associated. • Makes standardised baselines look worse.
Count all of the GHG-associated in the eligible materials in the reference scenario, even if it makes the baseline scenario a net emitter.	• Leads to the highest result for certifiable units in a project.	• Most complicated approach. • Mixes up theoretical avoided carbon emissions with actual carbon storage certification. • Final NCRB result could be dominated by the reference scenario more than the actual carbon storage in the project. • More scrutiny needed on reference scenario assumptions. • Makes standardised baselines look much worse.

Aspect	Remarks about 3.3 on activity-specific baseline GHG
Technical conclusions	For the sake of simplicity, the first option to disregard the GHG emissions associated with the structural or insulation materials used in a reference building could be supported. The second option is more balanced, to some extent accounted for GHG-associated in the baseline, but not to the point that they end up being direct sources of storage certificates. This approach is more complex because rules and assumptions would be needed for how to choose the source data for reference GHG-associated. Furthermore, the second option would create a disincentive for developing standardised baselines unless the same logic is applied to those baselines. The third option is extreme and could lead to an undesirable situation where the same building qualifies for different quantities of certified units depending on the assumptions behind GHG-associated of the reference materials. This might prompt competition among certification schemes to define the most pessimistic reference scenarios (i.e. with higher GHG-associated).
Next steps	To discuss as part of a focus group composed of experts involved with certification schemes for carbon storage in buildings.

4 Storage, monitoring and Liability

4.1 Introduction to the basic elements of storage, monitoring and liability

The basic elements of this concept are outlined in Article 6 of the CRCF Regulation, which is quoted in full below. Sections that directly relate to the activity of temporary storage of biogenic carbon in long-lasting building elements are quoted below, and key legal terms are also in bold and underlined font.

*"1. An **operator or group of operators** shall demonstrate that an **activity** […] is aimed at storing carbon over the long-term.*

*2. For the purposes of paragraph 1, an **operator or group of operators** shall be:*
*(a) subject to **monitoring rules** and rules on the mitigation of any identified risks of **reversal** occurring during the **monitoring period**;*
*(b) liable to address any reversal of the carbon captured and stored by an activity which occurs during the **monitoring period** for that activity through appropriate **liability mechanisms** in accordance with the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8.*

3. The monitoring rules referred to in paragraph 2, point (a), shall:
(a) […] ;
(b) […] ;
(c) for […] carbon storage in products, be set out and duly justified in accordance with the rules laid down in the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8.

*4. The **liability mechanisms** referred to in paragraph 2, point (b), shall:*
(a) […] ;
(b) […] ;
(c) for […] carbon storage in products, be set out and duly justified in the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8 and may include collective buffers or up-front insurance mechanisms.

*5. The carbon removed and subsequently stored by a carbon removal activity shall be considered released into the atmosphere at the end of the monitoring period, unless that **monitoring period** is prolonged through a new certification of the activity or the carbon is stored permanently pursuant to paragraph 3, points (a) and (b), and paragraph 4, points (a) and (b).*

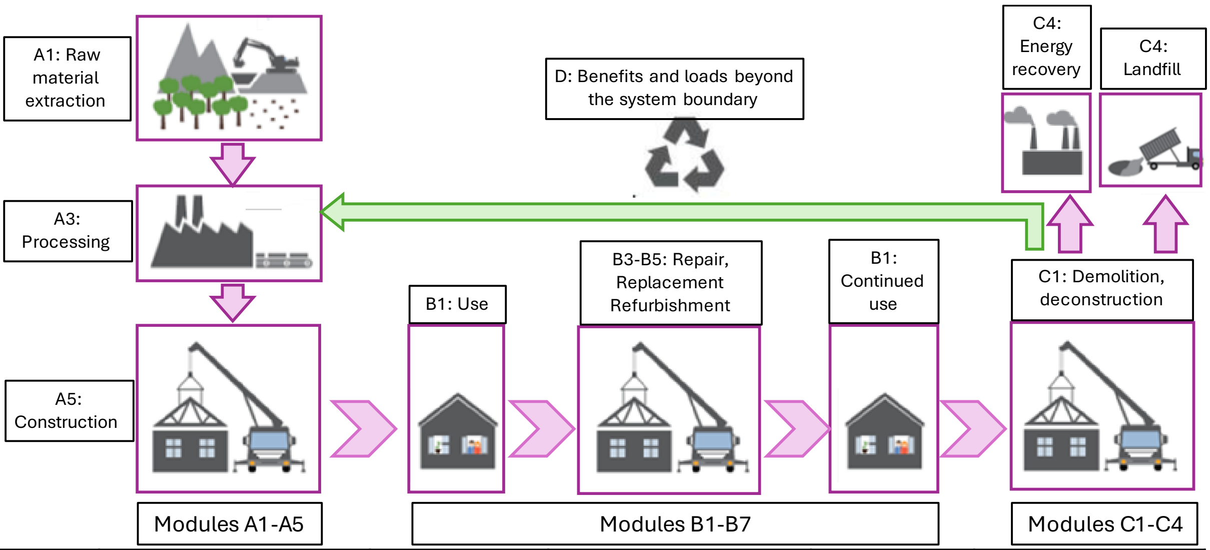
6. […] ."

Each of the terms highlighted and underlined will be examined in this chapter of the TAP. They are considered within the context of other definitions already provided in the CRCF Regulation, which are reproduced in the table below.

4.1.1 Who would the operators be?

The definitions of the terms "operator" and "group of operators" provide clear criteria for determining who the operator is. In the activity of temporary biogenic carbon storage in building elements, it is necessary to establish who effectively "controls the activity" and/or who has "decisive economic power over the technical functioning of the activity".

The operator is usually the building owner. However, because of the variety of real estate activities, this role can differ between different building projects. Additionally, the role may change within the same project over time due to the long lifespans of buildings.



Local authority	Planning permission, works licenses, safety and technical compliance checks.		Permission for major renovations.		Permission for deconstruction/ demolition.
Suppliers	Provide information on their products (i.e. EPDs).		Provide information on their products (i.e. EPDs).		May provide information on how to disassemble.
Contractor	Buys materials and installs them on site, can gather real A4 and A5 data.		Buys materials and installs them on site.		Conducts deconstruction, responsible for any material recovery.
Architect	Designs building, structural options and material specifications.		Give options for renovation.		May advise on recovery of higher value elements.
Building owner	Tells architect what they want in general, in public projects, direct dialogue with contractor.	Decisions to sell or rent or occupy or renovate.	Direct contact with architect/ contractor about what they want.	Decisions to sell or rent or occupy or demolish.	Decision to deconstruct or demolish, may specify certain methods to use.

Figure 9. Example of main roles and relevant actions for decisions over a building life cycle[13] that relate to CRCF certification and liability.

In all building projects and at every stage of the lifecycle, the building owner is effectively the operator, or at least has the option to be the operator.

However, with aspects of **module A**, there are various ways in which building projects take shape. This can influence who the operator will be at the critical design stage. For example:

- In a project for a single building designed and constructed for one private owner (e.g. a detached house for a family or an office block for one company), the prospective owner can decide, with advice from their architect and lead engineers, about options for the building structure, the structural materials, and the level and type of insulation to be used.
- When buying a new apartment or house in a new residential development, along with dozens of other buyers, individual buyers and prospective owners typically have no say at all in the choice of building structure and materials, whether structural or insulation-related. Instead, the architect, supported by the construction company that has already completed and costed the designs, acts as the operator.
- For social housing projects, the public authority or management company can have direct input into the options for the building structure and any minimum requirements concerning wood-based and bio-based materials. Such requirements could be included in call for tender documents or become part of a negotiated procedure with construction firms.
- From a broader real estate perspective, where funds are directed toward constructing multiple properties across a wide geographical area, the role of the ”operator' is usually delegated to regional representatives.

[13] Images adapted from: “Introduction to LifeCycle of Buildings; Trafik-Og Byggestyrelsen, Danish Transport and Construction Agency, 2016”.

Unless the building owner has a clear idea about the use of wood-based and bio-based materials, the most influential figure is often the architect, who, understanding the client's basic needs and project constraints, can propose a variety of architectural solutions.

Sometimes, the architect is also an employee of the construction contractor, while at other times, they are separate entities. In either case, close cooperation is essential to ensure the feasibility of the designs within site, time, and budget constraints. Moreover, the contractor must secure the availability of skilled labour and equipment needed to incorporate wood-based materials and bio-based insulation into the building as specified. It is also the contractor's responsibility to procure appropriate construction materials from suppliers. Any deviations at this final stage, such as procuring wood that does not meet sustainability standards, can completely undermine the original intentions or the units certified under CRCF.

Finally, the local authority plays a vital role in granting permits for construction activities, establishing health, safety, and waste management rules for site operations, and verifying that the completed building complies with relevant building codes. If, at the planning stage, the local authority mandates the use of structural wood or bio-based insulation through compulsory requirements, this will influence how the additionality principle is applied within the CRCF methodology.

In the **module B** aspects, which relate to building renovation, the individual building owner maintains control over renovation decisions unless mandated by the local authority (for example, to improve the building's energy rating or address safety risks). However, even when insulation is compulsory by the local authority, the owner still selects the type of insulation material to use. Typically, the owner communicates with an architect or directly with the contractor. In renovation projects, there is usually more flexibility with timing, allowing sufficient time to check the availability and secure sourcing of appropriate bio-based insulation materials.

Similar to module B, the individual building owner is the operator in **module C** and thus the person with the authority to decide whether to demolish or deconstruct the building. However, in some cases, the local authority may prevent demolition or deconstruction by refusing permission, or may actually demand demolition if irreparable damage is deemed to have occurred, if the building is considered structurally unsound, or if a major public infrastructure project necessitates its removal. Ultimately, what happens to the wood-based and bio-based materials at the end of their life lies with the contractor, who may be working according to clear requests from the old building owner or the requirements of a new owner planning to demolish and rebuild. Regardless, the fate of the wood-based and bio-based materials at end-of-life, especially if it occurs before the 35-year period, will determine whether there is liability for carbon reversal or not. In such a case, if reuse can be proven, then carbon reversal should be assumed.

Open questions for stakeholders:

Q10a. If structural wood or bio-based materials are reused or recycled into other products at the End-of-Life, should liability for reversal be waived?

Q10b. If you think the answer to Q10 should be yes, what type of proof of reuse or recycling would be needed?

A key cross-cutting element of the 'operator' role is that building ownership can change at any point during a building's lifetime. Typically, ownership changes occur during the use phase of the building lifecycle, but they can also be the primary factor influencing end-of-life decisions for the building. Ownership can also fully or partially transfer during the planning and construction stages, particularly in cases involving speculative investments in real estate portfolios and investment funds focused on real estate activities.

By default, the CM should consider that the building owner is regarded as the ”operator”. Consequently, purchasing a certified building also entails acquiring any associated certified units and liabilities.

4.1.2 Activity periods and monitoring periods

The activity period indicates the time during which carbon is stored in a specific building. The monitoring period, unless otherwise stated in the CM, will last 35 years. Any shortfall of the activity period compared to the monitoring period will have liability consequences. Several different examples are provided below.

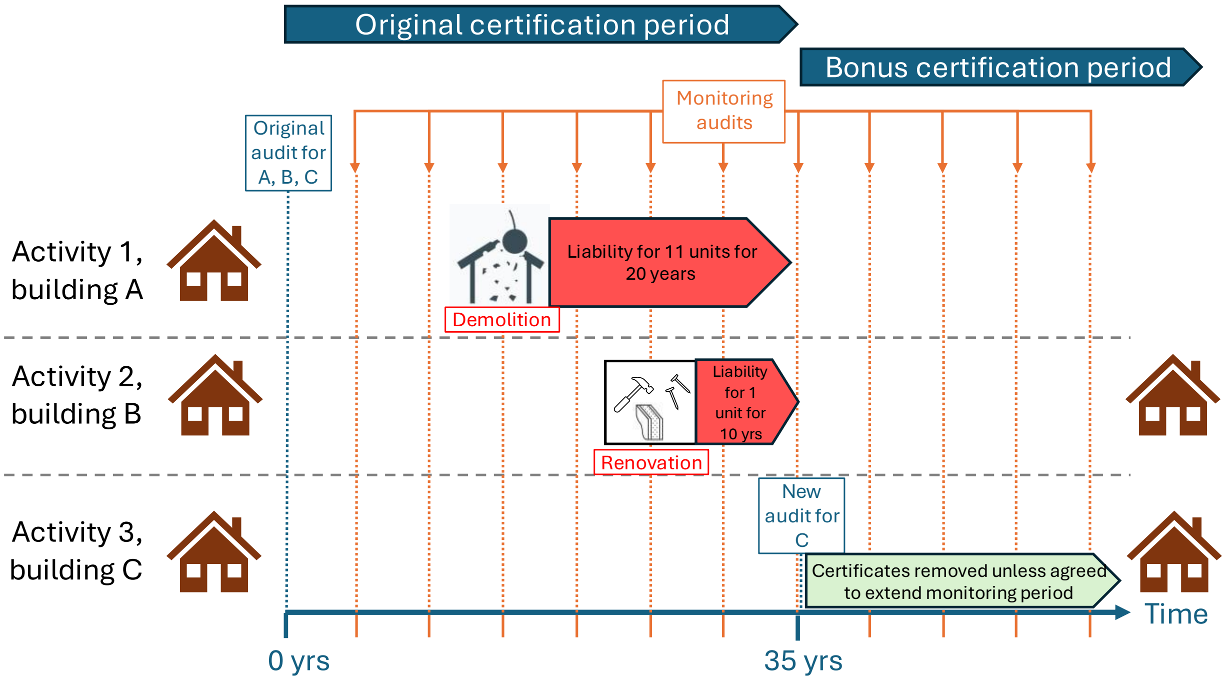


Figure 10. Three different examples of certified buildings and the consequences for certification and liability.

For simplicity, consider three identical buildings (A, B, and C) being evaluated for CRCF certifiable carbon storage. Since they are identical, they qualify for the same number of certified units (e.g. 10 units for structural timber and 1 unit for bio-based insulation). However, after their construction, the three buildings have different lifespans. Building A is demolished after 15 years due to irreparable damage caused by a local flood. Building B undergoes an energy renovation after 25 years, during which all the bio-based insulation is removed, although the structural wood remains. In building C, the same structural and insulation materials are used for well beyond 35 years (e.g. to 55+ years).

We propose that CRCF certificates for temporary carbon storage should signify a commitment to retain the net removed biogenic carbon for at least 35 years. Calculating liabilities is easiest when expressed in terms of t.year. Therefore, a building committed to storing 11t of biogenic carbon for 35 years has a total liability of 11 x 35 = **385 t.years**. The following would apply to buildings A, B, and C:

- Building A would be liable for 20 years of lost storage, which equates to (11t x 20yrs = 220 t.yrs). This lost storage **liability equates to around 57%** of the total commitment (i.e. 220/385 = 0.57). Therefore 57% of the original certified units would be liable (57% of 11t = 6.27t).
- Building B would be liable for 10 years of lost storage for a fraction (1/11) of the certified units (1t x 10yrs = 10 t.yrs). This lost storage **liability equates to around 2.6%** of the total commitment (i.e. 10/385 = 0.026). Therefore 2.6% of the original certified units would be liable (2.6% of 11t = 0.286t).
- Building C would have no liability, and the same certified units could potentially retained on the registry if it is agreed to extend the monitoring period.

In principle, the extension of certified units on the registry should also be possible for building B after year 35, but only for the 10 units that were originally part of the building and remain there at year 35. In the case that the new insulation material in Building B is biobased and qualifies for certification units in theory, it should not qualify for certified units precisely with Building B, because the reference scenario (i.e. of not doing the renovation in Building B) would mean keeping biobased insulation in the building – thus creating a very high activity-specific baseline.

4.1.3 Different roles and their relationships in the certification and monitoring process

Considering the definitions and general requirements in the CRCF Regulation, it is necessary to visualise how the different roles would function in the specific activity of storing net removed biogenic carbon in long-lasting construction products used in buildings. How some of the key actions and relationships could work are illustrated in the figure below.

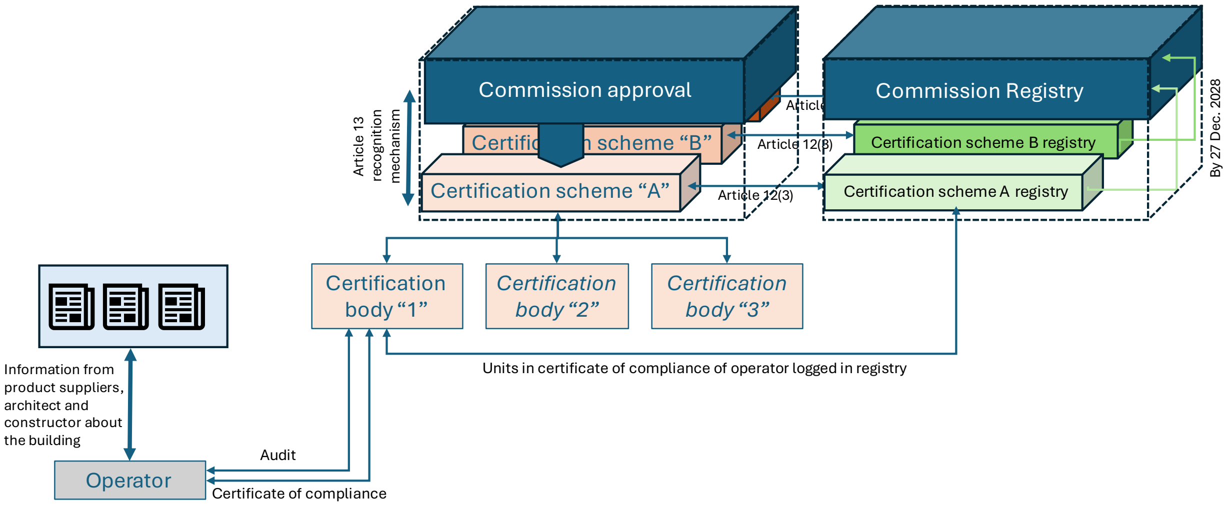


Figure 11. How the certification process could work for the activity of temporary carbon storage in buildings – example of a building being audited by certification body "1", under certification scheme "A".

At the core of the entire process is the European Commission, whose recognition is vital for any certification scheme to issue certificates of compliance with specified quantities of authenticated units for carbon storage. According to Article 2(23), 1 unit equals 1 metric tonne of net removed CO2eq. Certification schemes may be public or private and could potentially operate through their certification bodies within the same geographical area, although this would increase the risk of intentional or accidental double-counting of the same activity by different schemes.

Each certification scheme will keep its own register of issued units. However, according to Article 12(1) of the CRCF Regulation, these registers should eventually be linked to a single Union register by the end of 2028.

Within each approved certification scheme, there may be multiple certification bodies. These bodies carry out audits of operators and assess their information to decide if a certificate of compliance can be issued. They also determine how many units should be certified. It is recommended that units related to wood-based structures and bio-based insulation be reported separately on the certificate. Although these two types of units are of equal value and interchangeable, separate reporting will simplify future monitoring. It may also offer insights into the relative uptake of certification for these products and assist with future baseline estimates.

4.2 Liability mechanisms

Liability mechanisms are essential for any carbon certification scheme. They uphold the scheme’s credibility and ensure certificates are supported by trustworthy storage activities for net removed carbon. These mechanisms should be clearly outlined so operators understand the consequences of failing to meet storage requirements.

4.2.1 Collective buffers and upfront insurance

Article 6(4)(c) of the CRCF Regulation indicates the potential use of ”collective buffers” or ”upfront insurance” as liability mechanisms.

A collective buffer system is a risk management tool where a portion (e.g. 5%) of all credits issued is allocated to a buffer pool. This pool is then used to cover any subsequent real-life ”reversal” in a certified activity within the same certification scheme. In this way, the buffer pool functions as a guarantee, covering a specified amount of potential reversals that could occur in any certified activity under the same scheme.

Upfront insurance can operate in different ways. From the operator's point of view, insurance likely focuses on recovering financial losses or covering liabilities in the event of setbacks. Operator insurance would treat the stored carbon as an inherent part of the building's value. It takes into account fundamental risk factors that could lead to setbacks within the 35-year period, such as:

- Risk of irreversible damage from flooding, earthquakes, landslides, avalanches, wildfires, or storms: evaluate the building's location in relation to these physical and climate-related hazards.
- Risk of irreversible damage from internal fires: consider the nature of activities within the building, fire control systems in place, and the fireproofing of spaces, walls, and the structure.
- Risk of early demolition due to property development: assess the projected demand for higher density buildings in the same area over the next 35 years.

From the perspective of the certification scheme, insurance would primarily focus on covering actions and direct costs related to:

- Reversals that exceed those covered by any buffer pool,
- Situations where operators refuse to pay liabilities (such as declared bankruptcies, death, etc.), or
- The discovery of fraudulent claims related to issued certificates.

In an extreme scenario, insurance could cover the entire certification scheme if it were to fail, such as through bankruptcy or if its recognition is revoked by the Commission. Costs that may need to be covered by insurance could also include purchasing certificates from other schemes that oversee the same activity, to maintain investor confidence in the insured scheme.

4.2.2 What do established certification schemes do for liability mechanisms?

One particularly detailed approach is that of the **Rainbow** scheme. A discount factor between 0 and 10% is applied depending on the level of uncertainty in project data or the baseline scenario. This discounted carbon does not go to a buffer pool. The scheme also deducts a default 3% of carbon units to add to a buffer pool. This amount may increase to 6% depending on how the project compares against the scheme's risk assessment template.

Discussion with certification scheme representatives implied that the risk of reversals of carbon storage in buildings is very low for a minimum 35-year period if the scope of materials is limited to structural wood, and still low when including biobased insulation. A minimum buffer pool of 10% of the issued units was considered as an overly conservative approach to adopt in any official certification method under the CRCF Regulation for this activity.

The possibility of reducing the proportion of units sent to the buffer pool, depending on the risk of reversal in the project, could be sensible. However, this would increase the process's complexity and rely on a robust risk assessment framework. Such a framework could potentially be integrated into a future version of the CM once experience has been gained from auditing various projects. A simpler method might be to establish rules for buffer pool contributions based on how certified units are issued (see section 4.3) and whether the schemes claim only for the minimum 35 years of storage or for longer. The method of issuing the certified units significantly influences the risk of reversal. If certificates are issued only ex-post, then potentially no units need to be sent to a buffer pool.

4.2.3 What should the monitoring frequency be?

The Expert Group discussed how often to monitor carbon storage in long-lasting building products. Some believed annual checks were necessary. Others thought monitoring every five years was enough, the minimum frequency required by the CRCF Regulation (see Article 9(3)).

Buildings are stable and stationary, but each site can only hold a relatively limited quantity of certified units. The materials used, such as structural and insulation ones, tend to remain untouched. Therefore, five-year intervals seem appropriate. This differs from other activities under the CRCF Regulation, where quantities per operator and site can be much larger, but are also more dynamic, being able to fluctuate significantly over time.

It is also important to define what monitoring includes. According to Article 9(3), five-year intervals mean re-certification audits. Unlike other activities, the carbon stored in buildings is smaller. Buildings can be spread out geographically. Visiting every operator for audits is not practical or cost-effective.

Possible approaches include:

- Auditing a small, random sample of operators or projects that represent a minimum share of total issued certificates.
- Using digital media instead of onsite visits, like satellite images or video tours.
- Employing a digital building logbook to record renovations that could affect storage.
- Consulting local planning authorities for changes to buildings.

If mandatory digital pre-demolition audits are implemented in the future, then cross-checking these records should also be a part of the auditing process. Operators should also make periodic declarations on honour that the materials are still in place, or if there has been any change. A declaration should also be sent whenever the ownership of the building changes, so that the new operator can be contacted and registered with the certification scheme. Missing declarations could trigger a non-randomised remote audit and possibly a site visit.

4.2.4 Possible nuanced approach to minimum liability coverage

Certification schemes may use buffer pools, upfront insurance or other suitable mechanisms as safeguards against reversals. However, as part of the credibility of certified units via the CRCF Regulation, minimum shares of net carbon removal benefit should be covered by some form of liability mechanism.

For a given building project, the risk of reversal varies as a function of the claimed storage period and the form of certified unit issuance. These minimum shares to cover by buffer pools, upfront insurance etc. are initially proposed to vary as follows:

Table 16. Possible liability matrix for different minimum carbon storage claims and issuance arrangements

Form of certificate issuance	Minimum storage period claimed		
	35 years	50 years	100 years
Upfront (ex-ante) for the full period.	5%	7.5%	15%
Upfront (long-term, i.e. 10 to 20-year intervals)	3.5%	3.5%	3.5%
Upfront (mid-term, i.e. 5 to 10-year intervals)	2%	2%	2%
Upfront (short-term, i.e. 1 to 5-year intervals)	1%	1%	1%
Ex-post (any defined interval)	0%	0%	0%

Open questions for stakeholders:

Q11. Opinions about the liability matrix?

4.3 ”Upfront” versus ”termed” issuance of certified units and its effect on liability

Considering the information in sections 4.1 and 4.2, and the specific nature of the certification for temporary carbon storage, the manner in which certified units are issued can significantly influence the monitoring and liability aspects of the entire process. There are two main options regarding issuance:

- **Upfront**: where all the certified units are issued at the start of the activity, assuming that the certified carbon will still be stored there 35 years later.
- **Termed**: where only a fraction of the certified units are issued ex-ante at the start of the activity, and additional fractions are released only after recertification audits or other monitoring actions during the activity period. Termed issuance can also be ex-post, where only a fraction of certified units are issued after the same fraction of storage time has elapsed. In both cases, the proportion of further certified units released corresponds to the portion of the 35-year period that has elapsed.

The two graphs in the figure below illustrate how the issuance of certificates would look for the same three example buildings described in section 4.1.2. To clarify between lines, the construction of each of the three buildings has been offset by one year: building A in year 0, building B in year 1, and building C in year 2. The same events also apply, specifically:

- All 3 buildings contain 11 tonnes of certifiable carbon storage (10t as wood and 1t as bio-based insulation).
- Building A is demolished after 15 years.
- Building B has its bio-based insulation removed after 25 years.
- Building C retains all certified materials until year 35.

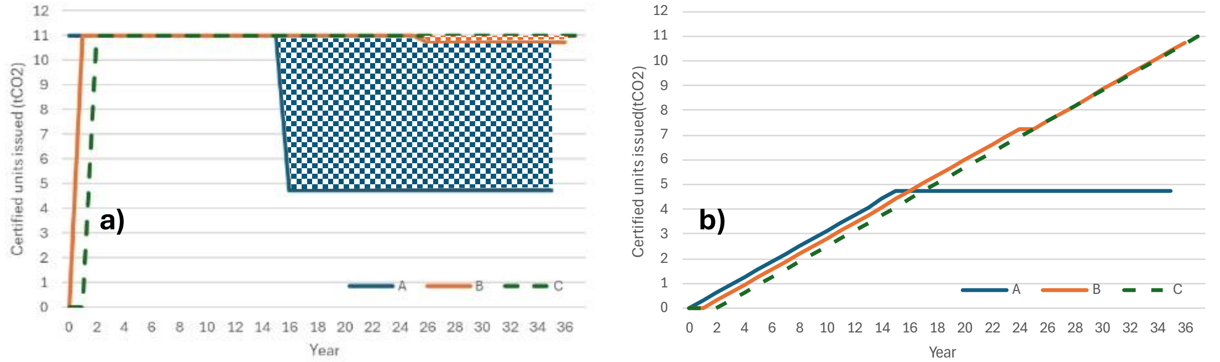


Figure 12. Comparison of unit issuing a) in an upfront manner, b) in an ex-post, annually-termed manner.

Examining the graphs above, if all the certificates are issued upfront (see graph a), then there are liability issues with building A (significant liability in blue shaded area) and with building B (minor liability issues in orange shaded area).

However, if certified units are issued gradually each year (see graph b), then the result is that certificates stop being issued for materials subject to carbon reversal – meaning there is ultimately no liability, only a lost income stream for the operator. Although less evident, building B in graph b) only reaches 10.71 tonnes of certified carbon after 35 years, not the 11 tonnes achieved by building C.

Intermediate approaches to ex-post and upfront issuance are also feasible, for example at 5-year intervals. In these cases, issuance would be directly tied to the recertification audit. Some liability might arise in the case of an upfront issuance, but only for a maximum of 5 years of certified storage.

Aspect	Remarks about 4.3 on upfront vs termed issuance of certified units
Issue	Committing to carbon storage for 35 years requires a long-term commitment that could potentially be broken, either intentionally or accidentally. The operators, usually the building owners, are responsible for any carbon reversal, but how their liability is managed depends on how the carbon storage certificates are initially issued. If all certificates are issued in advance (i.e. “upfront”), then the liability involves reimbursing the value of the share of certified units that have been “reversed”. If the certificates are issued gradually each year (i.e.

<table>
<tr><th>Aspect</th><th>Remarks about 4.3 on upfront vs termed issuance of certified units</th></tr>
<tr><td></td><td>“termed”), any reversal would lead to the cessation of certificate issuing from that point onward for the share of carbon that was “reversed”.</td></tr>
<tr><td>Objective</td><td>Compare the pros and cons of the “upfront” and “termed” approaches to the issuance of carbon storage certificates.</td></tr>
<tr><td>Existing, proven certification methodologies</td><td>The approaches to certification issuance in existing certification schemes are as follows:

<u>Rainbow</u>: Issues certificates ex-post, at the point of completion of the construction works for building-scale certification, or at the point of sale to a building developer for certification of material manufacturing.
C-sink standard (v0.91): Issues certificates upfront, at the point at which materials are incorporated into the construction.
<u>Climate Cleanup Foundation certification protocol</u> (v1.0): Appears to issue certificates upfront according to point 6 of the executive summary.
<u>Verra VCS</u>: Still under development, the draft methodology takes a termed approach to the issuance of certificates.
<u>Label Bas Carbone</u>: The certification process starts with a notification of the project owner to the administrative authority (DREAL) before the project starts. Files are submitted within 1 year and then certification takes place. An audit is conducted within two years of the building being completed and then certified units are registered and issued.
</td></tr>
<tr><td>Options</td><td>For issuing certificates at building-level:
<table>
<tr><th></th><th>Advantages</th><th>Disadvantages</th></tr>
<tr><td>Issue certificates upfront</td><td>Strong motivation to choose wood and bio-based materials.</td><td>Need to have more robust systems in place for liability.
Increases the size of the buffer pool needed.
Lack of reward for future building owners to comply with ongoing monitoring requirements.</td></tr>
<tr><td>Issue certificates on a 5-year basis ex-ante (termed issuance)</td><td>The system minimises the risk of payback liability.
Reduces the size of the buffer pool needed.
Passes on benefits to future building owners.</td><td>Reduced incentive to original building owners (i.e. only gradual benefits coming).</td></tr>
<tr><td>Issue certificates on an annual basis ex-post (the most conservative type of termed issuance)</td><td>The system completely avoids payback liability.
No buffer pool needed.
Benefits continue with future building owners.</td><td>Reduced incentive to original building owners (i.e. only gradual benefits coming).</td></tr>
</table></td></tr>
<tr><td>Technical conclusions</td><td>The termed approach is preferred because it inherently reduces or avoids liability and prevents situations where future building owners are burdened with monitoring requirements, but see no actual benefit if the previous owner sold all the certified carbon storage units beforehand.
An annual termed approach (either ex-post or ex-ante) is preferred because it helps the operator stay aware of the certification, especially in cases where building ownership might change multiple times.
However, it is also recognised that this approach will reduce the immediate incentive for decision makers and make the activity less attractive to investors because of its slow build-up of certified units.
Consequently, it is recommended to permit both approaches, so long as minimum liability coverage corresponds to the associated effects on reversal risks caused by the different modes of issuance.</td></tr>
</table>

Aspect	Remarks about 4.3 on upfront vs termed issuance of certified units
Next steps	To discuss with certification scheme representatives in a focus group about the different approaches to the issuance of certified units and the advantages of harmonisation with this approach.

5 Sustainability

5.1 Introduction to the basic elements of sustainability

The basic elements of this concept are laid out in Article 7 of the CRCF Regulation, which is quoted below with the most relevant parts for carbon storage in buildings underlined.

"*1. An activity shall do no significant harm to the environment and may generate co-benefits for one or more of the following sustainability objectives:*
(a) climate change mitigation beyond the net carbon removal benefit and net soil emission reduction benefit referred to in Article 4(1) and (2);
(b) climate change adaptation;
(c) the sustainable use and protection of water and marine resources;
(d) transition to a circular economy, including the efficient use of sustainably sourced bio-based materials;
(e) pollution prevention and control;
(f) protection and restoration of biodiversity and ecosystems, including soil health as well as avoidance of land degradation.

2. A carbon farming activity shall at least generate co-benefits for the sustainability objective referred to in paragraph 1, point (f).

3. For the purposes of paragraph 1 of this Article, an activity shall comply with the minimum sustainability requirements laid down in the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8. The minimum sustainability requirements shall:
(a) take into account the impact both within and outside the Union and local conditions;
(b) where appropriate, be consistent with the technical screening criteria for the 'do no significant harm' principle;
(c) promote the sustainability of forest and agriculture biomass raw material in accordance with the sustainability and greenhouse gas emissions saving criteria for biofuels, bioliquids and biomass fuels laid down in Article 29 of Directive (EU) 2018/2001.

4. Where an operator or group of operators reports co-benefits that contribute to the sustainability objectives referred to in paragraph 1 of this Article beyond the minimum sustainability requirements referred to in paragraph 3 of this Article, that operator or group of operators shall comply with the applicable certification methodologies set out in the delegated acts adopted pursuant to Article 8. Those certification methodologies shall include elements to incentivise as much as possible the generation of co-benefits going beyond the minimum sustainability requirements, in particular for the objective referred to in paragraph 1, point (f), of this Article."

The parts of Article 7(1) that are not highlighted and the entire Article 7(4) on co-benefits remain relevant, but they are not mandatory parts of the certification methodology. As discussed earlier in section 4.4, the extension of carbon storage could potentially be regarded as a type of time-dependent co-benefit.

Article 7(2) is considered less relevant because the activity in question does not qualify as carbon farming. Unlike farmers, the operators involved in construction activities do not have direct control over the generation of co-benefits for biodiversity and ecosystems linked to carbon farming activities.

Regarding parts (a) to (f) in Article 7(1), concerning do no significant harm (DNSH) criteria, the EU Taxonomy DNSH requirements for buildings are reproduced in Appendix VI, along with the accompanying technical screening criteria in Appendix VII.

5.2 Overview of EU Taxonomy

Both pieces of EU Taxonomy legislation cover activities related to the construction of new buildings and the renovation of existing ones, as shown in the table below.

Table 17. Overview of technical screening criteria (TSC) and DNSH criteria for buildings in the EU Taxonomy

Activity and legal reference	Environmental goal					
	Climate Change Mitigation	Climate Change Adaptation	Water resources	Circular economy	Pollution prevention & control	Biodiversity and ecosystems
New construction (Regulation 2021/2139)	TSC, DNSH	TSC, DNSH	DNSH	DNSH	DNSH	DNSH
Renovation (Regulation 2021/2139)	TSC, DNSH	TSC, DNSH	DNSH	DNSH	DNSH	-
New construction (Regulation 2023/2486)	DNSH	DNSH	DNSH	TSC	DNSH	DNSH
Renovation (Regulation 2023/2486)	DNSH	DNSH	DNSH	TSC	DNSH	-

The DNSH criteria in the EU Taxonomy provide a solid foundation for the sustainability aspects of this certification methodology. Using these criteria helps ensure consistency with existing regulations and facilitates harmonisation across sustainability standards. However, each requirement should be checked for its ease of implementation, taking into account both the context of the EU Omnibus Simplification Package and the practical needs for possible monitoring compliance over a 35-year period.

The full EU Taxonomy criteria are included only in Appendices V and VI because of their length. The DNSH and co-benefit proposals in the following sub-sections are based on the EU Taxonomy criteria. The text has been shortened and modified to make the requirements easier to understand for lay readers without prior experience of the EU Taxonomy. Remarks from the project team are also provided for each environmental goal to help contextualise the requirements and/or highlight benefits and issues. These proposals should be considered individually first, and then as a package.

5.2.1 Possible DNSH and requirements for the activity of carbon storage in buildings

Potential proposals for each of the CRCF DNSH requirements are presented in the table below, being generally based on the EU Taxonomy DNSH criteria.

Table 18. Possible DNSH criteria for the activity of new construction or renovation

Environmental goal	Proposed requirement	Remarks
a) Climate Change Mitigation (CCM)	(**for new construction or renovation**) 1. The building is not dedicated to the extraction, storage, transport or manufacture of fossil fuels. 2. The building will not generate onsite carbon emissions from fossil fuels during its normal use.	Point 1 is copy-paste from the EU Taxonomy. No reference to NZEB is made since this will be out of date eventually, being replaced by ZEB. Not yet clear what national definitions of ZEB will be, but one common element is referred to in point 2. Easy to verify.
b) Climate Change	(**for new construction only**) A screening exercise shall be conducted to assess the risk of climate hazards to the proposed building over a period of at least 35 years.	No requirements here for renovation.

Environmental goal	Proposed requirement	Remarks
Adaptation (CCA)	The building shall not be constructed on sites where risks from fluvial flooding, storm surges and coastal flood, coastal erosion, avalanches, landslides, subsidence, glacial lake, reservoir or lagoon outbursts are identified.	Considered useful to focus only on the climate hazards that could bring about a premature end-of-life of the building. Would need expertise from the insurance sectors for the risk assessment. How much risk is too much? Where to draw the line? In case there is no established method for the screening exercise or the CCVA, the assessment could follow the guidance set out in this Commission study.
c) Water resources (WAT)	(**for new construction or renovation, but only for non-residential buildings located in river basins that are under water stress or severe water stress**) Where installed, the specified water use for the following water appliances shall be attested by product datasheets, a building certification or an existing product label in the Union: (a) wash hand basin taps and kitchen taps have a maximum water flow of 6 litres/min; (b) showers have a maximum water flow of 8 litres/min; (c) WCs, including suites, bowls and flushing cisterns, have a full flush volume of a maximum of 6 litres and a maximum average flush volume of 3,5 litres; (d) urinals use a maximum of 2 litres/bowl/hour. Flushing urinals have a maximum full flush volume of 1 litre.	Can easily be determined if a building is located in a water-stressed river basin by using the Level(s) indicator 3.1 tool available here (specifically by selecting from two dropdown menus in cells B1 and B2 in the L2 estimate worksheet therein). However, in parts a) to d), some of these fittings could easily be changed to mislead auditors unless they intend to inspect the water appliances during audits.
d) Circular economy (CE)	(**for new construction only**) At least 70% (by weight) of the non-hazardous construction and demolition waste (excluding naturally occurring material referred to in category 17 05 04 in the European List of Waste established by Decision 2000/532/EC) generated on the construction site is prepared for reuse, recycling and other material recovery, including backfilling operations using waste to substitute other materials, in accordance with the waste hierarchy and the EU Construction and Demolition Waste Management Protocol[14].	A minimum of 70% of construction and demolition waste being prepared for reuse, recycled or recovered (incl. backfilling) is tantamount to complying with Article 11(2b) of the Waste Framework Directive. Doing it according to the EU CDW management protocol is to standardise how this is done across the EU.
e) Pollution prevention and control (PPC)	(**for new construction or renovation**) Bio-based insulation materials shall emit less than 0,06 mg of formaldehyde per m3 of material or component upon testing in accordance with the conditions specified in Annex XVII to Regulation (EC) No 1907/2006 and less than 0,001 mg of other categories 1A and 1B carcinogenic volatile organic compounds per m3 of test chamber air, upon testing in accordance with EN 16516 or ISO 16000-3:2011	Only applying the requirements on pollutant emissions directly to bio-based internal insulation materials, because the EU Taxonomy requirement applies only to materials that could come into contact with occupants.

[14] European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Oberender, A., Fruergaard Astrup, T., Frydkjær Witte, S., Camboni, M. et al., EU construction & demolition waste management protocol including guidelines for pre-demolition and pre-renovation audits of construction works – Updated edition 2024, Publications Office of the European Union, 2024, https://data.europa.eu/doi/10.2873/77980

Environmental goal	Proposed requirement	Remarks
	or other equivalent standardised test conditions and determination methods.	
f) Biodiversity and ecosystems (Bio-D)	(**for new construction only**) The new construction is not built on one of the following: (a) arable land and crop land with a moderate to high level of soil fertility and below ground biodiversity as referred to the EU LUCAS survey; (b) greenfield land of recognised high biodiversity value and land that serves as habitat of endangered species (flora and fauna) listed on the European Red List or the IUCN Red List; (c) land matching the definition of forest as set out in national law used in the national greenhouse gas inventory, or where not available, is in accordance with the FAO definition of forest.	The requirement for an Environmental Impact Assessment (EIA) or an EIA screening from the Taxonomy is not included here because it should be decided by local planning authorities if needed or not. If done only for CRCF DNSH, it would be a potentially significant added cost to smaller building projects.

Open questions for stakeholders:
Q12. What are your thoughts on each of the proposed mandatory DNSH criteria?

5.2.2 Possible optional co-benefit criteria for carbon storage in building activities

Proposals for each of the CRCF co-benefits are presented in the table below, being generally based on the EU Taxonomy DNSH and TSC criteria.

Table 19. Possible co-benefit criteria for the activity of new construction or renovation

Environmental goal	Proposed requirement	Remarks
a) Climate Change Mitigation (CCM)	(**for new construction only**) The energy performance is certified using an as-built Energy Performance Certificate (EPC) and is awarded an energy class A according to the national methodology. (**for renovation**) The renovation building meets the national definition of a cost-optimal energy renovation or has reduced its primary energy demand by at least 30% compared to before renovation.	Not the same as the Taxonomy TSC for new construction, which only makes reference to NZEB, which will soon be replaced by ZEB in the EPBD. The 30% reduction in primary energy demand might require two assessments, one immediately prior to renovation and one after.
b) Climate Change Adaptation (CCA)	(**for new construction only**) A screening exercise shall be conducted to assess the risk of climate hazards to the proposed building over a period of at least 35 years. Where the activity is assessed to be at risk from wildfires, storm damage and/or pluvial flooding, a climate risk and vulnerability assessment (CCVA) to assess the materiality of the physical climate risks on the economic activity shall be conducted, and adaptation solutions that can reduce the physical climate risk shall be identified. As necessary, the economic operator shall integrate the adaptation solutions that reduce the most important identified physical climate risks at the time of design and construction and shall implement them before the start of building use.	No requirements here for renovation. Considered useful to focus only on the climate hazards that could bring about a premature end-of-life of the building. Would need expertise from the insurance sectors for the risk assessment. How much risk is too much? Where to draw the line? In case there is no established method for the screening exercise or the CCVA, the assessment could follow the guidance set out in this Commission study.

Environmental goal	Proposed requirement	Remarks
	The adaptation solutions implemented shall not adversely affect the adaptation efforts or the level of resilience to physical climate risks of other people, of nature, of cultural heritage, of assets and of other economic activities; are consistent with local, sectoral, regional or national adaptation strategies and plans; and consider the use of nature-based solutions or rely on blue or green infrastructure to the extent possible.	
c) Water resources (WAT)	(**for new construction or renovation**) The building incorporates one or both of the following EN 16941-compatible design features: 1. Greywater collection and recycling system. 2. Rainwater harvesting system. Details of the collection, treatment, storage and intended end-use(s) of the systems should also be described.	Although no TSC was set for buildings for this environmental goal, all buildings can potentially recycle greywater. Many are also suitable for collecting rainwater and this can help reduce flood risks if well designed. Incorporation in renovation projects will depend on available space and the extent of the renovation.
d) Circular economy (CE)	(**for new construction only**) Construction designs and techniques support circularity via the incorporation of concepts for design for adaptability and deconstruction as outlined in Level(s) indicators 2.3 and 2.4 respectively. Compliance with this requirement is demonstrated by reporting on the Level(s) indicators 2.3 and 2.4 at Level 2 or 3.	This is just one of five requirements set in the TSC for contributions to CE in the EU Taxonomy. Excel tools are publicly available for both indicators. Indicator 2.3 is about answering a series of questions about the building design for adaptability. Indicator 2.4 is about estimating the future fate of the different materials and building elements used in the building at its end of life.
e) Pollution prevention and control (PPC)		Not applicable. No TSC set for this goal for buildings.
f) Biodiversity and ecosystems (Bio-D)		Not applicable. No TSC set for this goal for buildings.

5.3 The sustainability of forest and agriculture biomass raw material

Article 7(3c) of the CRCF Regulation stipulates that the minimum sustainability requirements of the certification methodology must promote the sustainability of forest and agriculture biomass raw material in accordance with the sustainability and greenhouse gas emissions saving criteria for biofuels, bioliquids and biomass fuels laid down in Article 29 of Directive (EU) 2018/2001 – the Renewable Energy Directive (hereafter referred to as RED).

The exact applicability of this provision to the certification methodologies on carbon storage in products could be discussed. First, most carbon storage in products applications are not energy related, while Article 29 of the Renewable Energy Directive specifically refers to sustainability criteria for fuels and bioliquids. Second, article 3(3) of the Renewable Energy Directive itself distinguishes between biomass-based products on the one hand and biofuels, bioliquids and biomass fuels on the other Moreover, within the principle of cascading use of biomass outlined in this article, wood-based products, such as construction wood, are placed at the top of the hierarchy. This may suggest that wood-based products could be considered sustainable within the context of the RED. Third, the application of specific RED-related sustainability requirements for certification under the CRCF, may hinder the possible future integration of the CRCF units with other policy instruments where these criteria are not applied, such as the EPBD whole life carbon accounting.

On the other hand, one could also argue that the article 29 criteria are applicable to all certification methods developed under the CRCF, including all carbon storage in product methodologies, regardless of their connection with biofuels. In the below sections the possible applicability of Article 29 RED provisions are assessed more in detail. This Article comprises 15 paragraphs and numerous subparagraphs, which are reproduced in Appendix VIII, with yellow highlights to help identify those sections relevant to wood and/or other biomass harvesting.

5.3.1 About biobased materials from agricultural activities

After reviewing Article 29, the potentially relevant RED-related sustainability requirements for **eligible bio-based insulation materials** can be summarised as:

1. Not be produced from raw material obtained from land or areas that were, in or after January 2008, designated as:
   - primary forest and other wooded land of native species, where there is no clearly visible indication of human activity and where ecological processes are not significantly disturbed (Art. 29(3)(a)).
   - old growth forests as defined in the country where the forest is located (Art. 29(3)(a)).
   - highly biodiverse forests and other wooded land that are species-rich, not degraded, and identified as highly biodiverse by the relevant competent authority (possible exemption if suitable proof of non-interference can be provided) (Art. 29(3)(b)).
   - areas designated by law or by the relevant competent authority for nature protection purposes (possible exemption if suitable proof of non-interference can be provided) (Art. 29(3)(c)(i)).
   - areas designated for the protection of rare, threatened, or endangered ecosystems or species recognised by international agreements or included in lists by intergovernmental organisations or the International Union for Conservation of Nature (possible exemption if suitable proof of non-interference can be provided) (Art. 29(3c(ii)).
   - highly biodiverse grassland exceeding one hectare in size (natural or non-natural, but with a possible exemption for non-natural grassland if suitable proof of non-interference can be provided) (Art. 29(3)(d)).
   - heathland (Art. 29(3)(e)).
2. Not be made from raw material obtained from land with any of the following high-carbon stock statuses in January 2008 and that no longer have that status:
   - wetlands (Art. 29(4)(a).
   - continuously forested areas, namely land spanning more than one hectare with trees higher than five metres and a canopy cover of more than 30 %, or trees able to reach those thresholds in situ (Art. 29(4)(b).
   - land spanning more than one hectare with trees higher than five metres and a canopy cover of between 10 % and 30 %, or trees able to reach those thresholds in situ (possible exemption to this) (Art. 29(4)(c)).
3. Not be made from raw material obtained from land that was peatland in January 2008, unless evidence is provided that the cultivation and harvesting of that raw material does not involve drainage of previously undrained soil (Art. 29(5)).

Table 20. Indicative list of voluntary schemes[15] for certification of agricultural biomass production

Name of scheme	Feedstocks covered	Geographical coverage	Chain of custody coverage
Austrian Agricultural Certification scheme (AACS). Website here.	Agricultural feedstocks and vegetable oils (including residues).	Austria (for the AACS.part) but global for AMA-approved inspection bodies.	Farm and initial processing (e.g. oil seed crushing) only.
Better Biomass. Website here.	Agricultural biomass (including wastes and residues)	Global	Full fuel chain (for biomethane up to the production unit)
Biomass biofuels voluntary scheme (2BS). Website here.	Agricultural biomass (including wastes and residues)	Global	Full fuel chain (for bio methane up to the production unit).
Bonsucro EU. Website here.	Sugar cane (including residues) to produce ethanol and solid biomass fuels from bagasse.	Global	Full, including compliance of the consignments of biofuels, bioliquids and biomass fuels with the low indirect land-use change-risk criteria set in Delegated Regulation (EU) 2019/807.
International Sustainability and Carbon Certification (ISCC). Website here.	Agricultural biomass, forest biomass, wastes and residues.	Global	Full fuel chain (for biomethane from the production unit up to the point of consumption), including compliance of the consignments of biofuels, bioliquids and biomass fuels with the low indirect land-use change-risk criteria set in Delegated Regulation (EU) 2019/807.
KZR INiG system. Website here.	Agricultural and forest biomass, wastes and residues.	Global (primarily Poland)	Full fuel chain (for biomethane up to the production unit)
REDcert. Website here.	Agricultural biomass (excluding high-ILUC risk feedstocks), waste and residues	Global (selected countries for which REDcert has adopted a “country profile”)	Full fuel chain (for biomethane from the production unit up to the point of consumption)
Round Table on Responsible Soy (RTRS). Website here.	Soy.	Global	Fuell fuel chain.
Roundtable on sustainable biomass (RSB). Website here.	Agricultural biomass, wastes and residues (forest biomass is excluded)	Global	Full fuel chain (for biomethane up to the production unit), including compliance of the consignments of biofuels, bioliquids and biomass fuels with the low indirect land-use change-risk criteria set in Delegated Regulation (EU) 2019/807.
Sustainable Biomass Program (SBP). Website here.	(a) ligno-cellulosic material derived from forest and non-forest land; (b) processing residues from forest and agriculture related industries (outside forest and agricultural	Global	Full fuel chain.

[15] List derived from this Commission website but excludes schemes that do not certify agricultural residues or wastes. The reason for excluding them is that most biobased materials that are used will not be the primary products of the agricultural activity (e.g. straw from wheat production, bagasse from sugar production, hemp shives from hemp fibre production etc.

Name of scheme	Feedstocks covered	Geographical coverage	Chain of custody coverage
	land). (c) Woody agricultural residues from agricultural land. Non-woody agricultural residues from agricultural land are excluded.		
Sustainable Resources (SURE). Website here.	Agricultural and forest biomass (including wastes and residues)	Global	Full fuel chain (for biomethane from the production unit up to point of consumption)

Open questions for stakeholders:

Q13a. The RED requirements for agricultural biomass are all based on the status of the land as it was in January 2008. Are the schemes listed in the table above appropriate for certifying compliance with the relevant parts of Article 29 of the RED?

Q13b. Are there other relevant schemes missing?

Q13c. What quantities of biobased materials are covered by these certifications?

5.3.2 About wood-based materials from forests

And the potentially relevant sustainability requirements for **eligible wood-based structural products** can be summarised as:

1. A series of requirements at national level (Article 29(6)(a), namely:
   - (i) the legality of harvesting operations;
   - (ii) forest regeneration of harvested areas;
   - (iii) that areas designated by international or national law or by the relevant competent authority for nature protection purposes, including in wetlands, grassland, heathland and peatlands, are protected with the aim of preserving biodiversity and preventing habitat destruction;
   - (iv) that harvesting is carried out considering maintenance of soil quality and biodiversity in accordance with sustainable forest management principles [...]
   - (v) that harvesting maintains or improves the long-term production capacity of the forest;
   - (vi) that forests in which the forest biomass is harvested do not stem from the lands that have the statuses referred to in paragraph 3, points (a), (b), (d) and (e), paragraph 4, point (a), and paragraph 5, respectively under the same conditions of determination of the status of land specified in those paragraphs; and
   - (vii) that installations producing biofuels, bioliquids and biomass fuels from forest biomass, issue a statement of assurance, underpinned by company-level internal processes, for the purpose of the audits conducted pursuant to Article 30(3), that the forest biomass is not sourced from the lands referred to in point (vi) of this subparagraph.

2. If evidence for the points listed above is not available, then Article 29(6)(b) permits evidence to be provided based on management systems implemented at the level of the forest sourcing area, as detailed below:
   - (i) the legality of harvesting operations;
   - (ii) forest regeneration of harvested areas;
   - (iii) that areas designated by international or national law or by the relevant competent authority for nature protection purposes, including in wetlands, grassland, heathland and peatlands, are protected with the aim of preserving biodiversity and preventing habitat destruction, unless evidence is provided that the harvesting of that raw material does not interfere with those nature protection purposes;
   - (iv) that harvesting is carried out considering maintenance of soil quality and biodiversity, in accordance with sustainable forest management principles [...]; and
   - (v) that harvesting maintains or improves the long-term production capacity of the forest.

3. Forest biomass shall meet the following land-use, land-use change and forestry (LULUCF) criteria (Art. 29(7)):
   - (a) the country or regional economic integration organisation of origin of the forest biomass is a Party to the Paris Agreement and:

- (i) it has submitted a nationally determined contribution (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC), covering emissions and removals from agriculture, forestry and land use which ensures that changes in carbon stock associated with biomass harvest are accounted towards the country's commitment to reduce or limit greenhouse gas emissions as specified in the NDC; or
    - (ii) it has national or sub-national laws in place, in accordance with Article 5 of the Paris Agreement, applicable in the area of harvest, to conserve and enhance carbon stocks and sinks, and provides evidence that reported LULUCF-sector emissions do not exceed removals;
- (b) where evidence referred to in point (a) of this paragraph is not available, the biofuels, bioliquids and biomass fuels produced from forest biomass shall be taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 if management systems are in place at forest sourcing area level to ensure that carbon stocks and sinks levels in the forest are maintained, or strengthened over the long term.

4. All of the requirements relating to forest areas are underpinned by the following Member State level requirement set out in Article 29(6(a)(vi-vii)):

- the country in which forest biomass was harvested has national or sub-national laws applicable in the area of harvest as well as monitoring and enforcement systems in place ensuring:
    - (vi) that forests in which the forest biomass is harvested do not stem from the lands that have the statuses referred to in paragraph 3, points (a), (b), (d) and (e), paragraph 4, point (a), and paragraph 5, respectively under the same conditions of determination of the status of land specified in those paragraphs; and
    - (vii) that installations producing biofuels, bioliquids and biomass fuels from forest biomass, issue a statement of assurance, underpinned by company-level internal processes, for the purpose of the audits conducted pursuant to Article 30(3), that the forest biomass is not sourced from the lands referred to in point (vi) of this subparagraph.

There were also requirements in Article 29(7)(a) and 29(7)(b) that refer to consistency with national energy and climate plans, but these points were considered to focus solely on bioenergy, not on the use of biomass for long-lasting construction products.

Articles 29(6) and 29(7) begin with national-level requirements in their respective section (a), but then in sections (b), they allow for similar evidence to be submitted at the source forest level when national evidence cannot be provided. This approach ensures that forest managers can benefit from effective forest management systems implemented in their forests.

The two most well-known forest certification schemes are FSC and PEFC, which not only audit forest management but also the chain of custody throughout the supply chain, providing labelling of final products to end consumers. These schemes offer five labels, which are as follows:

<table>
<tr><td>Sustainable virgin</td><td>100%</td><td rowspan="3">70-100%</td><td>0%</td><td>70-100%</td><td>0%</td></tr>
<tr><td>Post-consumer recycled</td><td rowspan="3">0%</td><td>70-100%</td><td rowspan="3">0-30%</td><td rowspan="2">100%</td></tr>
<tr><td>Pre-consumer recycled</td><td>0-30%</td></tr>
<tr><td>Controlled</td><td>0-30%</td><td>0%</td><td>0%</td></tr>
</table>

Figure 13. Overview of the 5 FSC and PEFC labels and the types of material that can be permitted.

As shown in the figure above, only the FSC 100% label guarantees the use of sustainable virgin material. Some doubts may arise about labels that allow controlled material, as they do not meet the same level of forest management standards as the sustainably certified virgin material.

Regarding recycled labels, these can be considered compliant with RED since no harvesting was necessary, but this may lead to other issues with the net carbon removal accounting outlined in the quantification section.

Open questions for stakeholders:
Q14a. Are the RED Article 29 sustainability requirements relevant to include in the certification methodology for carbon storage in construction materials?
Q14b. Is it fair to assume that FSC and PEFC controlled materials are also compliant with the relevant parts of RED Article 29?
Q14c. Are there any other schemes operating in the EU that can be considered as equivalent to FSC or equivalent to PEFC?
Q14d. Do any Member States have legislation in place that ensures all productive forests and plantations for wood harvesting meet the relevant requirements of Article 29 of RED?

Feedback from certification scheme representatives revealed that market coverage of third-party certification schemes for RED Article 29 compliance was very low or non-existent in many parts of the EU. In terms of sustainability requirements implemented by schemes, the following information was provided:

The effect of the EU Deforestation Regulation on this aspect is likely to be considerable but will not guarantee EU-wide compliance with Article 29 of RED. If the CRCF certification scheme is able to include the auditing of compliance of raw materials with Article 29, then this should also be permitted if the procedure is suitable.

- **Rainbow**: has several criteria in place to ensure sustainability of the project. An environmental and social do no harm risk analysis is mandatory, for which an assessment template is provided (notably biomass origin and certification is covered). Projects shall transparently evaluate the potential leakage risks. At least two co-benefits must be quantified and justified (using UN SDG reference)..
- **Climate Cleanup Foundation certification protocol** **(v1.0)**: Provides some different ways to demonstrate sustainability. The first option is to demonstrate compliance with national statutory requirements for building project sustainability and energy (e.g. MPG in the Netherlands). If national requirements are not in place, then an alternative would be to have the building certified by a green building certification scheme. Regarding the impact of biobased material production on biodiversity, FSC/PEFC or similar certification is sufficient for wood-based materials. For other biobased materials, there is no burden of proof on biodiversity impacts if the raw material is sourced from within the EU.
- **Label Bas Carbone**: Biobased materials must also be classified as A or A+ for VOC emissions. The origin of wood needs to be FSC/PEFC certified or, in the case of French wood, an alternative option is to be from holder of a sustainable forest management document for private forests (Management Plan, Code of Good Silvicultural Practices, etc.) or a forest management plan for public forests. French or European wood is considered as a co-benefit.

6 Certification Rules Report (CRR)

The TAP, which consists of sections 1 to 5 of this document and the associated appendices, forms the basis for the CRR, which in turn will form the skeleton around which the body of the CM is built up – as illustrated below.

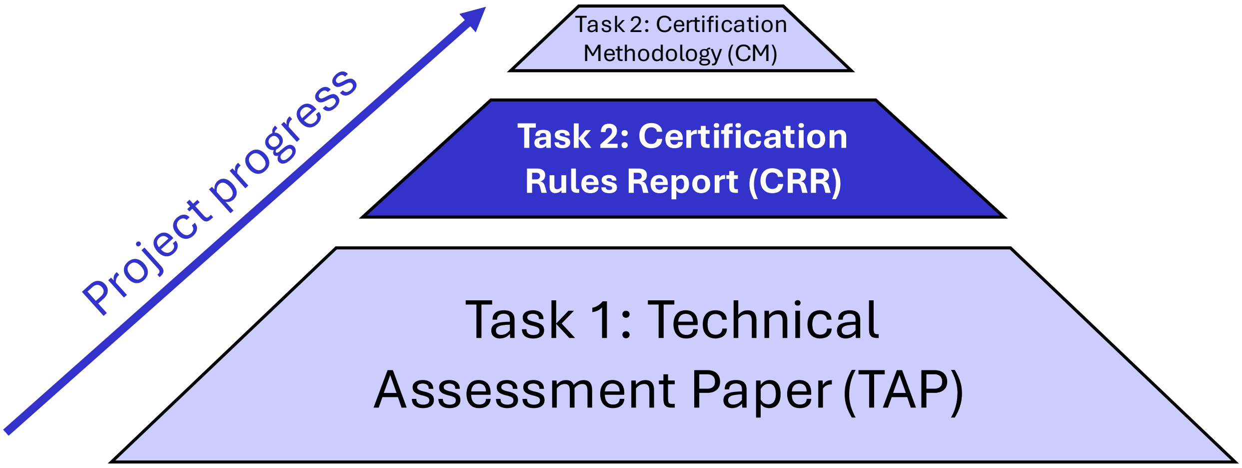


Figure 14. Illustration of the three main deliverables expected from the contracted work in this project.

The CRR is a very concise document that draws on all the specific areas where specific rules need to be defined for the activity in the CM. A summary of the areas where rules need to be defined in the CM, and relevant remarks about this are presented in the table below.

Table 21. Certification rules to be addressed in the methodology

General area	Specific area	Rules needed and remarks
Scope	Scope of building elements	Structural elements in buildings and thermal or acoustic insulation. ***Remarks**: these elements have expected lifespans of at least 35 years, do not require any maintenance and have a low risk of being removed or tampered with via DIY or minor renovations.*
	Scope of biobased materials	**For structural elements**: Solid wooden products such as planed timber, cross-laminated timber and glued laminated timber for use in building structures. **For insulation materials**: Wood-based and other biobased virgin raw materials or by-products from biomass that are generated by photosynthetic activity. ***Remarks**: The main wood-based structural materials are specifically mentioned.* *Other biomass has to be photosynthesis-based in order to guarantee the stored of atmospheric CO2. This initial methodology is limited to virgin raw materials or by-products because of the issue of initial carbon uptake having already been accounted for in the first life cycle of recycled materials (e.g. recycled paper, recycled cotton etc.).* *Either this EU certification methodology, or the certification schemes themselves, should provide default assumptions for different biobased materials within the scope that can be used when specific data is not available.*
	Scope of building typologies	All building types may be included in the scope. ***Remarks**: Certification schemes remain free to selectively restrict the scope for a variety of reasons. For example, it may not be economically viable to*

General area	Specific area	Rules needed and remarks
		certify individual small buildings with limited amounts of CRCF-eligible stored carbon. *Some buildings might not be eligible because their requested service lives are actually less than 35 years.* *A building might be incompatible with CRCF due to sustainability requirements. For example, the building use may be dedicated to the extraction, storage, transport or manufacture of fossil fuels. The building may generate onsite carbon emissions from fossil fuels during use or may be located in a floodplain.*
	Scope of building projects and project types	Both new construction and renovation projects may be included in the scope. ***Remarks**: Not considered realistic to define a standardised baseline for renovation projects, they must use activity-specific baselines, at least for structural wood.*
Quantification	Data sources	All CRCF-eligible products must be covered by a valid, third-party certified environmental product declaration in line with EN 15804 and any additional relevant product category rules defined in associated EN standards. When multiple sources are available, data shall be selected based on the following order: 1. Product data on climate change effects obtained from the declaration of performance and conformity (DoPC) under the Construction Products Regulation (CPR). 2. Product-specific data on global warming potential (GWP) in an EN 15804-compliant environmental product declaration (EPD). 3. Average data for a product group on GWP in an EN 15804-compliant EPD. Data in points 1 and 2 must be used for project calculations (CR-total and GHG-associated) while data in points 1, 2 or 3 may be used in baseline calculations. ***Remarks**: this is a more focused version of the same logic being applied in the draft DA for EPBD Annex III about life cycle assessments of new buildings.*
	Approaches to standardised baselines	Separate baseline mechanisms shall be used for structural wood and for biobased insulation. Standardised baselines for structural wood shall be determined by normalising quantities on a kgCO2/m2 floor area basis. Standardised baselines for biobased insulation shall be determined using a market-based approach related to % share of biobased insulation sold. Standardised baselines for structural wood should be elaborated at national level or more detailed regional level and should be assigned to specific building typologies. The GHG-associated with CRCF-eligible biogenic carbon storing elements shall be indirectly accounted for in quantification calculations by only counting GHG-associated with the additional CR in projects. ***Remarks**:* *Section 2.6.2 of the TAP explains the reasons for needing a separate approach to standardised baselines for structural wood and biobased insulation. Some worked examples are provided in sections 2.6.7 and 2.6.8. The worked examples are provided in order to make the approach for standardised baselines more prescriptive, thus ensuring better harmonisation amongst schemes.*

General area	Specific area	Rules needed and remarks
		The work in sections 2.6.5 and Appendix V of the TAP showed a high degree of variability in assumed data at national level and for different building typologies in the same country. *The indirect solution to counting GHG-associated in standardised baselines is the simplest approach. By only counting GHG-associated with the "additional" CRCF-elements, it is automatically assumed that the GHG-associated is the same in the relevant CRCF-baseline material. This is as good an assumption as any other, but avoids extra work for finding separate assumptions for GHG-associated in baseline materials (e.g. looking for generic EPD data).*
	Approaches to activity-specific baselines	Activity-specific baselines shall be based on hypothetical data for a reference construction or renovation activity. Assumptions on GHG-associated of CRCF-eligible biogenic carbon storing elements shall also be included in baseline calculations. ***Remarks**: The approach to activity-specific baselines is less prescriptive. Although it is clarified that GHG-associated if any relevant carbon storing elements is also included to ensure a fair comparison.*
	Calculation of CR-total	Calculations shall be based on the quantities of CRCF-eligible materials at the level of the building and multiplied by the declared stored biogenic carbon content on the associated DoPCs or product-specific EPDs. In case the functional units are not in terms of mass, the declared unit shall be converted to mass (tonnes) so that the stored carbon content can also be expressed in tonnes. ***Remarks**: More of an issue for biobased insulation than for wood. If the function unit is not in terms of mass, the DoPC or EPD should include other relevant information on the declared volume and density so that a conversion to mass can be conducted in a consistent manner.*
	Calculation of GHG-associated	The emissions of GWP-biogenic, GWP-fossil and GWP-LULUC from modules A1-A5 shall be counted for GHG-associated. This scope of counting shall be the same in the project accounting and in any GHG-associated incorporated into CR-baseline calculations. ***Remarks**: There may be some issues with the accounting of GWP-biogenic emissions in module A1 or, when reported as a group, modules A1-A3 due to the GWP-biogenic uptake overshadowing these numbers.*
	Uncertainties	Product data quality must be "fair" or better in terms of geographical representativeness, technical representativeness and time representativeness according to Annex E to EN 15804. ***Remarks**: The proposal here is the simplest option. A different approach might be recommended in the future depending on the outcome of efforts with the PEF experts in the Commission at coming up with a quantitative approach to scoring data quality, using a data quality index (DQI). Level(s) also provides a quantitative approach to DQI at building level, although this is for the whole LCA of a building.*
Additionality	Criteria for use of activity-specific baselines	Activity-specific baselines shall only be permitted in cases where: (i) no national- or regional-level standardised baseline is available, and (ii) when no directly relevant subsidies or statutory requirements are applicable at national or regional level. ***Remarks**: By "directly relevant", it is meant that the only the used of CRCF-eligible biobased elements is OBLIGATORY under the subsidy or statutory requirement.*

General area	Specific area	Rules needed and remarks
		The financial viability condition is deemed inappropriate for the activity of constructing or renovating buildings because the activity only applies to some of the many materials used in a project and that material costs overall are just one of many different costs in a project. Furthermore, changing from one structural material to another (in theory) would also have knock-on effects on the cost of locally available labour, equipment needs, construction times and possibly the size of foundations needed.
Long-term storage, monitoring and liability	Minimum activity and storage periods.	The activity period for quantification purposes shall correspond to modules A1-A5 of a biobased construction product and building lifecycle, as defined in EN 15804 and EN 15978. Schemes may also opt to use higher storage periods (e.g. 50 years, 60 years etc.). ***Remarks****: There were different opinions about the minimum storage period and certification schemes take different approaches. By keeping a minimum of 35 years but allowing schemes to claim for longer, we do not exclude any schemes and we allow for different levels of storage certification that can be adapted to the demands in the voluntary carbon market.*
	Minimum monitoring frequency and requirements	After the initial certification, operators are obliged to notify the certification scheme of any change in building ownership and provide contact details of the new owner (operator). In order to reduce the risk of reversal, the certification scheme should share relevant information with the new operator about the stored carbon in the building that is covered by CRCF certification and timings of re-certification audits for the remaining liability period. The re-certification audit shall be considered as synonymous with the monitoring activity. Re-certification audits shall be conducted at intervals of every 5 years, or more frequently. The monitoring process shall entail contacting the operator and obtaining a signed declaration that the CRCF-eligible materials remain in place or notifying of any partial or complete reversal of certified units. Re-certification methodology may rely on remote inspections using satellite images, video tours and consultation of local authority registers for any renovation or demolition permits. If deemed necessary by the certification scheme, onsite visits may also form part of the monitoring and re-certification auditing process. ***Remarks****: The approach to monitoring is non-prescriptive, but inserts a minimum re-certification audit frequency of every 5 years because this is stipulated in recital 29 and Article 9(3) of the CRCF Regulation.*
	Issuance of certified units	The initial certification audit cannot be completed until the construction or renovation project has been completed. Certification schemes shall request adequate proof and explanations of the quantities of CRCF-eligible materials used in the building. Invoices or a supplier declaration shall be provided to ensure that the materials used match up with the EPDs or DoPCs used in CR-total and GHG-associated calculations. Certification schemes may issue the CRCF-certified units for the entire monitoring period upfront or in portions at fixed intervals, and either in an ex-ante (upfront) or ex-post manner. ***Remarks****: This is a polarising issue between different schemes and so it makes sense to allow for both possibilities. However, the extra liability risk that is inherent with upfront issuance should be reflected in minimum liability requirements.*
	Minimum liability coverage	Certification schemes may use buffer pools, upfront insurance or other suitable mechanisms as safeguards against reversals. The minimum shares

<table>
<tr><th>General area</th><th>Specific area</th><th>Rules needed and remarks</th></tr>
<tr><td></td><td></td><td>of net carbon removal benefit for a project shall vary as a function of the claimed storage period and the form of certified unit issuance. These minimum shares to cover by buffer pools, upfront insurance etc. shall vary as follows:
<table>
<tr><th rowspan="2">Form of certificate issuance</th><th colspan="3">Minimum storage period claimed</th></tr>
<tr><th>35 years</th><th>50 years</th><th>100 years</th></tr>
<tr><td>Upfront (ex-ante) for the full period.</td><td>5%</td><td>7.5%</td><td>15%</td></tr>
<tr><td>Upfront (short-term, i.e. 1 to 5-year intervals)</td><td>1%</td><td>1%</td><td>1%</td></tr>
<tr><td>Upfront (mid-term, i.e. 5 to 10-year intervals)</td><td>2%</td><td>2%</td><td>2%</td></tr>
<tr><td>Upfront (long-term, i.e. 10 to 20-year intervals)</td><td>3.5%</td><td>3.5%</td><td>3.5%</td></tr>
<tr><td>Ex-post (any defined interval)</td><td>0%</td><td>0%</td><td>0%</td></tr>
</table>

Remarks: The percentage shares are arbitrary, but increase as reversal risk increases. Discussions with certification schemes generally implied that 10% would already be a very high percentage, although this discussion took place in the context of 35-year storage periods.
Certification schemes are free to have higher percentages to cover liabilities according to their own reversal risk evaluation criteria.</td></tr>
<tr><td>Sustainability</td><td>DNSH</td><td>Prior to completion of the initial certification audit, compliance with the sustainability DNSH requirements shall be verified.
<table>
<tr><th>Environmental goal</th><th>Requirement</th></tr>
<tr><td>Climate change mitigation</td><td>(for new construction or renovation)
1. The building is not dedicated to the extraction, storage, transport or manufacture of fossil fuels.
2. The building will not generate onsite carbon emissions from fossil fuels during its normal use.</td></tr>
<tr><td>Climate change adaptation</td><td>(for new construction only)
A screening exercise shall be conducted to assess the risk of climate hazards to the proposed building over a period of at least 35 years.
The building shall not be constructed on sites where risks from fluvial flooding, storm surges and coastal flood, coastal erosion, avalanches, landslides, subsidence, glacial lake, reservoir or lagoon outbursts are identified.</td></tr>
<tr><td>Water resources</td><td>(for new construction or renovation, but only for non-residential buildings located in river basins that are under water stress or severe water stress)
Where installed, the specified water use for the following water appliances shall be attested by product datasheets, a building certification or an existing product label in the Union:</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>General area</th><th>Specific area</th><th colspan="2">Rules needed and remarks</th></tr>
<tr><td rowspan="5"></td><td rowspan="5"></td><td></td><td>(a) wash hand basin taps and kitchen taps have a maximum water flow of 6 litres/min;
(b) showers have a maximum water flow of 8 litres/min;
(c) WCs, including suites, bowls and flushing cisterns, have a full flush volume of a maximum of 6 litres and a maximum average flush volume of 3,5 litres;
(d) urinals use a maximum of 2 litres/bowl/hour. Flushing urinals have a maximum full flush volume of 1 litre.</td></tr>
<tr><td>Circular economy</td><td>(for new construction only)
At least 70% (by weight) of the non-hazardous construction and demolition waste (excluding naturally occurring material referred to in category 17 05 04 in the European List of Waste established by Decision 2000/532/EC) generated on the construction site is prepared for reuse, recycling and other material recovery, including backfilling operations using waste to substitute other materials, in accordance with the waste hierarchy and the EU Construction and Demolition Waste Management Protocol[16].</td></tr>
<tr><td>Pollution prevention and control</td><td>(for new construction or renovation)
Bio-based insulation materials shall emit less than 0,06 mg of formaldehyde per m3 of material or component upon testing in accordance with the conditions specified in Annex XVII to Regulation (EC) No 1907/2006 and less than 0,001 mg of other categories 1A and 1B carcinogenic volatile organic compounds per m3 of test chamber air, upon testing in accordance with EN 16516 or ISO 16000-3:2011 or other equivalent standardised test conditions and determination methods.</td></tr>
<tr><td>Biodiversity and ecosystems</td><td>(for new construction only)
The new construction is not built on one of the following:
(a) arable land and crop land with a moderate to high level of soil fertility and below ground biodiversity as referred to the EU LUCAS survey;
(b) greenfield land of recognised high biodiversity value and land that serves as habitat of endangered species (flora and fauna) listed on the European Red List or the IUCN Red List;
(c) land matching the definition of forest as set out in national law used in the national greenhouse gas inventory, or where not available, is in accordance with the FAO definition of forest.</td></tr>
<tr><td colspan="2">Remarks: On CCM: Point 1 is copy-paste from the EU Taxonomy. No reference to NZEB is made since this will be out of date eventually, being replaced by ZEB. Not yet clear what national definitions of ZEB will be, but one common element is referred to in point 2. Easy to verify.
On CCA: Considered useful to focus only on the climate hazards that could bring about a premature end-of-life of the building.</td></tr>
</table>

[16] European Commission: Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Oberender, A., Fruergaard Astrup, T., Frydkjær Witte, S., Camboni, M. et al., EU construction & demolition waste management protocol including guidelines for pre-demolition and pre-renovation audits of construction works – Updated edition 2024, Publications Office of the European Union, 2024, https://data.europa.eu/doi/10.2873/77980

<table>
<tr><th>General area</th><th>Specific area</th><th>Rules needed and remarks</th></tr>
<tr><td></td><td></td><td>On Water: Possible to use the Level(s) water calculator to easily determine if a building is located in a water-stressed river basin by using the Level(s) indicator 3.1 tool <u>available here</u> (specifically by selecting from two dropdown menus in cells B1 and B2 in the L2 estimate worksheet therein).
However, in parts a) to d), some of these fittings could easily be changed to mislead auditors unless they intend to inspect the water appliances during audits.
On CE: A minimum of 70% of construction and demolition waste being prepared for reuse, recycled or recovered (incl. backfilling) is tantamount to complying with Article 11(2b) of the Waste Framework Directive. Doing it according to the EU CDW management protocol is to standardise how this is done across the EU.
On PPC: Only applying the requirements on pollutant emissions directly to bio-based internal insulation materials, because the EU Taxonomyrequirement applies only to materials that could come into contact with occupants.
On Bio-D: The requirement for an Environmental Impact Assessment (EIA) or an EIA screening from the Taxonomy is not included here because it should be decided by local planning authorities if needed or not. If done only for CRCF DNSH, it would be a potentially significant added cost to smaller building projects. The potential applicability of the RED Article 29 sustainability requirements shall be further assessed.</td></tr>
<tr><td></td><td>Co-benefits</td><td>Certified units may also carry co-benefit claims when associated with building projects that meet the criteria defined below:
<table>
<tr><th>Environmental goal</th><th>Requirement</th></tr>
<tr><td>Climate change mitigation</td><td>(for new construction only)
The energy performance is certified using an as-built Energy Performance Certificate (EPC) and is awarded an energy class A according to the national methodology.
(for renovation)
The renovation of the building meets the national definition of a cost-optimal energy renovation or has reduced its primary energy demand by at least 30% compared to before renovation.</td></tr>
<tr><td>Climate change adaptation</td><td>(for new construction only)
A screening exercise shall be conducted to assess the risk of climate hazards to the proposed building over a period of at least 35 years.
Where the activity is assessed to be at risk from wildfires, storm damage and/or pluvial flooding, a climate risk and vulnerability assessment (CCVA) to assess the materiality of the physical climate risks on the economic activity shall be conducted, and adaptation solutions that can reduce the physical climate risk shall be identified.
As necessary, the economic operator shall integrate the adaptation solutions that reduce the most important identified physical climate risks at the time of design and construction and shall implement them before the start of building use.
The adaptation solutions implemented shall not adversely affect the adaptation efforts or the level of resilience to</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>General area</th><th>Specific area</th><th>Rules needed and remarks</th></tr>
<tr><td></td><td></td><td>
<table>
<tr><td></td><td>physical climate risks of other people, of nature, of cultural heritage, of assets and of other economic activities; are consistent with local, sectoral, regional or national adaptation strategies and plans; and consider the use of nature-based solutions or rely on blue or green infrastructure to the extent possible.</td></tr>
<tr><td>Water resources</td><td>(for new construction or renovation)
The building incorporates one or both of the following EN 16941-compatible design features:
1. Greywater collection and recycling system.
2. Rainwater harvesting system.
Details of the collection, treatment, storage and intended end-use(s) of the systems should also be described.</td></tr>
<tr><td>Circular economy</td><td>(for new construction only)
Construction designs and techniques support circularity via the incorporation of concepts for design for adaptability and deconstruction as outlined in Level(s) indicators 2.3 and 2.4 respectively. Compliance with this requirement shall be demonstrated by reporting on the Level(s) indicators 2.3 and 2.4 at Level 2 or 3.</td></tr>
</table>

Remarks: On CCM: Not the same as the Taxonomy TSC for new construction, which only makes reference to NZEB, which will soon be replaced by ZEB in the EPBD. The 30% reduction in primary energy demand might require two assessments, one immediately prior to renovation and one after.
On CCA: Considered useful to focus only on the climate hazards that could bring about a premature end-of-life of the building.
Would need expertise from the insurance sectors for the risk assessment. How much risk is too much? Where to draw the line?
In case there is no established method for the screening exercise or the CCVA, the assessment could follow the guidance set out in <u>this Commission study</u>.
On Water: Although no TSC was set for buildings for this environmental goal, all buildings can potentially recycle greywater. Many are also suitable for collecting rainwater and this can help reduce flood risks if well designed. Incorporation in renovation projects will depend on available space and the extent of the renovation.
On CE: This is just one of five requirements set in the TSC for contributions to CE in the EU Taxonomy. Excel tools are publicly available for both indicators. Indicator 2.3 is about answering a series of questions about the building design for adaptability. Indicator 2.4 is about estimating the future fate of the different materials and building elements used in the building at its end of life.
</td></tr>
</table>

Appendix I: CPR product families and priorities

In the table below are stated each of the 36 product families listed in Annex XVII to the CPR as well as the priority for updated harmonised standards in line with the new CPR and their current status (as of February 2025). It should be noted that the code 4 family has been split into two separate priorities, one for thermal insulation products, and the other for composite insulating kits/systems.

Table 22. List of CPR product families, their priorities and status regarding standardisation

CPR code and product family name	Priority	Status
1. PRECAST NORMAL/LIGHTWEIGHT/AUTOCLAVED AERATED CONCRETE PRODUCTS.	1st	Standardisation request adopted
20. STRUCTURAL METALLIC PRODUCTS AND ANCILLARIES.	2nd	
16. REINFORCING AND PRESTRESSING STEEL FOR CONCRETE (AND ANCILLARIES). POST TENSIONING KITS.	3rd	CPR Aquis work ongoing
2. DOORS, WINDOWS, SHUTTERS, GATES AND RELATED BUILDING HARDWARE.	4th	
15. CEMENT, BUILDING LIMES AND OTHER HYDRAULIC BINDERS.	5th	Standardisation request adopted
4. THERMAL INSULATION PRODUCTS AND COMPOSITE INSULATING KITS/SYSTEMS.	6th	CPR Aquis work ongoing
13. STRUCTURAL TIMBER PRODUCTS/ELEMENTS AND ANCILLARIES. **14.** WOOD BASED PANELS AND ELEMENTS.	7th 16th	
26. PRODUCTS RELATED TO CONCRETE, MORTAR AND GROUT.	8th	
17. MASONRY AND RELATED PRODUCTS. MASONRY UNITS, MORTARS, AND ANCILLARIES.	9th	
24. AGGREGATES.	10th	
10. FIXED FIRE FIGHTING EQUIPMENT (FIRE ALARM/DETECTION, FIXED FIREFIGHTING, FIRE AND SMOKE CONTROL AND EXPLOSION SUPPRESSION PRODUCT).	11th	CPR Aquis work due to start Dec. 2025
23. ROAD CONSTRUCTION PRODUCTS.	12th	
19. FLOORINGS.	13th	
9. CURTAIN WALLING/CLADDING/STRUCTURAL SEALANT GLAZING.	15th	Fast track ongoing
5. STRUCTURAL BEARINGS. PINS FOR STRUCTURAL JOINTS.	17th	
34. BUILDING KITS, UNITS, AND PREFABRICATED ELEMENTS.	18th	
21. INTERNAL & EXTERNAL WALL AND CEILING FINISHES. INTERNAL PARTITION KITS.	19th	
27. SPACE HEATING APPLIANCES.	20th	
22. ROOF COVERINGS, ROOF LIGHTS, ROOF WINDOWS, AND ANCILLARY PRODUCTS. ROOF KITS.	21st	
12. CIRCULATION FIXTURES: ROAD EQUIPMENT.	22nd	
18. WASTE WATER ENGINEERING PRODUCTS.	23rd	
25. CONSTRUCTION ADHESIVES.	24th	
7. GYPSUM PRODUCTS.	25th	
33. FIXINGS.	26th	
3. MEMBRANES, INCLUDING LIQUID APPLIED AND KITS (FOR WATER AND/OR WATER VAPOUR CONTROL).	27th	
30. FLAT GLASS, PROFILED GLASS AND GLASS BLOCK PRODUCTS.	28th	Fast track ongoing

CPR code and product family name	Priority	Status
8. GEOTEXTILES, GEOMEMBRANES, AND RELATED PRODUCTS.	29th	CPR Aquis work not started yet
11. SANITARY APPLIANCES.	30th	
28. PIPES-TANKS AND ANCILLARIES NOT IN CONTACT WITH WATER INTENDED FOR HUMAN CONSUMPTION.	31st	
31. POWER, CONTROL AND COMMUNICATION CABLES.	32nd	
6. CHIMNEYS, FLUES AND SPECIFIC PRODUCTS.	33rd	Fast track ongoing excluding self-standing chimneys
32. SEALANTS FOR JOINTS.	34th	CPR Aquis work not started yet
35. FIRE STOPPING, FIRE SEALING AND FIRE PROTECTIVE PRODUCTS. FIRE RETARDANT PRODUCTS.	35th	
29. CONSTRUCTION PRODUCTS IN CONTACT WITH WATER INTENDED FOR HUMAN CONSUMPTION.	36th	
36. ATTACHED LADDERS.	37th	

Appendix II: Default building element service lifetimes

The scope and hierarchy of building elements developed for the Union framework for the assessment of new building life cycle assessment is reproduced below for the parts that pertain to the actual building (shell and core). Areas that are potentially suitable for CRCF certification are highlighted in yellow. The three main criteria to apply are: (i) likely service life in excess of 35 years; (ii) likely to use wood or biobased materials, and (iii) low need for monitoring / risk of unforeseen future removal.

Table 23. Typical service lives for the scope of building elements in the EPBD Union framework

Tier 1	Tier 2	Tier 3 (example)	Tier 4 (examples)	Service life
Shell	Substructure	Foundation piling and underpiling	Permanent piles and caisson	60+ years
			Underpinning	
		Foundations	Lateral supports	60+ years
			Raft footings, pile caps, column bases, wall footings; strap beams, tie beams	
			Substructure walls and columns	
			Ground floor slabs and beams (when the building includes a basement, basement bottom slabs should be counted within the relevant Tier 3 "Basement elements")	
			Lift pits (slabs and walls)	
		Basement elements	Basement lateral supports	60+ years
			Basement bottom slabs and blinding	
			Retaining walls	
			Basement structural walls, braces and columns	
			Basement beams, joists, braces and slabs	
			Basement staircases and ramps	
			Vertical waterproof tanking, drainage blanket, drains and skin wall	
			Horizontal waterproof tanking, drainage blanket, drains and topping slab	
			Basement insulation	
			Basement lift pits, sump pits, sleeves	
		Composite work, prefabricated work and sundries for 'Substructure'[17]		
	Structure	Frames and slabs (above top of ground floor slabs)	Structural walls, braces and columns	60 years
			Upper floor beams, joists, braces and slabs	
			Roof beams, joists, braces and slabs	
			Staircases (forming part of the structure)	
			Fireproofing to steel structure	
		Tanks, pools and sundries	Only when located within the building envelope (otherwise included in external works)	
		Composite work, prefabricated work and sundries for 'Structure'[18]		
	External architectural works (non-structural)	Façade	Non-structural external walls and features	30-35 years
			External wall finishes except cladding	10 years (paint), 30 years (render)
			Facade cladding and curtain walls	30 years
			External windows	
			External doors	
			External shop fronts	
			Roller shutters and fire shutters	
		Roof	Roof finishes	30 years
			Skylights	
			Waterproofing	
			Insulation	
			Roof landscaping (hard and soft)	

[17] Insulation, waterproofing, screed, connections, fittings or elements for drainage, elements for services that are inserted or applied together with the substructural works but which are not already counted under specific entries in this Table **Error! Reference source not found.Error! Reference source not found.** or elsewhere.

[18] Fireproofing, insulation, waterproofing, screed, connections, fittings, ramps, permanent formwork, mezzanine structures, supports for tiered seating, maintenance routes or other elements that are inserted or applied together with the structural works but which are not already counted under specific entries in this Table **Error! Reference source not found.Error! Reference source not found.** or elsewhere.

Tier 1	Tier 2	Tier 3 (example)	Tier 4 (examples)	Service life
		Composite work, prefabricated work and sundries for 'External architectural works (non-structural)'[19]		
Core	Internal or under cover architectural works (non-structural)	Internal divisions	Non-structural internal walls and partitions	30 years
			Insulation	
			Internal shop fronts	
			Toilet cubicles	
			Moveable partitions	
			Cold rooms	
			Internal doors	
			Internal windows	
			Roller shutters and fire shutters	
			Sundry concrete work	
		Fittings and sundries	Balustrades, railings and handrails	30 years
			Staircases and catwalk not forming part of the structure, cat ladders	
			Built-in[20] cabinets, cupboards, storage, lockers, seating, shelves, counters, benches	10-30 years
			Built-in decorative features	
			Access panels	
		Finishes under cover	Floor finishes (internal and external (that is to say, under cover or in balconies))	10 years (coatings) 20-30 years finishes
			Internal wall finishes and cladding	
			Ceiling finishes and false ceilings (internal or external)	
			Insulation	
		Composite work, prefabricated work and sundries for 'Internal or under cover architectural works (non-structural)'[21]		10-30 years
	Building services and equipment: Water and wastewater-related systems	Sanitaryware	Toilets, cisterns, shower trays, bathtubs, taps, controls, shower heads, basin units, sinks, instant hot water heaters	20 years
		Cold water systems	Thermostat, thermal meters, cold water meters, pumps/booster set, other meters, pipework, pipe insulation, support/hanger, frost protection and trace heating equipment	25 years (10 years for pumps)
		Cold water storage	Storage tank plus any treatment and filtration system for water quality control	
		Surface water / rainwater / foul water drainage	Piping, insulation, support, rainwater storage tank, attenuation, outlets, pumps, downpipes, sewage piping, condensate piping, insulation, support, cistern, traps, pump, drain	
		Water reuse systems	Grey water / rainwater harvesting storage tank, pipework and treatment equipment inside the building line	
	Building services and equipment: Heating systems	Heat and hot water generation equipment	Gas/electric boiler, air/water/ground heat pumps, chiller, local water heater, wood burning stove, biomass boiler, solar thermal heating and hot water systems. Communal heating systems located within the building's footprint are included in this scope up to the point of the meter. Beyond the meter, these systems are considered part of the distribution network. Pit and manifold shall be included even if outside the building footprint. Plate heat exchanger that connects to a district heating network. Hot water generation equipment (for example, calorifier) shall also be included.	20 years (heating plant and distribution) 30 years (radiators)
		Heat and hot water distribution, control, ancillaries, emitters, exchangers/ terminal units	Electric radiator, wet radiator, underfloor heating, heat interface unit, plate heat exchanger, pumps, mechanical switchboard, pressurisation unit, dosing pot, branch circuit (BC) controller, dehumidifier, vibration mounts, thermostat, thermal meters, hot water meter, pipework, pipe insulation, support/hanger, frost protection and trace heating equipment	
		Heat storage equipment	Hot water store, buffer vessel, expansion vessel	

19 Fireproofing, insulation, waterproofing, screed, connections and fixings to the structure, fittings, ramps, shading devices, louvres, eaves, insect protection, grilled assemblies, parapets, railings, green walls, chimneys or other elements that are inserted or applied together with the external architectural works but which are not already counted under specific entries here or elsewhere.

20 Built-in refers to the incorporation of the relevant building features during the construction stage and prior to handover of the building to the owner.

21 Fireproofing, insulation, waterproofing, screed, connections and fixings to the structure or maintenance routes, framing, sealing, adhesives, floating floors, sprung floors, finishes, line markings, trim, skirting, fittings, ramps, grilled assemblies, parapets, railings, fireplaces or other elements that are inserted or applied together with the internal architectural works but which are not already counted under specific entries in this Table **Error! Reference source not found.** or elsewhere.

Tier 1	Tier 2	Tier 3 (example)	Tier 4 (examples)	Service life
	Building services and equipment: Dedicated cooling systems (if a system does both heating and cooling, it shall fall within the scope of 'heating systems' only)	Cooling generation equipment	Cooling tower, fan coil units, air conditioner.	15 years (cooling plant and distribution)
		Cooling emitter, exchangers/ terminal units, ancillaries and control, distribution, storage	Cold water store, buffer vessel, expansion vessel for cooling, pumps, mechanical switchboard, pressurisation unit, dosing pot, BC controller, dehumidifier, vibration mounts, thermostat, thermal meters, cold water meter, pipework, pipe insulation, support/hanger, frost protection and trace heating equipment	
	Building services and equipment: Ventilation systems	Air movement	Fans, mechanical ventilation with heat recovery, air handling units, ceiling fans, kitchen ventilation, air curtains	20 years (air handling units) 30 years ductwork and distribution
		Air terminals	Diffusers, grilles, variable air volume systems, constant air volume systems, louvre	
		Ductwork and ancillaries	Ductwork, insulation, support, fire-rated ductwork, support	
		Control dampers, attenuation and fire safety related to ventilation equipment	Variable air volume damper, volume control damper, fire damper, fume and smoke extraction, motorised fire smoke damper, staircase pressurisation, fire-rated fans, pressure relief dampers, controls, louvres, gas extract, acoustic attenuation	
	Building services and equipment: Lighting systems	Internal lighting	Internal light fixtures, outlet, junction box, socket, light control, cable, switch	5-10 years (light fittings) 15-20 years (light controls)
		External lighting (building mounted)	Lamps/ poles/ supports etc. that are building mounted. External light fixtures, outlet, junction box, socket, light control, cable, switch	
		Emergency lighting	Emergency lights, controls, cable, switch	
		Other lighting	Task lighting, stage/entertainment lighting, retail display lighting, architectural lighting including associated light fixture, outlet, junction box, socket, light control, cable, switch	
	Building services and equipment: Electrical services for power, communications, security, IT and fire detection	Electrical power	Includes internal and building-mounted installations. Power cable, cable trays containment, panel board/ distribution, back up equipment, busbar, transformer, sockets/ switches, floor boxes, sensors, high voltage, low voltage, small power, containment	15-30 years
		Extra Low Voltage (ELV)/ communications/security	ELV: Low, medium and high voltage systems. Security: Closed circuit Television (CCTV) equipment, security sensors and alarms. Communications and audiovisual equipment	10-15 years
		IT and data	IT equipment: anything related to data, for example, Wi-Fi equipment, server, backbone and structured cabling, computers, printers, data cabinets, patch panels	5-10 years
		Building management system (BMS)	BMS/controllers on fan coils, outstation, main controller system with computer (headend), cabling required, control valves, sensors for temperature statistics	5-15 years
		Electricity backup generation	Uninterruptible power supply (UPS), backup generation, battery supply, standby generators within the building line	10-20 years
		Fire detection and alarm	Fire alarm systems including detection, cabling, firefighting panel and final call unit	
	Building services and equipment: On-site renewable energy generation	Renewable energy - electrical generation on-site and building mounted	Solar photovoltaic (PV) panel, inverter, wind turbine, water turbine building mounted or within building footprint	15-30 years
		Renewable energy - storage on-site	Battery within building footprint	10-20 years
	Building services and equipment: Life safety, fuel and movement system installations	Sprinkler system	Pipes, heads, valves, tank, hose, pumps	15 years (pumps) 30 years
		Firefighting systems	Dry and wet riser, hydrant, within designated building footprint, automatic opening vent (AOV) controls/sensors, fire suppression system	5 years (hydrant) 30 years
		Lightning protection / earth bonding	Lightning conductor, earth rods	60 years
		Fuel installations	All fuel supplies other than electric, anything pumped or pressurised. Gas equipment: connection, gas meter, pressure regulator, pipes, valves.	30 years

Tier 1	Tier 2	Tier 3 (example)	Tier 4 (examples)	Service life
			Fuel storage tank on site, dry stores. Augers.	
		Lift, stair lift, lifting platform	Systems for lift, stair lift, lifting platform shall be included. Power to those systems shall be included in electrical installations	30 years
		Escalators and moving walkways	Systems for escalators and moving walkways shall be included. Power to those systems shall be included in electrical installations	30 years
	Building services and equipment: Waste disposal systems	Specialised and communal waste disposal	Refuse, incinerators and any systems for specialist/ communal waste streams and disposal installation	20 years
		Composite work, prefabricated work and sundries for 'Building services and equipment'[22]		10-30 years

[22] Any other fixtures, fittings or other elements that are inserted or applied together with the building services, system and infrastructure but which are not already counted under specific entries in this Table **Error! Reference source not found.** or elsewhere.

Appendix III: Review of selected EPDs for construction products

A selection of EPDs related to wood-based or bio-based construction products was reviewed. These products are considered compatible with potential temporary biogenic carbon storage certificates for buildings. The review focused on the available data and the actual numbers. Specifically, it compared CR-total to GHG-associated at the level of individual products. In total, 12 EPDs were examined. This included 6 for structural wood products and 6 for bio-based insulation materials.

All of these EPDs either directly reported the stored biogenic carbon in the product (11 out of 12) or allowed it to be inferred (1 out of 12). All of these EPDs also distinguished between GWP-total, GWP-biogenic, GWP-fossil, and GWP-LULUC. However, some EPDs (2 of 12) did not report on modules A4 and A5, and only some of them (5 of 12) reported A1, A2, and A3 separately.

Life cycle module data availability

Data reported regarding which life-cycle modules were covered for GWP data is summarised below. In the table, cells in green indicate that GWP data was reported. Cells left blank indicate that ”n.d” was entered, which stands for ”not declared”. Cells in yellow indicate that zero was actively declared for all GWP categories within that life cycle module.

Table 24. EN 15804 life cycle module coverage by selected construction products (A1-A3, C1-C4 and D are mandatory).

<table>
<tr><th>Type of product</th><th>Manufacturer</th><th>A1</th><th>A2</th><th>A3</th><th>A4</th><th>A5</th><th>B1</th><th>B2</th><th>B3</th><th>B4</th><th>B5</th><th>B6</th><th>B7</th><th>C1</th><th>C2</th><th>C3</th><th>C4</th><th>D</th></tr>
<tr><td rowspan="3">Wood (Planed timber)</td><td>Stora Enso</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>Valbo Tra</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td>Nordlund</td><td colspan="3">X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X*</td><td>?</td><td>X*</td></tr>
<tr><td rowspan="2">Wood (Glulam)</td><td>Holzwerk Gebr. Schneider</td><td colspan="3">X</td><td>X</td><td>X</td><td>?</td><td>?</td><td></td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>Stora Enso</td><td colspan="3">X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>Wood (CLT)</td><td>KLH</td><td colspan="3">X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
<tr><td rowspan="6">Insulation material</td><td>ISOCELL</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>Cyclin B.V.</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>Holzwerk Gebr. Schneider</td><td colspan="3">X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>Havnens Haender</td><td colspan="3">X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>X</td><td>X</td><td>X</td><td>?</td><td>X</td></tr>
<tr><td>FASBA</td><td colspan="3">X</td><td>X</td><td>X</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X*</td><td>?</td><td>X*</td></tr>
<tr><td>SNaB</td><td>X</td><td>X</td><td>X</td><td>?</td><td>?</td><td>?</td><td>?</td><td>?</td><td></td><td></td><td></td><td></td><td>?</td><td>X</td><td>X</td><td>X</td><td>X</td></tr>
</table>

* *Module C3 and D data for the Nordlund and FASBA products seemed to have been split up, but not clear exactly why because the text of the EPDs were in Danish and German, respectively.*

Although all the mandatory modules were declared in the 12 EPDs listed above, 10 of them had at least one mandatory module declared at zero GWP, while 5 of them had two mandatory modules declared as zero. In all cases, the zero declarations for mandatory modules involved either module C1 (deconstruction/demolition of the building) or C4 (final disposal of waste from building deconstruction/demolition).

Regarding the A modules, all 12 EPDs reported data on modules A1-A3, although 7 of the 12 EPDs chose to combine the data for these modules into single values. For the optional A4-A5 modules, 9 of the 12 EPDs provided data, which could be important for GHG-associated calculations.

Significance of stored carbon in packaging compared to the product

Declarations of biogenic carbon storage in EPDs should also distinguish between any biogenic carbon stored in the product and in packaging materials. Both of these numbers would otherwise appear in module A1 and not be distinguishable simply by reading the data table in the EPD.

Distinguishing any carbon storage in packaging is important for two reasons. First, packaging is not a long-lasting product and therefore does not qualify for CRCF units. Second, the disposal of packaging will typically result in GHG emissions that should be counted as GHG-associated in module A5.

The table below compares declarations of stored biogenic carbon in the product and its packaging. Results are sometimes given in units of kgC and other times in kgCO2eq. Both values are listed, with the difference being the multiplication factor of 44/12. In half of the cases (6 out of 12), biogenic carbon storage in packaging was zero. In the other half, the share of stored biogenic carbon in packaging ranged from 0.05% to 6.3%.

Table 25. Declared biogenic carbon contents in selected construction product EPDs

Type of product	Manufacturer	Functional unit	Stored biogenic carbon (product)	Stored biogenic carbon (packaging)
Wood (Planed timber)	Stora Enso	1m3 planed timber with 12% moisture content (average density 460 kg/m3)	205.4 kgC/m3 753 kgCO2eq./m3 *	0 kgC/m3 [0%] 0 kgCO2eq./m3 *
	Valbo Tra	1m3 at an average density of 471 kg/m3	196 kgC/m3 718.7 kgCO2eq./m3 *	1.4 kgC/m3 [0.7%] 5.1 kgCO2eq./m3 *
	Nordlund	1m3 dried and planed timber with 16% moisture content (average density 453 kg/m3)	204 kgC/m3 748 kgCO2eq./m3 *	0.49 kgC/m3 [0.2%] 1.8 kgCO2eq./m3 *
Wood (Glulam)	Holzwerk Gebr. Schneider	1m3 glued laminated timber with 10% moisture content (average density 430 kg/m3)	194.5 kgC/m3 713.1 kgCO2eq./m3 *	0 kgC/m3 [0%] 0 kgCO2eq./m3 *
	Stora Enso	1m3 glued laminated timber posts and beams at an average density of 500 kg/m3	220 kgC/m3 806.7 kgCO2eq./m3 *	< 0.1 kgC/m3 [0.05%]
Wood (CLT)	KLH	1m3 cross laminated timber with 12% moisture content (average density 470 kg/m3)	208.2 kgC/m3 ** 763.5 kgCO2eq./m3 *	0 kgC/m3 [0%] 0 kgCO2eq./m3 *
Insulation material	ISOCELL	1kg of loose fill cellulose insulation (converted to 1m3 using a density of 45kg/m3)	18.3 kgC/m3 67.1 kgCO2eq./m3 *	0.783 kgC/m3 [4.3%] 2.87 kgCO2eq./m3 *
	Cyclin B.V.	1m2 of recycled newspaper-based insulation (60mm thick) with a density of 5.25 kg/m2. (converted to 1m3 multiplying by 1000/60).	40.0 kgC/m3 147 kgCO2eq./m3 *	0 kgC/m3 [0%] 0 kgCO2eq./m3 *
	Holzwerk Gebr. Schneider	1m3 of wood fiber insulation board with a representative density of 160 kg/m3.	68.59 kgC/m3 251.5 kgCO2eq./m3 *	0.23 kgC/m3 [0.3%] 0.84 kgCO2eq./m3 *
	Havnens Haender	1m2 of waste grass-based insulation with thermal resistance of 2.5 m2.K/W for 60 years (converted to 1m3 multiplying by 100/1000)	19.2 kgC/m3 70.5 kgCO2eq./m3 *	1.2 kgC/m3 [6.3%] 4.4 kgCO2eq./m3 *
	FASBA	1m3 of building straw with a density of 100kg/m3.	40.64 kgC/m3 149.0 kgCO2eq./m3 *	0 kgC/m3 [0%] 0 kgCO2eq./m3 *
	SNaB	1m3 straw bale with a density of 100kg/m3 and a moisture content of 20%.	35.25 kgC/m3 129.2 kgCO2eq./m3 *	0 kgC/m3 [0%] 0 kgCO2eq./m3 *

* *taking a value in kgC and multiplying by 3.67 to get kgCO2 (3.67 = 44/12)*

** *value for the KLH product was not directly stated in the EPD, but was calculated by the authors based on relevant information provided in the EPD, which was the moisture content (12%), carbon content by dry mass (50%), glue content (4kg glue / 470kg product, all per 1m3).*

Declared CO2 storage, GHG-associated and net CR storage at product level

An analysis of data from the same twelve EPDs is presented below, where stored biogenic carbon in the product (blue columns) is compared to GHG-associated (yellow columns). The GHG-associated encompasses all fossil and LULUC emissions from A1 to A5, along with any available biogenic emissions from A2 to A5. If A1-A3 are not separated, only biogenic emissions from A4 and A5 are included. These two columns are then combined to produce a net storage result (green columns). A negative green column indicates net carbon storage in the building's product. Note that this analysis does not account for any potential baseline.

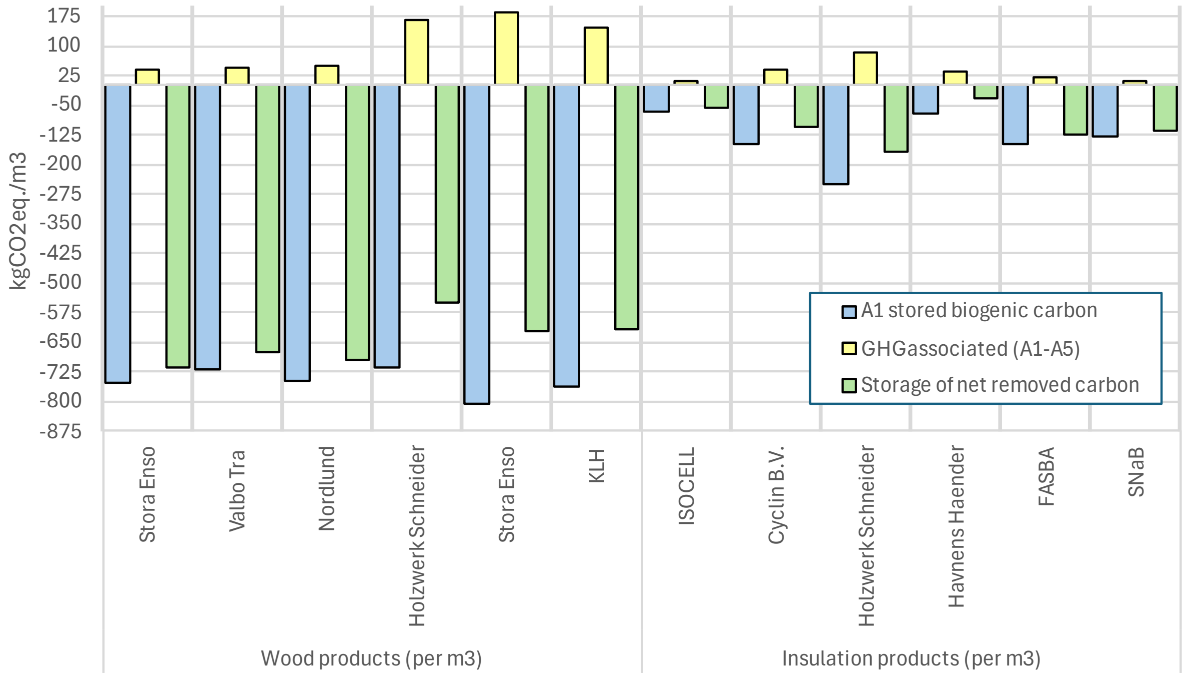


Figure 15. Comparison of declared biogenic carbon storage in EPDs (blue) with GHG-associated (orange) and how the two numbers combine (green) for 12 different construction products.

Results for biobased insulation products were originally reported in different functional units of per kg, per 1m2, and per 1m3. To facilitate comparison, results have been adjusted so that all are now expressed on a per 1m3 basis.

Looking at wood-based structural products, which are the first six products on the left side of the figure above, all six have similar carbon storage, ranging from 700 to 800 kgCO2/m3. However, there is a clear difference in GHG-associated (yellow columns) between products 1 to 3 and products 4 to 6, which then leads to differences in the net removed carbon results (green columns). Products 1 to 3 are made of planed timber and have lower GHG-associated, resulting in a relatively high net carbon storage (93 to 95% remains). Products 4 to 6, which are glulam and cross-laminated timber, exhibit significantly higher GHG-associated values due to the more intense production process, involving resins/adhesives and pressing the wood under stable curing conditions. Although still relatively high, the net carbon storage results are lower (75 to 80% remains).

On a per cubic meter basis, bio-based insulation materials store considerably less biogenic carbon than structural wood materials (i.e. 60-250 kg/m3 versus 700-800 kg/m3). This difference mainly stems from variations in the densities of these materials. The relative importance of GHG-associated was generally higher in bio-based insulation materials than in planed timber. However, the significance varied greatly among different insulation materials, with between 46% and 90% of net stored carbon remaining.

Significance of modules A4 and A5 to GHG-associated?

Of the twelve product EPDs evaluated, nine provided data on the optional modules A4 and A5. One of the CRCF methodological options is to decide whether data from modules A4 and A5 should be included in GHG-associated calculations. Therefore, in this section, data from the nine applicable EPDs are compared to assess the significance of GHG emissions in modules A4 and A5 relative to those in modules A1 to A3.

Table 26. Comparison of module A4 and A5 emissions with A1-A3 emissions

	Product name	A1-A3 values	A1-A5 values	A4+A5 values only	A4-A5 as a % of A1-A3
Wood products (per m3)	Nordlund	-706.6	-699.1	7.5	-1.1%
	Holzwerk Schneider	-554.4	-548.3	6.2	-1.1%
	Stora Enso	-726.5	-621.7	104.8	-14.4%
	KLH	-674.6	-616.1	58.5	-8.7%
Insulation products (per m3)	ISOCELL	-63.9	-58.2	5.7	-8.9%
	Cyclin B.V.	-59.6	-53.3	6.3	-10.6%
	Holzwerk Schneider	-173.4	-168.8	4.5	-2.6%
	Havnens Haender	-49.7	-37.0	12.7	-25.6%
	FASBA	-129.1	-125.3	3.8	-3.0%

Emissions from modules A4 and A5 in three of the nine products shown in the figure above affect the A1-A3 results by 10% or more. A further two products show an A4 and A5 effect of almost 9%. In the remaining products, the contribution of A4 and A5 can be considered as negligible (3.0% or less).

Two conclusions can be drawn from this limited analysis. First, the potential significance of modules A4 and A5 justifies including these modules in calculations of GHG-associated. Second, the considerable variability in the contributions of modules A4 and A5 highlights the need for clear and detailed rules regarding assumptions in these modules.

Appendix IV: Examples of building typologies

As referred to in section 2.4, some examples of building typologies are provided here in the table below.

Table 27. Different categories of building typology according to two different sources

Building categories according to CPV codes[23]	Building categories according to EN ISO 52000 series[24]
44211000-2. Prefabricated buildings	
44211100-3. Modular and portable buildings	Mobile home [BLDNGCAT_RES_MOBIL] (repeated)
45211100-0. Construction work for houses (codes for sub-types: sheltered housing; houses) 45211340-4. Multi-dwelling buildings construction work (code for sub-type: flats) 45211350-7. Multi-functional buildings construction work	**Residential buildings:** Single family houses of different types [BLDNGCAT_RES_SINGLE] Apartment block [BLDNGCAT_RES_APPBLOCK] Homes for elderly and disabled people [BLDNGCAT_RES_ELDER] Residence for collective use [BLDNGCAT_RES_COLL] Mobile home [BLDNGCAT_RES_MOBIL] (repeated) Holiday home [BLDNGCAT_RES_HOL]
45212000-6. Construction work for buildings relating to leisure, sports, culture, lodging and restaurants (various codes for different sub-types therein)	**Non-residential buildings:** Hotels and restaurants [BLDNGCAT_HOTEL]
45212200-8. Construction work for sports facilities (codes for sub-types: ”single-purpose” and ”multi-purpose” and various example like ice-rink, swimming pool, gynnasium, stadium etc.	Sport facilities [BLDNGCAT_SPORT]
45212300-9. Construction work for art and cultural buildings (codes for various sub-types: art gallerys; exhibition centres; museums; historical monuments; memorials)	
45212320-5. Construction work for buildings relating to artistic performances (codes for various sub-types: auditoriums; theatres; libraries).	
45212340-1. Lecture hall construction work	
45212360-7. Religious buildings construction work	
45212400-0. Accommodation and restaurant buildings (codes for sub-types: lodging buildings; hotels; hostels; short-stay accommodation; restaurants; canteens; cafeterias)	
45213100-4. Construction work for commercial buildings (codes for sub-types: shops; shopping centre; ship units; post office; bank; market; office block)	Office buildings [BLDNGCAT_OFF] Wholesale and retail trade services buildings [BLDNGCAT_RETAIL]
45213200-5. Construction work for warehouses and industrial buildings (codes for sub-types: cold storage; warehouses; warehouse store; abattoir)	Wholesale and retail trade services buildings [BLDNGCAT_RETAIL] (repeated)
45213240-7. Agricultural buildings construction work (codes for sub-types: barns; cowsheds)	Non-residential agricultural buildings [BLDNGCAT_AGRIC]
45213250-0. Construction work for industrial buildings (codes for sub-types: industrial units; workshops)	Industrial sites [BLDNGCAT_INDUS] Workshops [BLDNGCAT_WORKS]
45213260-3. Stores depot construction work	
45213270-6. Construction works for recycling station	
45213280-9. Construction works for compost facility	
45213300-6. Buildings associated with transport (codes for sub-types like bus stations; car parks; service areas; bus garage; railway station; rail terminal building; airports; airport control towers; ferry terminals	
45214000-0. Construction work for buildings relating to education and research (codes for sub.types:	Educational buildings [BLDNGCAT_EDUC]

[23] As defined in Regulation (EC) No 213/2008 on the Common Procurement Vocabulary (CPV). Available here.
[24] Taken from Table B3 in EN ISO 52000-1:2017

Building categories according to CPV codes[23]	Building categories according to EN ISO 52000 series[24]
kindergartens; schools; colleges; universities; lecture theatres	
45214600-6. Construction work for research buildings (codes for sub-types: laboratories; research & testing facilities; scientific installations; cleanrooms; meterological stations)	
45214700-7. Construction work for halls of residence	
45214800-8. Training facilities building	
45215000-7. Construction work for buildings relating to health and social services, for crematoriums and public conveniences (codes for sub-types: spas; clinics; hospitals; social services buildings; subsidised residential accommodation; retirement homes; nursing homes etc.)	Hospitals [BLDNGCAT_HOSP]
45215300-0. Construction work for crematoriums	
45216000-4. Construction work for buildings relating to law and order or emergency services and for military buildings (codes for various sub-types)	

Appendix V: Default EU standardised baseline

This section presents how the information from the DG GROW study on life cycle GHG emissions from the EU building sector has been used to create a default EU standardised baseline. The data in the report and the first layer of underlying data was not sufficiently detailed to be able to estimate baselines. Any assumptions and gap filling necessary are also explained here in this section.

Report with quantitative baseline figures for whole life carbon and stored biogenic carbon removals

Section 3.1.3.2 of the DG GROW study[25] presents the quantification of baseline CO_2 removal fluxes, disaggregated into biobased and mineral-based removals at the archetype level. The quantification is based on carbon content and moisture content data for each biobased material considered, primarily sourced from the Ecoinvent database (v 3.9.1) and robust proxy datasets from TU Graz. These values serve as inputs to the EN 16449 formula used to convert carbon content of the biobased material into CO_2. Importantly, the quantification excludes packaging materials. Results are reported across the following dimensions:

- Regions: Nordic (NOR), Mediterranean (MED), Continental (CON), Oceanic (OCE)
- Building subtypes: single-family homes (SFH), multi-family homes (MFH), office buildings (OFF)
- Energy performance classes: standard (STD), advanced (ADV)
- Project types: existing buildings (EXB), refurbishments (REF), new buildings (NEW)
- Life cycle phases: before use (A), use (B), after use (C)

While the reported data provides useful insights, there are methodological constraints that limit its direct use as a CRCF baseline. First, the aggregated fluxes include products such as boards, flooring, window frames, and doors. Since these are outside the CRCF scope, their inclusion will inflate biogenic carbon storage values and create a misalignment with the CRCF's focus on long-lived structural and insulation materials. Furthermore, for CRCF purposes and following the proposed approach in this report, separate flux values for structural wood and for biobased insulation products are needed. To improve alignment with CRCF methodological requirements, the proposed regional baseline values in the DG GROW study should either:

- exclude non-qualifying product categories (e.g. cladding) from the biogenic carbon removal values in a future iteration, or
- apply a default correction factor, subtracting an estimated share of biogenic carbon attributable to out-of-scope products. This correction can be derived from national representative LCA datasets, where available.

Furthermore, even if excluding the non-relevant product categories, to meet the proposed approach in the present report for separate baselines for structural and insulation biogenic removals, further disaggregation would be necessary. In the absence of detailed material-function mapping, a conservative interim assumption is proposed: for the first iteration of the baseline, the biogenic insulation share may be based on current market data or be assumed to be negligible (≈ 0 %) based on the likely dominance of mineral-based insulation in the present market. These are simplifications intended as temporary assumptions and should be revisited in future baseline updates as more granular product-level data becomes available.

Scenario explorer

The DG GROW reports are accompanied by a Whole Life Carbon Scenario Explorer developed by KU Leuven, an interactive tool that enables users to explore various whole life carbon scenarios, including the "business-as-usual" (BAU) scenario, which can serve as the baseline reference for CRCF specification. Within the dashboard, users can access more granular data than what is available in the published reports. The tool allows filtering by multiple dimensions, including:

- **indicator**: GWP-total, GWP-fossil, GWP-biogenic. GWP-luluc and material mass
- **geographical scope**: the four EU climatic regions and each EU-27 Member State

[25] Röck et al. (2023). Analysis of Life Cycle Greenhouse Gas Emissions of Buildings and Construction: D2.1 – Report with Quantitative Baseline Figures for Whole Life Carbon and Carbon Removals. Recipient: European Commissions (DG GROW). Available at: https://ec.europa.eu/docsroom/documents/58196

- **building type and subtype**: residential and non-residential categories, further divided into nine subtypes (e.g. single-family house, multi-family house, education, office, etc.)
- **element class**: including but not limited to structural elements such as: substructure, storey floors, external walls, internal walls, external openings, and internal openings
- **material class**: including the two key classes "wood" and "insulation" (note: insulation is not further divided into biobased and conventional insulation)
- **building stock activity**: existing, new, refurbishment
- **life cycle stages**: construction EC and renovation EC are the relevant ones for CRCF
- **life cycle modules**: with A1-3 values aggregated

Although the Scenario Explorer provides a significantly more detailed view than the summary reports, some methodological limitations remain: inflated values due to the aggregated fluxes including products such as boards, flooring, window frames, and doors, which are outside the CRCF scope. It is important to note that these values do not distinguish between structural and non-structural wood products, such as flooring or wood cladding, due to the granularity of the underlying data.

After discussions with the experts involved in gathering the underlying data, it is important to note that the quality of underlying data varied significantly between different Member States. We have therefore selected three countries representing which had good quality underlying data, which represent different parts of Europe, and which highlight the heterogeneity that can exist between Member States. These three options are:

- a central European case, Germany (DE),
- a Southern European case, Italy (IT), and
- a Northern European case, Sweden (SWE).

The wood mass is shown for six relevant element categories and aggregated under "all elements". Table 28 presents the wood mass per floor area (kg/m²) in A1–A3 life-cycle modules for new constructions under the business-as-usual (BAU) scenario for 2020-2025. It covers three building types—single-family houses (SFH), multi-family houses (MFH), and office buildings (OFF).

Table 28: wood mass per useful floor area (kg/m²) in six different building elements, in all elements aggregated, and when correction is applied to only include the structural wood mass (last column).

Country	Building type	Building element and normalised masses (kg/m2)						All wood elements (kg/m2)	CRCF-eligible wood mass (based on assumptions in Table 29)
		Floor storeys (FS)	External walls (EW)	Common walls (CW)	Roofs (RO)	Stair-cases (SC)	Sub-structure (SS)		
DE	SFH	0.0	0	0.0	9.5	1.0	0.0	10.5	8.0
	MFH	0.0	0.0	0.0	0.0	0.0	0.0	0	0
	OFF	0.0	0.0	0.0	0.0	0.0	0.0	0	0
IT	SFH	0.0	0.0	0.0	1.4	1.5	0.0	2.9	0
	MFH	0.0	0.0	0.0	0.9	0.0	0.0	0.9	0
	OFF	0.0	0.0	0.0	0.0	0.0	0.0	0	0
SE	SFH	0.00	26.4[26]	0.0	17.6	0.8	14.2	85.4	12.0 (EW) + 17.6 (RO) ≈ 30.0
	MFH	9.2	9.3[19]	0.0	5.9	0.0	4.7	38.3	6.1 (EW) + 5.9 (RO) = 12.0
	OFF	0.0	0.0	0.0	0.0	0.0	0.0	0	0

The table indicates that for office buildings, and by extension most non-residential buildings, an initial zero value can be assumed for the standardized baseline. For Southern European countries, this baseline can also apply to residential buildings, as very low amounts of wood are observed in roof and staircase elements. Given that concrete

[26] It should be noted that the corresponding value shown in the Scenario Explorer is twice as high, which appears to result from a system error

values for these elements range from 25–45 kg/m², it is reasonable to assume that any wood present is non-structural.

For the three remaining cases—SFH in central and northern Europe, and MFH in northern Europe—countries may either:

- Use the “all elements” values as a very conservative baseline, or
- Apply default correction factors to subtract estimated shares of wood mass attributable to out-of-scope products.

Such corrections can be derived from nationally representative archetypes used for LCA or other relevant studies where available. In the absence of these national correction factors, rules of thumb can be applied, as shown in Table V.2 for roof and external wall elements. For storey floors, staircases, and substructure, the proportion of wood relative to typically structural materials (concrete, brick) is low and can generally be considered non-structural.

Table 29: Default assumptions used in this report to distinguish between the structural and non-structural wood mass in the relevant elements of the archetypes presented in Table 28

<table>
<tr><th>Country</th><th>Building type</th><th>Element</th><th>Correction factor for % share of structural part per element</th></tr>
<tr><td>DE</td><td>SFH</td><td>RO</td><td>For estimating the structural wood content in typical pitched roofs, the following assumptions are used:

Rafters / Trusses
• A standard rafter cross-section of 0.20 m × 0.045 m is assumed.
• With rafters spaced at 600 mm = roughly 1.5 rafters per m^2 of roof area.
• This equates to approximately 0.0135 m³ of timber per m² of roof section, or <u>6.8 kg/m²</u> (based on a timber density of 400 kg/m³).

Battens
• Battens are assumed to measure 0.038 m × 0.025 m each.
• At a spacing of 400 mm, this results in about 2.5 battens per m².
• This equals around 0.0024 m³ per m², or <u>1.0 kg/m²</u> of wood (based on a timber density of 400 kg/m³).

Total Structural Wood Mass (ratio)
• Assumption: 1.02 m² flat roof / 1 m² useful floor (single storey)
• 6.8 + 1.0 = 7.8 kg structural wood / 1 m² flat roof
The structural wood per m² useful floor is: 7.8 x 1.02 = 8.0 kg/m^2</td></tr>
<tr><td rowspan="2">SE</td><td rowspan="2">SFH</td><td>RO</td><td>For estimating the structural wood content in typical pitched roofs, the following assumptions are used:

Rafters / Trusses
• A standard rafter cross-section of 0.20 m × 0.045 m is assumed.
• With rafters spaced at 600 mm = roughly 1.5 rafters per m^2 of roof area.
• This equates to approximately 0.0135 m³ of timber per m² of roof section, or <u>6.8 kg/m²</u> (based on a timber density of 400 kg/m³).

Battens
• Battens are assumed to measure 0.038 m × 0.025 m each.
• At a spacing of 400 mm, this results in about 2.5 battens per m².
• This equals around 0.0024 m³ per m², or <u>1.0 kg/m²</u> of wood (based on a timber density of 400 kg/m³).

Roof Boarding
It is assumed that a roof boarding (e.g., OSB, plywood, or timber planks) is used, of 15 mm thickness, at 600 kg/m³ density = approx. <u>9 kg/m²</u>.

Total Structural Wood Mass (ratio)
• Assumption: 1.1 m² pitched roof / 1 m² useful floor ((single storey, 120 m^2 useful floor area, 25° roof inclination)
• 6.8 + 1.0 + 9 = 16.8 kg structural wood / 1 m² pitched roof
• The structural wood per m² useful floor is: 16.8 x 1.1 = 18.5 kg/m^2 (since larger than the entire wood mass in DG GROW archetype, it can be assumed the full amount given in the Explorer is structural in this case)</td></tr>
<tr><td>EW</td><td>For estimating the structural wood content in typical external wall sections, the following assumptions are used:</td></tr>
</table>

		Studs: • Stud cross-section default: 45 × 145 mm (0.045 × 0.145 m). • Stud spacing default: 600 mm (0.6 m) = roughly 1.5 stud per m² of wall area. • This equates to 0,0098 m³ of wood per m² of wall section, or <u>3.9 kg/m²</u> (based on a timber density of 400 kg/m³). **Plates:** • Typical Swedish timber-frame dimensions: 45 × 45 mm plates • One top and one bottom plate (two in total) • Standard storey height: 2.4 m • Wood density: ~400 kg/m³ (softwood) • Because a 2.4-metre-high wall contains 0.42 m of plate length per m² of wall area (1 m² ÷ 2.4 m height), the combined volume of the two plates is ≈0.0017 m³ of timber per m² of wall ≈ <u>0.7 kg/m²</u> structural wood **OSB boarding:** 15 mm thickness, at 600 kg/m³ density = approx. <u>9 kg/m²</u>. **Total Structural Wood Mass (ratio)** • Assumption: 0.88 m² external wall / 1 m² useful floor (single storey, 120 m² useful floor area) • 3.9 + 0.7 + 9 = 13.6 kg structural wood / 1 m² external wall • The structural wood per m² useful floor is: 13.6 x 0.88 = **12.0** kg/m²
	SC + SS	Very low amount of wood mass compared to the concrete amount strongly signals that it is non-structural in both elements, e.g. for substructure: having 671 kg/m² concrete vs. 14 kg/m² wood makes the wood ≈2% of the mass
MFH	RO	See assumptions for SFH in terms of rafters/trusses, battens and roof boarding. **Total Structural Wood Mass (ratio)** • Assumption: 0.35 m² pitched roof / 1 m² useful floor (3 storey building) • 6.8 + 1.0 + 9 = 16.8 kg structural wood / 1 m² pitched roof The structural wood per m² useful floor is: 16.8 x 0.35 = **5.9** kg/m²
	EW	See assumptions for SFH in terms of studs, plates and OSB boarding. **Total Structural Wood Mass (ratio)** • Assumption: 0.45 m² external wall / 1 m² useful floor (3 storey building) • 3.9 + 0.7 + 9 = 13.6 kg structural wood / 1 m² external wall • The structural wood per m² useful floor is: 13.6 x 0.45 = **6.1** kg/m²
	FS + SS	Very low amount of wood mass compared to the concrete amount strongly signals that it is non-structural in both elements, e.g. for floor storeys 9 kg/m² is assumed to be for flooring

The detailed quantities of different types of materials in the archetypes are provided for information below:

Table 30. Quantities of main material types in the three Italian archetypes kg/m2 of useful floor area

IT_SFH	Aluminum	Brick	Concrete	Glass	Plastic	Steel	Wood	Total
External openings	0.0	0.0	0.0	2.4	0.0	0.0	3.7	6.1
External walls	0.0	359.0	113.3	0.0	0.0	0.1	0.0	472.5
Internal openings	0.0	0.0	0.0	0.0	0.0	0.0	2.9	2.9
Internal walls	0.0	95.8	15.7	0.0	0.0	0.0	0.0	111.5
Roofs	0.0	88.3	44.1	0.0	1.3	3.8	1.4	139.0
Staircases	0.0	0.0	26.6	0.0	0.1	2.3	1.5	30.6
Storey floors	0.0	39.1	35.5	0.0	1.1	1.6	0.0	77.4
Substructure	0.0	0.0	923.5	0.1	19.4	24.2	0.0	967.2
Technical services	0.0	0.0	9.7	0.0	1.1	1.2	0.0	12.1
Total	**0.0**	**582.3**	**1168.6**	**2.5**	**23.2**	**33.2**	**9.6**	**1819.4**
IT_MFH	**Aluminum**	**Brick**	**Concrete**	**Glass**	**Plastic**	**Steel**	**Wood**	**Total**
External openings	0.0	0.0	0.0	1.9	0.0	0.0	3.2	5.2
External walls	0.0	174.5	50.2	0.2	0.0	0.1	0.0	225.0
Internal openings	0.0	0.0	0.0	0.0	0.0	0.0	3.3	3.3
Internal walls	0.0	107.5	19.9	0.0	0.0	0.0	0.0	127.4

Roofs	0.0	58.9	29.4	0.0	0.9	2.5	0.9	92.7
Staircases	0.0	0.0	14.9	0.0	0.0	1.8	0.0	16.8
Storey floors	0.0	101.2	73.1	0.0	1.6	4.3	0.0	180.1
Substructure	0.0	0.0	539.6	0.0	9.9	13.8	0.0	563.3
Technical services	0.0	0.0	0.0	0.0	0.0	1.2	0.0	1.2
Total	**0.0**	**442.2**	**727.0**	**2.2**	**12.5**	**23.6**	**7.4**	**1214.9**
IT_OFF	**Aluminum**	**Brick**	**Concrete**	**Glass**	**Plastic**	**Steel**	**Wood**	**Total**
External openings	1.1	0.0	0.0	3.3	0.0	0.0	0.0	5.2
External walls	0.0	224.6	70.9	0.0	0.0	0.1	0.0	225.0
Internal openings	0.0	0.0	0.0	0.1	0.0	0.0	1.5	3.3
Internal walls	0.0	32.4	5.9	0.5	0.0	0.9	0.0	127.4
Roofs	0.0	1.3	1,7	0.0	0.9	0.1	0.0	92.7
Storey floors	0.0	133.5	96.2	0.0	1.6	5.6	0.0	180.1
Substructure	0.0	0.0	301.8	0.0	9.9	4.8	0.0	563.3
Technical services	0.1	0.0	2.5	3.3	0.0	19.6	0.0	1.2
Total	**1.2**	**391.8**	**479.1**	**3.9**	**9.2**	**31.1**	**1.5**	**1214.9**

Table 31. Quantities of main material types in the three German archetypes kg/m2 of useful floor area

DE_SFH	Aluminum	Brick	Concrete	Glass	Plastic	Steel	Wood	Total
External openings	0.0	0.0	0.0	3.7	7.3	0.0	0.0	11.0
External walls	0.0	158.6	45.7	0.2	0.1	0.1	0.0	204.7
Internal openings	0.0	0.0	0.0	0.0	0.0	0.0	2.9	2.9
Internal walls	0.0	44.3	8.0	0.0	0.0	0.0	1.5	54.0
Roofs	0.0	0.0	0.0	0.0	0.4	0.0	4.8	5.2
Staircases	0.0	0.0	17.5	0.0	0.1	1.5	1.0	20.1
Storey floors	0.0	34.0	30.8	0.0	1.0	1.4	0.0	67.2
Substructure	0.0	0.0	372.0	0.1	4.6	11.0	0.0	387.4
Technical services	0.0	0.0	9.7	0.0	1.1	0,6	0.0	11.5
Total	**0.0**	**582.3**	**1168.6**	**2.5**	**23.2**	**33.2**	**9.6**	**1819.4**
DE_MFH	**Aluminum**	**Brick**	**Concrete**	**Glass**	**Plastic**	**Steel**	**Wood**	**Total**
External openings	0.0	0.0	0.0	2.9	5.8	0.0	0.0	8.7
External walls	0.0	120.3	34.7	0.2	0,0	0,0	0.0	155.3
Internal openings	0.0	0.0	0.0	0.0	0,0	0.0	3.3	3.3
Internal walls	0.0	107.5	19.9	0.0	0.0	0.0	0.0	127.4
Roofs	0.0	0.0	155.4	0.0	2,9	0.0	0.0	158.3
Staircases	0.0	0.0	7.6	0.0	0,0	0,9	0.0	8.6
Storey floors	0.0	117.9	84.9	0.0	0.0	4,9	0.0	207.8
Substructure	0.0	0.0	283.1	0.0	1.9	11.2	0.0	296.3
Technical services	0.0	0.0	0.0	0.0	2,2	1,3	0.0	3.5
Total	**0.0**	**345.8**	**585.6**	**3.1**	**12.9**	**18.4**	**3.3**	**969.1**
DE_OFF	**Aluminum**	**Brick**	**Concrete**	**Glass**	**Plastic**	**Steel**	**Wood**	**Total**
External openings	2.2	0.0	0.0	3.4	0.0	0.0	0.0	5.6
External walls	0.0	285.1	0.0	0.1	0.0	10.3	0.0	295.5
Internal openings	0.0	0.0	0.0	0.1	0.0	0.0	1.5	1.6
Internal walls	0.0	47.9	0.0	0.5	0.0	1.0	0.0	49.4

Roofs	0.0	86.9	0.0	0.0	0.5	0.0	0.0	87.4
Storey floors	0.0	505.1	0.0	0.0	2.8	0.0	0.0	507.9
Substructure	0.0	587.7	0.0	0.0	4.3	12.7	0.0	604.7
Technical services	0.1	2.5	0.0	0.0	2.6	19.6	0.0	24.8
Total	**2.3**	**1515.1**	**0.0**	**4.1**	**10.2**	**43.6**	**1.5**	**1576.8**

Table 32. Quantities of main material types in the three Swedish archetypes kg/m2 of useful floor area

SE_SFH	Aluminum	Brick	Concrete	Glass	Plastic	Steel	Wood	Total
External openings	0.0	0.0	0.0	3.2	0.0	0.0	5.4	8.6
External walls	0.0	77.3	22.3	0.1	0.3	0.2	26.4	126.6
Internal openings	0.0	0.0	0.0	0.0	0.0	0.0	3.0	3.0
Internal walls	0.0	14.8	2.7	0.0	0.0	0.0	1.7	19.2
Roofs	0.0	0.0	0.0	0.0	0.9	0.4	18.0	19.3
Staircases	0.0	0.0	14.6	0.0	0.0	1.2	0.8	16.7
Substructure	0.0	0.0	617.5	0.0	4.9	17.9	14.2	654.5
Technical services	0.0	0.0	0.0	0.0	1.1	0.3	0.0	1.4
Total	**0.0**	**92.1**	**657.0**	**3.4**	**7.3**	**20.0**	**69.7**	**849.5**
SE_MFH	**Aluminum**	**Brick**	**Concrete**	**Glass**	**Plastic**	**Steel**	**Wood**	**Total**
External openings	0.0	0.0	0.0	2.1	0.0	0.0	3.7	5.8
External walls	0.0	27.2	7.8	0.0	0.1	0.1	9.3	44.5
Internal openings	0.0	0.0	0.0	0.0	0.0	0.0	3.3	3.3
Internal walls	0.0	35.8	6.6	0.0	0.0	0.0	0.7	43.2
Roofs	0.0	0.0	0.0	0.0	0.3	0.1	6.2	6.7
Staircases	0.0	0.0	5.1	0.0	0.0	0.6	0.0	5.8
Storey floors	0.0	32.6	115.9	0.0	0.4	1.4	9.2	159.5
Substructure	0.0	0.0	388.3	0.0	1.6	15.7	4.7	410.3
Technical services	0.0	0.0	0.0	0.0	0.1	6.6	0.0	6.7
Total	**0.0**	**95.6**	**523.8**	**2.2**	**2.7**	**24.5**	37.0	685.8
SE_OFF	**Aluminum**	**Brick**	**Concrete**	**Glass**	**Plastic**	**Steel**	**Wood**	**Total**
External openings	0.6	0.0	0.0	1.9	1.2	0.0	0.0	3.7
External walls	0.0	378.1	108.6	0.0	0.0	17.7	0.0	504.4
Internal openings	0.0	0.0	0.0	0.1	0.0	0.0	1.8	1.8
Internal walls	0.0	32.4	5.9	0.5	0.0	1.0	0.0	39.7
Roofs	0.0	3.5	4.6	0.0	0.5	0.1	0.0	8.8
Staircases	0.0	0.0	56.1	0.0	0.3	6.8	0.0	63.2
Storey floors	0.0	133.3	96.1	0.0	2.8	5.6	0.0	237.9
Substructure	0.0	0.0	321.4	0.0	1.1	6.9	0.0	329.4
Technical services	0.0	0.0	2.5	0.0	2.6	19.6	0.0	24.8
Total	**0.74**	**547.3**	**595.3**	**2.6**	**8.5**	**57.7**	**1.8**	**1213.9**

Appendix VI: DNSH requirements for new and renovated buildings in the EU Taxonomy

The DNSH requirements associated with the construction and renovation of buildings are reproduced below together with references to their sources. The use of the sign * indicates a footnote, but due to the extensive nature of the footnotes, they are not included here. Instead, readers should refer to the original legal texts here and here.

Table 33. EU Taxonomy DNSH requirements for buildings

Environmental goals and legal references	Requirement
Climate Change Mitigation In Annex II to Regulation (EU) 2021/2139: Activity 7.1: Construction of new buildings Activity 7.2: renovation of existing buildings Annex II to Regulation (EU) 2023/2486: Activity 3.1: Construction of new buildings Activity 3.2: renovation of existing buildings	The building is not dedicated to extraction, storage, transport or manufacture of fossil fuels. The Primary Energy Demand (PED)* setting out the energy performance of the building resulting from the construction does not exceed the threshold set for the nearly zero-energy building (NZEB) requirements in national regulation implementing Directive 2010/31/EU of the European Parliament and of the Council*. The energy performance is certified using an as built Energy Performance Certificate (EPC). [the part in yellow is DNSH for new construction only and does not apply to renovation activities)
Climate Change Adaptation In Annex I to Regulation (EU) 2021/2139): Activity 7.1: new construction of buildings, and Activity 7.2: renovation of existing buildings. In Annex II to Regulation (EU) 2023/2486: Activity 3.1: new construction of buildings. Activity 3.2: renovation of existing buildings.	The activity complies with the criteria set out in Appendix A to this Annex. **Appendix A.** *The physical climate risks that are material to the activity have been identified from those listed in the table in Section II of this Appendix by performing a robust climate risk and vulnerability assessment with the following steps:* *(a) screening of the activity to identify which physical climate risks from the list in Section II of this Appendix may affect the performance of the economic activity during its expected lifetime;* *(b) where the activity is assessed to be at risk from one or more of the physical climate risks listed in Section II of this Appendix, a climate risk and vulnerability assessment to assess the materiality of the physical climate risks on the economic activity;* *(c) an assessment of adaptation solutions that can reduce the identified physical climate risk.* *The climate risk and vulnerability assessment is proportionate to the scale of the activity and its expected lifespan, such that:* *(a) for activities with an expected lifespan of less than 10 years, the assessment is performed, at least by using climate projections at the smallest appropriate scale;* *(b) for all other activities, the assessment is performed using the highest available resolution, state-of-the-art climate projections across the existing range of future scenarios(1) consistent with the expected lifetime of the activity, including, at least, 10 to 30 year climate projections scenarios for major investments.* *The climate projections and assessment of impacts are based on best practice and available guidance and take into account the state-of-the-art science for vulnerability and risk analysis and related methodologies in line with the most recent Intergovernmental Panel on Climate Change reports(2), scientific peer-reviewed publications, and open source(3) or paying models.*

Environmental goals and legal references	Requirement
	For existing activities and new activities using existing physical assets, the economic operator implements physical and non-physical solutions ('adaptation solutions'), over a period of time of up to five years, that reduce the most important identified physical climate risks that are material to that activity. An adaptation plan for the implementation of those solutions is drawn up accordingly. *For new activities and existing activities using newly-built physical assets, the economic operator integrates the adaptation solutions that reduce the most important identified physical climate risks that are material to that activity at the time of design and construction and has implemented them before the start of operations.* *The adaptation solutions implemented do not adversely affect the adaptation efforts or the level of resilience to physical climate risks of other people, of nature, of cultural heritage, of assets and of other economic activities; are consistent with local, sectoral, regional or national adaptation strategies and plans; and consider the use of nature- based solutions (4) or rely on blue or green infrastructure (5) to the extent possible.*

	Temperature-related	Wind-related	Water-related	Solid-mass related
Chronic	Changing temperature (air, freshwater, marine water)	Changing wind patterns	Changing precipitation patterns and types (rain, hail, snow/ice)	Coastal erosion
	Heat stress		Precipitation or hydrological variability	Soil degradation
	Temperature variability		Ocean acidification	Soil erosion
	Permafrost thawing		Saline intrusion	Solifluction
			Sea level rise	
			Water stress	
Acute	Heat wave	Cyclone, hurricane, typhoon	Drought	Avalanche
	Cold wave/frost	Storm (including blizzards, dust and sandstorms)	Heavy precipitation (rain, hail, snow/ice)	Landslide
	Wildfire	Tornado	Flood (coastal, fluvial, pluvial, ground water)	Subsidence
			Glacial lake outburst	

Sustainable use and protection of water and marine resources	Where installed, except for installations in residential building units, the specified water use for the following water appliances are attested by product datasheets, a building certification or an existing product label in the Union, in accordance with the technical specifications laid down in Appendix E to this Annex:

Environmental goals and legal references	Requirement
Annex I to Regulation (EU) 2021/2139): Activity 7.1: new construction of buildings, Activity 7.2: renovation of existing buildings (part in yellow not included). In Annex II to Regulation (EU) 2023/2486: Activity 3.1: new construction of buildings Activity 3.2: renovation of existing buildings (part in yellow not included).	(a) wash hand basin taps and kitchen taps have a maximum water flow of 6 litres/min; (b) showers have a maximum water flow of 8 litres/min; (c) WCs, including suites, bowls and flushing cisterns, have a full flush volume of a maximum of 6 litres and a maximum average flush volume of 3,5 litres; (d) urinals use a maximum of 2 litres/bowl/hour. Flushing urinals have a maximum full flush volume of 1 litre. To avoid impact from the construction site, the activity complies with the criteria set out in Appendix B to this Annex. **Appendix E / Appendix B**: *Environmental degradation risks related to preserving water quality and avoiding water stress are identified and addressed with the aim of achieving good water status and good ecological potential as defined in Article 2, points (22) and (23), of Regulation (EU) 2020/852, in accordance with Directive 2000/60/EC of the European Parliament and of the Council (1) and a water use and protection management plan, developed thereunder for the potentially affected water body or bodies, in consultation with relevant stakeholders.* *Where an Environmental Impact Assessment is carried out in accordance with Directive 2011/92/EU of the European Parliament and of the Council (2) and includes an assessment of the impact on water in accordance with Directive 2000/60/EC, no additional assessment of impact on water is required, provided the risks identified have been addressed.*
Transition to a circular economy Annex I to Regulation (EU) 2021/2139): Activity 7.1: new construction of buildings. Activity 7.2: renovation of existing buildings.	At least 70 % (by weight) of the non-hazardous construction and demolition waste (excluding naturally occurring material referred to in category 17 05 04 in the European List of Waste established by Decision 2000/532/EC) generated on the construction site is prepared for reuse, recycling and other material recovery, including backfilling operations using waste to substitute other materials, in accordance with the waste hierarchy and the EU Construction and Demolition Waste Management Protocol (287). Operators limit waste generation in processes related to construction and demolition, in accordance with the EU Construction and Demolition Waste Management Protocol and taking into account best available techniques and using selective demolition to enable removal and safe handling of hazardous substances and facilitate reuse and high-quality recycling by selective removal of materials, using available sorting systems for construction and demolition waste. Building designs and construction techniques support circularity and in particular demonstrate, with reference to ISO 20887 (288) or other standards for assessing the disassembly or adaptability of buildings, how they are designed to be more resource efficient, adaptable, flexible and dismantleable to enable reuse and recycling.
Pollution prevention and control Annex I to Regulation (EU) 2021/2139): Activity 7.1 new construction of buildings. Activity 7.2: renovation of existing buildings. In Annex II to Regulation (EU) 2023/2486: Activity 3.1: new construction of buildings. Activity 3.2: renovation of existing buildings.	Building components and materials used in the construction comply with the criteria set out in Appendix C to this Annex. **Appendix C:** *The activity does not lead to the manufacture, placing on the market or use of:* *(a) substances, whether on their own, in mixtures or in articles, listed in Annexes I or II to Regulation (EU) 2019/1021 of the European Parliament and of the Council (1), except in the case of substances present as an unintentional trace contaminant;* *(b) mercury and mercury compounds, their mixtures and mercury-added products as defined in Article 2 of Regulation (EU) 2017/852 of the European Parliament and of the Council (2);* *(c) substances, whether on their own, in mixture or in articles, listed in Annexes I or II to Regulation (EC) No 1005/2009 of the European Parliament and of the Council (3);* *(d) substances, whether on their own, in mixtures or in an articles, listed in Annex II to Directive 2011/65/EU of the European Parliament and of the Council (4), except where there is full compliance with Article 4(1) of that Directive;*

Environmental goals and legal references	Requirement
	(e) substances, whether on their own, in mixtures or in an article, listed in Annex XVII to Regulation (EC) 1907/2006 of the European Parliament and of the Council (5), except where there is full compliance with the conditions specified in that Annex; *(f) substances, whether on their own, in mixtures or in an article, meeting the criteria laid down in Article 57 of Regulation (EC) 1907/2006 and identified in accordance with Article 59(1) of that Regulation, except where their use has been proven to be essential for the society;* *(g) other substances, whether on their own, in mixtures or in an article, that meet the criteria laid down in Article 57 of Regulation (EC) 1907/2006, except where their use has been proven to be essential for the society.* Building components and materials used in the construction that may come into contact with occupiers (289) emit less than 0,06 mg of formaldehyde per m 3 of material or component upon testing in accordance with the conditions specified in Annex XVII to Regulation (EC) No 1907/2006 and less than 0,001 mg of other categories 1A and 1B carcinogenic volatile organic compounds per m 3 of material or component, upon testing in accordance with CEN/EN 16516 (290) or ISO 16000-3:2011 (291) or other equivalent standardised test conditions and determination methods (292). Where the new construction is located on a potentially contaminated site (brownfield site), the site has been subject to an investigation for potential contaminants, for example using standard ISO 18400 (293). Measures are taken to reduce noise, dust and pollutant emissions during construction or maintenance works.
Protection and restoration of biodiversity and ecosystems Annex I to Regulation (EU) 2021/2139): Activity 7.1: new construction of buildings. In Annex II to Regulation (EU) 2023/2486: Activity 3.1: new construction of buildings	The activity complies with the criteria set out in Appendix D to this Annex. **Appendix D**: *An Environmental Impact Assessment (EIA) or screening (1) has been completed in accordance with Directive 2011/92/EU (2).* *Where an EIA has been carried out, the required mitigation and compensation measures for protecting the environment are implemented.* *For sites/operations located in or near biodiversity-sensitive areas (including the Natura 2000 network of protected areas, UNESCO World Heritage sites and Key Biodiversity Areas, as well as other protected areas), an appropriate assessment (3), where applicable, has been conducted and based on its conclusions the necessary mitigation measures (4) are implemented.* The new construction is not built on one of the following: (a) arable land and crop land with a moderate to high level of soil fertility and below ground biodiversity as referred to the EU LUCAS survey (294); (b) greenfield land of recognised high biodiversity value and land that serves as habitat of endangered species (flora and fauna) listed on the European Red List (295) or the IUCN Red List (296); (c) land matching the definition of forest as set out in national law used in the national greenhouse gas inventory, or where not available, is in accordance with the FAO definition of forest (297).

Appendix VII: Technical Screening Criteria for new and renovated buildings in the EU Taxonomy

The technical screening criteria for the construction and renovation of buildings are reproduced below together with references to their sources.

Table 34. EU Taxonomy Technical Screening Criteria for buildings

Environmental goals and legal references	Requirement
Climate Change Mitigation Activity 7.1: new construction of buildings, In Annex I to Regulation (EU) 2021/2139)	Constructions of new buildings for which: 1. The Primary Energy Demand (PED) (281), defining the energy performance of the building resulting from the construction, is at least 10 % lower than the threshold set for the nearly zero-energy building (NZEB) requirements in national measures implementing Directive 2010/31/EU of the European Parliament and of the Council (282). The energy performance is certified using an as built Energy Performance Certificate (EPC). 2. For buildings larger than 5000 m2 (283), upon completion, the building resulting from the construction undergoes testing for air-tightness and thermal integrity (284), and any deviation in the levels of performance set at the design stage or defects in the building envelope are disclosed to investors and clients. As an alternative; where robust and traceable quality control processes are in place during the construction process this is acceptable as an alternative to thermal integrity testing. 3. For buildings larger than 5000 m 2 (285), the life-cycle Global Warming Potential (GWP) (286) of the building resulting from the construction has been calculated for each stage in the life cycle and is disclosed to investors and clients on demand.
Climate Change Mitigation Activity 7.2: renovation of existing buildings. In Annex I to Regulation (EU) 2021/2139)	The building renovation complies with the applicable requirements for major renovations (298). Alternatively, it leads to a reduction of primary energy demand (PED) of at least 30% (299). ***298 says****: As set in the applicable national and regional building regulations for 'major renovation' implementing Directive 2010/31/EU. The energy performance of the building or the renovated part that is upgraded meets cost-optimal minimum energy performance requirements in accordance with the respective directive.* ***299 says****: The initial primary energy demand and the estimated improvement is based on a detailed building survey, an energy audit conducted by an accredited independent expert or any other transparent and proportionate method, and validated through an Energy Performance Certificate. The 30 % improvement results from an actual reduction in primary energy demand (where the reductions in net primary energy demand through renewable energy sources are not taken into account), and can be achieved through a succession of measures within a maximum of three years.*
Climate Change Adaptation Activity 7.1: new construction of buildings, and Activity 7.2: renovation of existing buildings. In Annex II to Regulation (EU) 2021/2139)	1. The economic activity has implemented physical and non-physical solutions ('adaptation solutions') that substantially reduce the most important physical climate risks that are material to that activity. 2. The physical climate risks that are material to the activity have been identified from those listed in Appendix A to this Annex by performing a robust climate risk and vulnerability assessment with the following steps: (a) screening of the activity to identify which physical climate risks from the list in Appendix A to this Annex may affect the performance of the economic activity during its expected lifetime; (b) where the activity is assessed to be at risk from one or more of the physical climate risks listed in Appendix A to this Annex, a climate risk and vulnerability assessment to assess the materiality of the physical climate risks on the economic activity; (c) an assessment of adaptation solutions that can reduce the identified physical climate risk.

Environmental goals and legal references	Requirement
	The climate risk and vulnerability assessment is proportionate to the scale of the activity and its expected lifespan, such that: (a) for activities with an expected lifespan of less than 10 years, the assessment is performed, at least by using climate projections at the smallest appropriate scale; (b) for all other activities, the assessment is performed using the highest available resolution, state-of-the-art climate projections across the existing range of future scenarios (566) consistent with the expected lifetime of the activity, including, at least, 10 to 30 year climate projections scenarios for major investments. 3. The climate projections and assessment of impacts are based on best practice and available guidance and take into account the state-of-the-art science for vulnerability and risk analysis and related methodologies in line with the most recent Intergovernmental Panel on Climate Change reports (567), scientific peer-reviewed publications and open source (568) or paying models. 4. The adaptation solutions implemented: (a) do not adversely affect the adaptation efforts or the level of resilience to physical climate risks of other people, of nature, of cultural heritage, of assets and of other economic activities; (b) favour nature-based solutions (569) or rely on blue or green infrastructure (570) to the extent possible; (c) are consistent with local, sectoral, regional or national adaptation plans and strategies; (d) are monitored and measured against pre-defined indicators and remedial action is considered where those indicators are not met; (e) where the solution implemented is physical and consists in an activity for which technical screening criteria have been specified in this Annex, the solution complies with the do no significant harm technical screening criteria for that activity. **Appendix A:** *The physical climate risks that are material to the activity have been identified from those listed in the table in Section II of this Appendix by performing a robust climate risk and vulnerability assessment with the following steps:* *(a) screening of the activity to identify which physical climate risks from the list in Section II of this Appendix may affect the performance of the economic activity during its expected lifetime;* *(b) where the activity is assessed to be at risk from one or more of the physical climate risks listed in Section II of this Appendix, a climate risk and vulnerability assessment to assess the materiality of the physical climate risks on the economic activity;* *(c) an assessment of adaptation solutions that can reduce the identified physical climate risk.* *The climate risk and vulnerability assessment is proportionate to the scale of the activity and its expected lifespan, such that:* *(a) for activities with an expected lifespan of less than 10 years, the assessment is performed, at least by using climate projections at the smallest appropriate scale;* *(b) for all other activities, the assessment is performed using the highest available resolution, state-of-the-art climate projections across the existing range of future scenarios(1) consistent with the expected lifetime of the activity, including, at least, 10 to 30 year climate projections scenarios for major investments.* *The climate projections and assessment of impacts are based on best practice and available guidance and take into account the state-of-the-art science for vulnerability and risk analysis and related methodologies in line with the most recent Intergovernmental Panel on Climate Change reports(2), scientific peer-reviewed publications, and open source(3) or paying models.* *For existing activities and new activities using existing physical assets, the economic operator implements physical and non-physical solutions (‘adaptation solutions’), over a period of time of up to five years, that reduce the most important identified physical climate risks that are material*

Environmental goals and legal references	Requirement
	to that activity. An adaptation plan for the implementation of those solutions is drawn up accordingly. *For new activities and existing activities using newly-built physical assets, the economic operator integrates the adaptation solutions that reduce the most important identified physical climate risks that are material to that activity at the time of design and construction and has implemented them before the start of operations.* *The adaptation solutions implemented do not adversely affect the adaptation efforts or the level of resilience to physical climate risks of other people, of nature, of cultural heritage, of assets and of other economic activities; are consistent with local, sectoral, regional or national adaptation strategies and plans; and consider the use of nature- based solutions (4) or rely on blue or green infrastructure (5) to the extent possible.*

	Temperature-related	Wind-related	Water-related	Solid-mass related
Chronic	Changing temperature (air, freshwater, marine water)	Changing wind patterns	Changing precipitation patterns and types (rain, hail, snow/ice)	Coastal erosion
	Heat stress		Precipitation or hydrological variability	Soil degradation
	Temperature variability		Ocean acidification	Soil erosion
	Permafrost thawing		Saline intrusion	Solifluction
			Sea level rise	
			Water stress	
Acute	Heat wave	Cyclone, hurricane, typhoon	Drought	Avalanche
	Cold wave/frost	Storm (including blizzards, dust and sandstorms)	Heavy precipitation (rain, hail, snow/ice)	Landslide
	Wildfire	Tornado	Flood (coastal, fluvial, pluvial, ground water)	Subsidence
			Glacial lake outburst	

Environmental goals and legal references	Requirement
Transition to a circular economy Activity 3.1: new construction of buildings. Annex I to Regulation (EU) 2021/2139)	1. All generated construction and demolition waste is treated in accordance with Union waste legislation and with the full checklist of the EU Construction and Demolition Waste Management Protocol, in particular by setting sorting systems and pre-demolition audits(74). The preparing for re-use(75) or recycling(76) of the non-hazardous construction and demolition waste generated on the construction site is at least 90 % (by mass in kilogrammes), excluding backfilling(77). This excludes naturally occurring material referred to in category 17 05 04 in the European List of Waste established by Decision 2000/532/EC. The operator of the activity demonstrates

Environmental goals and legal references	Requirement
	compliance with the 90 % threshold by reporting on the Level(s) indicator 2.2(78) using the Level 2 reporting format for different waste streams. 2. The life-cycle Global Warming Potential (GWP) of the building resulting from the construction has been calculated for each stage in the life cycle and is disclosed to investors and clients on demand(79). 3. Construction designs and techniques support circularity via the incorporation of concepts for design for adaptability and deconstruction as outlined in Level(s) indicators 2.3 and 2.4 respectively. Compliance with this requirement is demonstrated by reporting on the Level(s) indicators 2.3(80) and 2.4(81) at Level 2. 4. The use of primary raw material in the construction of the building is minimised through the use of secondary raw materials(82). The operator of the activity ensures that the three heaviest material categories used to construct the building, measured by mass in kilogrammes, comply with the following maximum total amounts of primary raw material used: (a) for the combined total of concrete(83), natural or agglomerated stone, a maximum of 70 % of the material comes from primary raw material; (b) for the combined total of brick, tile, ceramic, a maximum of 70 % of the material comes from primary raw material; (c) for bio-based materials(84), a maximum of 80 % of the total material comes from primary raw material; (d) for the combined total of glass, mineral insulation, a maximum of 70 % of the total material comes from primary raw material; (e) for non-biobased plastic, a maximum of 50 % of the total material comes from primary raw material; (f) for metals, a maximum of 30 % of the total material comes from primary raw material; (g) for gypsum, a maximum of 65 % of the material comes from primary raw material. The thresholds are calculated by subtracting the secondary raw material from the total amount of each material category used in the works measured by mass in kilogrammes. Where the information on the recycled content of a construction product is not available, it is to be counted as comprising 100 % primary raw material. In order to respect the Waste Hierarchy and thereby favour re-use over recycling, re-used construction products, including those containing non-waste materials reprocessed on site, are to be counted as comprising zero primary raw material. Compliance with this criterion is demonstrated by reporting in accordance with the Level(s) indicator 2.1(85). 5. The operator of the activity uses electronic tools to describe the characteristics of the building as built, including the materials and components used, for the purpose of future maintenance, recovery, and reuse, for example using EN ISO 22057:2022 to provide Environmental Product Declarations(86). The information is stored in a digital format and is made available to investors and clients on demand. In addition, the operator ensures the long-term preservation of this information beyond the useful life of the building by using the information managing systems provided by national tools, such as cadastre or public register.
Activity 3.2: renovation of existing buildings. Annex I to Regulation (EU) 2021/2139)	1. All generated construction and demolition waste is treated in accordance with Union waste legislation and the full checklist of the EU Construction and Demolition Waste Management Protocol, in particular by setting sorting systems and pre-demolition audits(98). The preparing for re-use(99) or recycling(100) of the non-hazardous construction and demolition waste generated on the construction site is at least 70 % (by mass in kilogrammes), excluding backfilling(101). This excludes naturally occurring material referred to in category 17 05 04 in the European List of Waste established by Decision 2000/532/EC. The operator of the activity demonstrates compliance with the 70 % threshold by reporting on the Level(s) indicator 2.2(102) using the Level 2 reporting format for different waste streams.

Environmental goals and legal references	Requirement
	2. The life cycle Global Warming Potential (GWP)(103) of the building's renovation works has been calculated for each stage in the life cycle, from the point of renovation, and is disclosed to investors and clients on demand. 3. Construction designs and techniques support circularity via the incorporation of concepts for design for adaptability and deconstruction as outlined in Level(s) indicators 2.3 and 2.4 respectively. The operator of the activity demonstrates compliance with this requirement by reporting on the Level(s) indicators 2.3(104) and 2.4(105)at Level 2. 4. At least 50 % of the original building is retained. This is to be calculated based on the gross external floor area retained from the original building using the applicable national or regional measurement methodology, alternatively using the definition of 'IPMS 1' contained in the International Property Measurement Standards(106). 5. The use of primary raw material in the renovation of the building is minimised through the use of secondary raw materials(107). The operator of the activity ensures that the three heaviest material categories that have been newly added to the building in the renovation of the building, measured by mass in kilogrammes, comply with the following thresholds regarding the maximum amount of primary raw material used: (a) for the combined total of concrete(108), natural or agglomerated stone, a maximum of 85 % of the material comes from primary raw material; (b) for the combined total of brick, tile, ceramic, a maximum of 85 % of the material comes from primary raw material; (c) for bio-based materials(109), a maximum of 90 % of the material comes from primary raw material; (d) for the combined total of glass, mineral insulation, a maximum of 85 % of the material comes from primary raw material; (e) for non-biobased plastic, a maximum of 75 % of the material comes from primary raw material; (f) for metals, a maximum of 65 % of the material comes from primary raw material; (g) for gypsum, a maximum of 83 % of the material comes from primary raw material. The thresholds are calculated by subtracting the secondary raw material from the total amount of each material category used in the works measured by mass in kilogrammes. Where the information on the recycled content of the construction product is not available, it is to be counted as comprising 100 % primary raw material. In order to respect the Waste Hierarchy and thereby favour re-use over recycling, re-used construction products, including those containing non-waste materials reprocessed on site, are to be counted as comprising zero primary raw material. Compliance with this criterion is demonstrated by reporting in accordance with the Level(s) indicator 2.1(110).

Appendix VIII: Article 29 of the Renewable Energy Directive

Article 29 of the RED consists of 15 paragraphs and many more subparagraphs which are reproduced in Appendix VII, together with yellow highlights to try and identify those parts that are relevant to wood and/or other biomass harvesting.

"1. Energy from biofuels, bioliquids and biomass fuels shall be taken into account for the purposes referred to in points (a), (b) and (c) of this subparagraph only if they fulfil the sustainability and the greenhouse gas emissions saving criteria laid down in paragraphs 2 to 7 and 10:

(a) contributing towards the renewable energy shares of Member States and the targets set in Article 3(1), Article 15a(1), Article 22a(1), Article 23(1), Article 24(4), and Article 25(1);

(b) measuring compliance with renewable energy obligations, including the obligation laid down in Article 25;

(c) eligibility for financial support for the consumption of biofuels, bioliquids and biomass fuels.

However, biofuels, bioliquids and biomass fuels produced from waste and residues, other than agricultural, aquaculture, fisheries and forestry residues, are required to fulfil only the greenhouse gas emissions saving criteria laid down in paragraph 10 in order to be taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of this paragraph. In the case of the use of mixed wastes, Member States may require operators to apply mixed waste sorting systems that aim to remove fossil materials. This subparagraph shall also apply to waste and residues that are first processed into a product before being further processed into biofuels, bioliquids and biomass fuels.

Electricity, heating and cooling produced from municipal solid waste shall not be subject to the greenhouse gas emissions saving criteria laid down in paragraph 10.

Biomass fuels shall fulfil the sustainability and greenhouse gas emissions saving criteria laid down in paragraphs 2 to 7 and 10 if used:

(a) in the case of solid biomass fuels, in installations producing electricity, heating and cooling with a total rated thermal input equal to or exceeding 7,5 MW;

(b) in the case of gaseous biomass fuels, in installations producing electricity, heating and cooling with a total rated thermal input equal to or exceeding 2 MW;

(c) in the case of installations producing gaseous biomass fuels with the following average biomethane flow rate:

(i) above 200 m3 methane equivalent/h measured at standard conditions of temperature and pressure, namely 0 °C and 1 bar atmospheric pressure;

(ii) if biogas is composed of a mixture of methane and non-combustible other gas, for the methane flow rate, the threshold set out in point (i), recalculated proportionally to the volumetric share of methane in the mixture.

Member States may apply the sustainability and greenhouse gas emissions saving criteria to installations with lower total rated thermal input or biomethane flow rate.

The sustainability and the greenhouse gas emissions saving criteria laid down in paragraphs 2 to 7 and 10 shall apply irrespective of the geographical origin of the biomass.

2. Biofuels, bioliquids and biomass fuels produced from waste and residues derived not from forestry but from agricultural land shall be taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 only where operators or national authorities have monitoring or management plans in place in order to address the impacts on soil quality and soil carbon. Information about how those impacts are monitored and managed shall be reported pursuant to Article 30(3).

3. Biofuels, bioliquids and biomass fuels produced from agricultural biomass taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 shall not be made from raw material obtained from land with a high biodiversity value, namely land that had one of the following statuses in or after January 2008, irrespective of whether the land continues to have that status:

(a) primary forest and other wooded land, namely forest and other wooded land of native species, where there is no clearly visible indication of human activity and the ecological processes are not significantly disturbed; and old growth forests as defined in the country where the forest is located;

(b) highly biodiverse forest and other wooded land which is species-rich and not degraded, and has been identified as being highly biodiverse by the relevant competent authority, unless evidence is provided that the production of that raw material did not interfere with those nature protection purposes;

(c) areas designated:

(i) by law or by the relevant competent authority for nature protection purposes, unless evidence is provided that the production of that raw material did not interfere with those nature protection purposes; or

(ii) for the protection of rare, threatened or endangered ecosystems or species recognised by international agreements or included in lists drawn up by intergovernmental organisations or the International Union for the Conservation of Nature, subject to their recognition in accordance with Article 30(4), first subparagraph, unless evidence is provided that the production of that raw material did not interfere with those nature protection purposes;

(d) highly biodiverse grassland spanning more than one hectare that is:

(i) natural, namely grassland that would remain grassland in the absence of human intervention and that maintains the natural species composition and ecological characteristics and processes; or

(ii) non-natural, namely grassland that would cease to be grassland in the absence of human intervention and that is species-rich and not degraded and has been identified as being highly biodiverse by the relevant competent authority, unless evidence is provided that the harvesting of the raw material is necessary to preserve its status as highly biodiverse grassland; or

(e) heathland.

Where the conditions set out in paragraph 6, points (a)(vi) and (vii), are not met, the first subparagraph of this paragraph, with the exception of point (c), also applies to biofuels, bioliquids and biomass fuels produced from forest biomass.

The Commission may adopt implementing acts further specifying the criteria by which to determine which grassland is to be covered by the first subparagraph, point (d), of this paragraph. Those implementing acts shall be adopted in accordance with the examination procedure referred to in Article 34(3).

4. Biofuels, bioliquids and biomass fuels produced from agricultural biomass taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 shall not be made from raw material obtained from land with high-carbon stock, namely land that had one of the following statuses in January 2008 and no longer has that status:

(a) wetlands, namely land that is covered with or saturated by water permanently or for a significant part of the year;

(b) continuously forested areas, namely land spanning more than one hectare with trees higher than five metres and a canopy cover of more than 30 %, or trees able to reach those thresholds in situ;

(c) land spanning more than one hectare with trees higher than five metres and a canopy cover of between 10 % and 30 %, or trees able to reach those thresholds in situ, unless evidence is provided that the carbon stock of the area before and after conversion is such that, when the methodology laid down in Part C of Annex V is applied, the conditions laid down in paragraph 10 of this Article would be fulfilled.

This paragraph shall not apply if, at the time the raw material was obtained, the land had the same status as it had in January 2008.

Where the conditions set out in paragraph 6, points (a)(vi) and (vii), are not met, the first subparagraph of this paragraph, with the exception of points (b) and (c), and the second subparagraph of this paragraph also apply to biofuels, bioliquids and biomass fuels produced from forest biomass.

5. Biofuels, bioliquids and biomass fuels produced from agricultural biomass taken into account for the purposes referred to in paragraph 1, first subparagraph, points (a), (b) and (c), shall not be made from raw material obtained *from land that was peatland in January 2008, unless evidence is provided that the cultivation and harvesting of that raw material does not involve drainage of previously undrained soil. Where the conditions set out in paragraph 6, points (a)(vi) and (vii), are not met, this paragraph also applies to biofuels, bioliquids and biomass fuels produced from forest biomass.*

6. Biofuels, bioliquids and biomass fuels produced from forest biomass taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 shall meet the following criteria to minimise the risk of using forest biomass derived from unsustainable production:

(a) the country in which forest biomass was harvested has national or sub-national laws applicable in the area of harvest as well as monitoring and enforcement systems in place ensuring:

(i) the legality of harvesting operations;

(ii) forest regeneration of harvested areas;

(iii) that areas designated by international or national law or by the relevant competent authority for nature protection purposes, including in wetlands, grassland, heathland and peatlands, are protected with the aim of preserving biodiversity and preventing habitat destruction;

(iv) that harvesting is carried out considering maintenance of soil quality and biodiversity in accordance with sustainable forest management principles, with the aim of preventing any adverse impact, in a way that avoids harvesting of stumps and roots, degradation of primary forests, and of old growth forests as defined in the country where the forest is located, or their conversion into plantation forests, and harvesting on vulnerable soils, that harvesting is carried out in compliance with maximum thresholds for large clear-cuts as defined in the country where the forest is located and with locally and ecologically appropriate retention thresholds for deadwood extraction and that harvesting is carried out in compliance with requirements to use logging systems that minimise any adverse impact on soil quality, including soil compaction, and on biodiversity features and habitats;

(v) that harvesting maintains or improves the long-term production capacity of the forest;

(vi) that forests in which the forest biomass is harvested do not stem from the lands that have the statuses referred to in paragraph 3, points (a), (b), (d) and (e), paragraph 4, point (a), and paragraph 5, respectively under the same conditions of determination of the status of land specified in those paragraphs; and

(vii) that installations producing biofuels, bioliquids and biomass fuels from forest biomass, issue a statement of assurance, underpinned by company-level internal processes, for the purpose of the audits conducted pursuant to Article 30(3), that the forest biomass is not sourced from the lands referred to in point (vi) of this subparagraph.

(b) when evidence referred to in point (a) of this paragraph is not available, the biofuels, bioliquids and biomass fuels produced from forest biomass shall be taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 if management systems are in place at forest sourcing area level ensuring:

(i) the legality of harvesting operations;

(ii) forest regeneration of harvested areas;

(iii) that areas designated by international or national law or by the relevant competent authority for nature protection purposes, including in wetlands, grassland, heathland and peatlands, are protected with the aim of preserving biodiversity and preventing habitat destruction, unless evidence is provided that the harvesting of that raw material does not interfere with those nature protection purposes;

(iv) that harvesting is carried out considering maintenance of soil quality and biodiversity, in accordance with sustainable forest management principles, with the aim of preventing any adverse impact, in a way that avoids harvesting of stumps and roots, degradation of primary forests, and of old growth forests as defined in the country where the forest is located, or their conversion into plantation forests, and harvesting on vulnerable soils, that harvesting is carried out in compliance with maximum thresholds for large clear-cuts as defined in the country where the forest is located, and with locally and ecologically appropriate retention thresholds for deadwood extraction and that harvesting is carried out in compliance with requirements to use logging systems

that minimise any adverse impact on soil quality, including soil compaction, and on biodiversity features and habitats; and

(v) that harvesting maintains or improves the long-term production capacity of the forest.

7. Biofuels, bioliquids and biomass fuels produced from forest biomass taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 shall meet the following land-use, land-use change and forestry (LULUCF) criteria:

(a) the country or regional economic integration organisation of origin of the forest biomass is a Party to the Paris Agreement and:

(i) it has submitted a nationally determined contribution (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC), covering emissions and removals from agriculture, forestry and land use which ensures that changes in carbon stock associated with biomass harvest are accounted towards the country's commitment to reduce or limit greenhouse gas emissions as specified in the NDC; or

(ii) it has national or sub-national laws in place, in accordance with Article 5 of the Paris Agreement, applicable in the area of harvest, to conserve and enhance carbon stocks and sinks, and provides evidence that reported LULUCF-sector emissions do not exceed removals;

(b) where evidence referred to in point (a) of this paragraph is not available, the biofuels, bioliquids and biomass fuels produced from forest biomass shall be taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 if management systems are in place at forest sourcing area level to ensure that carbon stocks and sinks levels in the forest are maintained, or strengthened over the long term.

7a. The production of biofuels, bioliquids and biomass fuels from domestic forest biomass shall be consistent with Member States' commitments and targets laid down in Article 4 of Regulation (EU) 2018/841 of the European Parliament and of the Council (24) and with the policies and measures described by the Member States in their integrated national energy and climate plans submitted pursuant to Articles 3 and 14 of Regulation (EU) 2018/1999.

7b. As part of their final updated integrated national energy and climate plan to be submitted by 30 June 2024 pursuant to Article 14(2) of Regulation (EU) 2018/1999, Member States shall include all of the following:

(a) an assessment of the domestic supply of forest biomass available for energy purposes in 2021-2030 in accordance with the criteria laid down in this Article;

(b) an assessment of the compatibility of the projected use of forest biomass for the production of energy with the Member States' targets and budgets for 2026 to 2030 laid down in Article 4 of Regulation (EU) 2018/841; and

(c) a description of the national measures and policies ensuring compatibility with those targets and budgets.

Member States shall report to the Commission on the measures and policies referred in the first subparagraph, point (c), of this paragraph as part of their integrated national energy and climate progress reports submitted pursuant to Article 17 of Regulation (EU) 2018/1999.

8. By 31 January 2021, the Commission shall adopt implementing acts establishing the operational guidance on the evidence for demonstrating compliance with the criteria laid down in paragraphs 6 and 7 of this Article. Those implementing acts shall be adopted in accordance with the examination procedure referred to in Article 34(3).

9. By 31 December 2026, the Commission shall assess whether the criteria laid down in paragraphs 6 and 7 effectively minimise the risk of using forest biomass derived from unsustainable production and address LULUCF criteria, on the basis of the available data.

The Commission shall, if appropriate, submit a legislative proposal to amend the criteria laid down in paragraphs 6 and 7 for the period after 2030.

10. The greenhouse gas emission savings from the use of biofuels, bioliquids and biomass fuels taken into account for the purposes referred to in paragraph 1 shall be:

(a) at least 50 % for biofuels, biogas consumed in the transport sector, and bioliquids produced in installations in operation on or before 5 October 2015;

(b) at least 60 % for biofuels, biogas consumed in the transport sector, and bioliquids produced in installations starting operation from 6 October 2015 until 31 December 2020;

(c) at least 65 % for biofuels, biogas consumed in the transport sector, and bioliquids produced in installations starting operation from 1 January 2021;

(d) for electricity, heating and cooling production from biomass fuels used in installations that started operating after 20 November 2023, at least 80 %;

(e) for electricity, heating and cooling production from biomass fuels used in installations with a total rated thermal input equal to or exceeding 10 MW that started operating between 1 January 2021 and 20 November 2023, at least 70 % until 31 December 2029, and at least 80 % from 1 January 2030;

(f) for electricity, heating and cooling production from gaseous biomass fuels used in installations with a total rated thermal input equal to or lower than 10 MW that started operating between 1 January 2021 and 20 November 2023, at least 70 % before they have been operating for 15 years, and at least 80 % after they have been in operation for 15 years;

(g) for electricity, heating and cooling production from biomass fuels used in installations with a total rated thermal input equal to or exceeding 10 MW that started operating before 1 January 2021, at least 80 % after they have been operating for 15 years, at the earliest from 1 January 2026 and at the latest from 31 December 2029;

(h) for electricity, heating and cooling production from gaseous biomass fuels used in installations with a total rated thermal input equal to or lower than 10 MW that started operating before 1 January 2021, at least 80 % after they have been operating for 15 years and at the earliest from 1 January 2026.

An installation shall be considered to be in operation once the physical production of biofuels, biogas consumed in the transport sector and bioliquids, and the physical production of heating and cooling and electricity from biomass fuels has started.

The greenhouse gas emission savings from the use of biofuels, biogas consumed in the transport sector, bioliquids and biomass fuels used in installations producing heating, cooling and electricity shall be calculated in accordance with Article 31(1).

11. Electricity from biomass fuels shall be taken into account for the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 only if it meets one or more of the following requirements:

(a) it is produced in installations with a total rated thermal input below 50 MW;

(b) for installations with a total rated thermal input from 50 to 100 MW, it is produced applying high-efficiency cogeneration technology, or, for electricity-only installations, meeting an energy efficiency level associated with the best available techniques (BAT-AEELs) as defined in Commission Implementing Decision (EU) 2017/1442 (25);

(c) for installations with a total rated thermal input above 100 MW, it is produced applying high-efficiency cogeneration technology, or, for electricity-only installations, achieving an net-electrical efficiency of at least 36 %;

(d) it is produced applying Biomass CO2 Capture and Storage.

For the purposes of points (a), (b) and (c) of the first subparagraph of paragraph 1 of this Article, electricity-only-installations shall be taken into account only if they do not use fossil fuels as a main fuel and only if there is no cost-effective potential for the application of high-efficiency cogeneration technology according to the assessment in accordance with Article 14 of Directive 2012/27/EU.

For the purposes of points (a) and (b) of the first subparagraph of paragraph 1 of this Article, this paragraph shall apply only to installations starting operation or converted to the use of biomass fuels after 25 December 2021. For the purposes of point (c) of the first subparagraph of paragraph 1 of this Article, this paragraph shall be without prejudice to support granted under support schemes in accordance with Article 4 approved by 25 December 2021.

Member States may apply higher energy efficiency requirements than those referred in the first subparagraph to installations with lower rated thermal input.

The first subparagraph shall not apply to electricity from installations which are the object of a specific notification by a Member State to the Commission based on the duly substantiated existence of risks for the security of supply of *electricity. Upon assessment of the notification, the Commission shall adopt a decision taking into account the elements included therein.*

12. For the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1 of this Article, and without prejudice to Articles 25 and 26, Member States shall not refuse to take into account, on other sustainability grounds, biofuels and bioliquids obtained in compliance with this Article. This paragraph shall be without prejudice to public support granted under support schemes approved before 24 December 2018.

13. For the purposes referred to in point (c) of the first subparagraph of paragraph 1 of this Article, Member States may derogate, for a limited period of time, from the criteria laid down in paragraphs 2 to 7 and 10 and 11 of this Article by adopting different criteria for:

(a) installations located in an outermost region as referred to in Article 349 TFEU to the extent that such facilities produce electricity or heating or cooling from biomass fuels and bioliquids or produce biofuels; and

(b) biomass fuels and bioliquids used in the installations referred to in point (a) of this subparagraph and biofuels produced in those installations, irrespective of the place of origin of that biomass, provided that such criteria are objectively justified on the grounds that their aim is to ensure, for that outermost region, access to safe and secure energy and a smooth phase-in of the criteria laid down in paragraphs 2 to 7 and 10 and 11 of this Article and thereby incentivise the transition from fossil fuels to sustainable biofuels, bioliquids and biomass fuels.

The different criteria referred to in this paragraph shall be subject to a specific notification by the relevant Member State to the Commission.

14. For the purposes referred to in points (a), (b) and (c) of the first subparagraph of paragraph 1, Member States may establish additional sustainability criteria for biomass fuels.

By 31 December 2026, the Commission shall assess the impact of such additional criteria on the internal market, accompanied, if necessary, by a proposal to ensure harmonisation thereof.

15. Until 31 December 2030, energy from biofuels, bioliquids and biomass fuels may also be taken into account for the purposes referred to in paragraph 1, first subparagraph, points (a), (b) and (c), of this Article, where:

(a) support was granted before 20 November 2023, in accordance with the sustainability and greenhouse gas emissions saving criteria set out in Article 29 in its version in force on 29 September 2020; and

(b) support was granted in the form of a long-term support for which a fixed amount has been determined at the start of the support period and provided that a correction mechanism to ensure the absence of overcompensation is in place."